记忆 VLA 全景调研：从 EventVLA、TRACE、SAI 到 103 篇文献 —— awesome_memory_vla 全文报告

总览图 · 趋势与洞见 · 三篇核心论文深读 · 设计空间矩阵 · 研究机会清单 · 数字口径账本 · 103 篇解读按家族合订

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> 本报告由 `scripts/build_pdfs.py` 从仓库内的趋势报告、三份深读报告、设计空间矩阵与 103 篇论文笔记自动合订。各章的权威版本仍是仓库中的对应 Markdown 文件。所有数字只取自原文或摘要；不同论文的数字建立在各自的任务集与判定口径上，不可直接比大小。

目录

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0. 总览图

图 1 · 103 项工作的时间线：按十个家族分泳道、按 arXiv 提交年月定位，★ 为三篇核心论文，橙色竖带为 2026–06 汇合月

Figure 1 · Timeline of 103 works: one lane per family, positioned by arXiv month; ★ = the three core papers; orange band = the June–2026 convergence

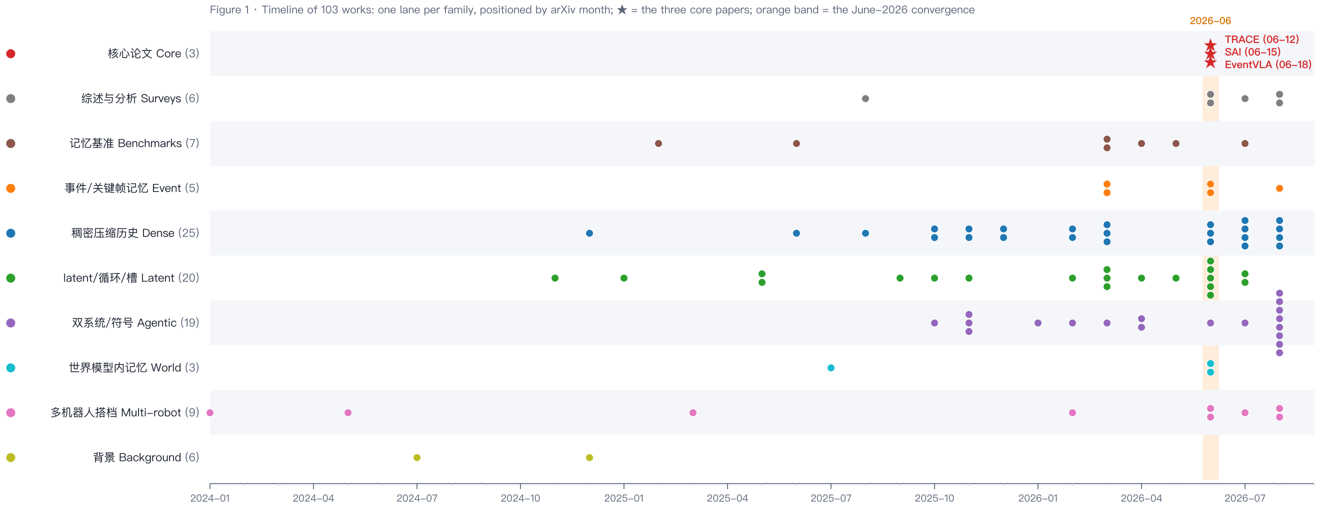

图 2 · 记忆 VLA 的设计空间：十个家族、二十六个子类——与 README 第 1–10 节一一对应

Figure 2 · The design space of memory for VLA policies: ten families, twenty–six sub–classes, matching README sections 1–10

1. 趋势与洞见

> 本报告是 awesome_memory_vla 的综合判断部分。三篇核心论文的逐篇深读见 EventVLA、TRACE、SAI；全部 103 篇条目见 README。所有数字只取自原文或摘要，不同论文的数字建立在各自的任务集和判定口径上，不能直接比大小。

1. 这个领域正在发生什么

1.1 时间线：2026 年 6–8 月的爆发

按提交月份统计仓库里 97 篇非背景文献：2025 年全年 24 篇，2026 年 1–5 月 23 篇，**2026 年 6 月 20 篇、7 月 10 篇、8 月 17 篇**。三篇核心论文全部出自 6 月中旬（TRACE 6 月 12 日、SAI 6 月 15 日、EventVLA 6 月 18 日），和 KEMO（6 月 22 日）、μVLA（6 月 10 日）、MemoryWAM（6 月 18 日）、WeaveLA（6 月 16 日）、PRISM/ReMemBench（6 月 15 日）、长上下文扩散策略研究（6 月 15 日）挤在同一个月。这不是巧合：RMBench（3 月 1 日）、RoboMME（3 月 4 日）、RoboMemArena（5 月 11 日）三个记忆基准在春天相继落地，给了方法论文一个能对齐比较的靶子，夏天的论文就是对这些靶子的第一轮回应。

1.2 一个共同的出发点

几乎每篇论文的第一段都在说同一件事：VLA 是马尔可夫的（`a_t = π(o_t, l)`），而操作任务不是。差别在于各家对”为什么不是”的诊断，这决定了他们的记忆设计：

诊断	典型表述	代表工作	记忆设计倾向
证据消失	掀盖看一眼就盖上；线索出现后离开视野	EventVLA、TRACE、KEMO、UniMem	稀疏、显式、可寻址
状态混叠	视觉相同的状态对应不同阶段（第 2 次和第 3 次按按钮）	KEMO、TFP、ChainVLA、WeaveLA、CAMP	任务进度信念、事件计数

诊断	典型表述	代表工作	记忆设计倾向
空间遗忘	单腕部相机看不到离开视野的物体	AtlasVLA、AnchorVLA4D、SAM2Act+	持久空间状态、初始锚帧
搭档不可观	对方处于哪个阶段只能从历史推	SAI、Duet	历史 token + 训练分布覆盖
缺乏进度追踪	冻结 VLA 开环执行动作块，不知道自己做到哪了	AGM、HELM、BATON、HyMeS	外部符号状态 + 验证器

1.3 五个技术家族

把 97 篇按”记忆放在哪、以什么形式存在”分成五类（README 的第 3–7 节）：

1. **事件 / 关键帧记忆**（EventVLA、KEMO、Keyframe-Chaining、UniMem、WeaveLA、Bi-HIL、MemoryWAM 的锚帧层、MemER 的高层选帧）：只在”发生了什么”的时刻写，存的是原图或原图的 token。
2. **稠密压缩历史**（MemoryVLA / MemoryVLA++、ContextVLA、CronusVLA、HAMLET、NativeMEM、StreamPI、TempoFit、DySta、FibVLA、PRISM、HALO、LaMem-VLA、Remember Smarter）：每帧都进，靠压缩、token 化、KV 复用或采样控制成本。
3. **latent 状态 / 循环 / 槽记忆**（TRACE、μVLA、VPWEM、VQ-Memory、MemoAct、RB-VLA、MTIL、DSSP、Chronos、TFP、CAMP、GMP、ELMUR、AEM、FM-VLA）：历史被压成固定大小的向量或槽，随时间递归更新。
4. **双系统 / 智能体 / 符号记忆**（MEM、MemER、MAP-VLA、Notes-to-Self、显式语言记忆、CodeGraphVLP、HyMeS、HELM、Harness VLA、AGM、PonderPounce、BATON、OnEvoMemory、ChainVLA）：高层用 VLM / LLM / 代码维护文本、图或进度指针，低层 VLA 执行。
5. **世界模型内的记忆**（MemoryWAM、MemoryVAM、TriVLA、MemoryVLA++ 的想象分支）：记忆服务于预测未来，再由预测服务于动作。

第六个方向——**多机器人的搭档记忆**（SAI、HATS、Duet、Tri-Manual、AutoIntervene）——目前只有零星工作，但它和前五类共享同一个数学问题。

2. 设计空间：六个必须回答的问题

任何记忆 VLA 都要回答下面六个问题。把三篇核心论文和代表性周边工作填进去，能看清各自站在哪里：

问题	EventVLA	TRACE	SAI	其他典型答案
存什么（内容）	原始图像帧	视觉+状态的 512 维 latent	GRU 隐状态	单 token/帧（NativeMEM）、K/V（TempoFit）、文本（MEM、Notes-to-Self）、语义图（CodeGraphVLP）、力序列（FM-VLA）、体素状态（AtlasVLA）
何时写（触发）	学到的未来关键帧概率 + NMS + 冷却	每步写，门控决定写多少	每步	运动学+视觉规则（KEMO）、事件分类器（UniMem）、子目标完成（WeaveLA）、物理验证后才推进（AGM）、价值引导（OnEvoMemory）
写到哪 / 从哪读（寻址）	时间顺序拼接，注意力自找	机器人轨迹的路径签名	无（单向量）	内容相似度（MemoryVLA、HALO）、进度感知查询（Keyframe-Chaining）、轨迹相似度检索演示（MAP-VLA）、LRU（ELMUR）
存多少（容量）	5 帧 + 3–4 锚帧	K = 4/6 槽	30 步	固定 token 数（VPWEM、μVLA 的 m 个记忆 token）、随历史线性增长（帧堆叠、StreamPI）、文本无界（MEM）

问题	EventVLA	TRACE	SAI	其他典型答案
接在哪（集成点）	VLM 视觉编码器输入	动作头前的适配器	解码器输入	动作专家的条件（TFP、WeaveLA、FM-VLA）、KV 缓存（TempoFit）、提示（MAP-VLA、PonderPounce）
靠什么学（监督）	Qwen3-VL 自动关键帧标签 + 软标签 + 师生课程	无标签，三项稳定器	数据课程 + 干预	时间对比（HAMLET）、过去 token 预测（PTP）、VQA 蒸馏（HALO）、TBPTT（μVLA）、只靠模仿损失（多数）

读表的方式：两个极端已经被同一个月的两篇论文占住——EventVLA 在"何时写"上下最重的注，TRACE 在"从哪读"上下最重的注。中间地带（既学写入时机、又学寻址）还是空的。

3. 跨基准的证据

三个 2026 年基准上目前公开的数字（各自口径，按论文自报）：

RMBench（9 任务，RoboTwin 2.0）

方法	平均	备注
$\pi_{0.5}$（无记忆）	10.4–11.2	EventVLA 与 Chronos 各自报告
MemoryVLA (QwenOFT)	41.7	EventVLA 复现
Mem-0（RMBench 自带模块化策略）	42.0	
ChainVLA	62.8	1.2B，去掉 Motion Tail 掉到 11.2，去掉 Progress Context 掉到 3.0
EventVLA（仅锚帧）	67.8	未启用 KEM
Chronos	73.6	参数量比 $\pi_{0.5}$ 少 10 倍，比 Mem-0 少 30 倍

RoboMME（16 任务，时间 / 空间 / 物体 / 过程记忆，$\pi_{0.5}$ 骨干）

方法	数字
$\pi_{0.5}$ 当前观测	17.93
FrameSamp+Modul（RoboMME 自带最强变体）	44.51（9× 数据时 57.88）
PonderPounce 0.8B / 9B	50.04 / 60.83（9× 数据时 75.54）
WeaveLA	最难重复切片 SwingXtimes N=3 从 0% 到 47.8%，单次执行任务不变
AGM	Counting 子集上超过最强记忆基线（PickXTimes、BinFill）

RoboMemArena（26 任务，平均 >1000 步，68.9% 子任务依赖记忆）

方法	累计成功 / 任务成功
$\pi_{0.5}$	52.5 / 41.3
PrediMem（基准自带双系统）	比 HyMeS 低 4.5 / 14.5
HyMeS（技能在权重、记忆在代码）	66.2 / 60.1
BATON	相对 SoTA +14.9 / +11.6

MIKASA-Robo（32 任务，RL 出身）

方法	数字
MemoryVLA	41.2（比 CogACT/π_0 +11.8）
VPWEM	比扩散策略与 VLA 高 20% 以上
μVLA（最小循环记忆）	五个训练任务 0.42 → 0.84，留出任务 0.07 → 0.23
ELMUR	23 个任务里 21 个最好，聚合成功率比此前最好高约 70%

真机延迟证据 / 瞬态证据

论文	数字
EventVLA（ARX ACONE，4 任务 × 20）	90 / 60 / 90 / 75 vs π_MEM 50 / – / 30 / 40
TRACE（5 任务 × 25）	ACT 25.50 → 69.23，DP 25.00 → 59.53，$\pi_{0.5}$ 50.47
KEMO（双臂，830–2846 步）	任务成功 +23.6，阶段完成 +34.1（相对 $\pi_{0.5}$）
NativeMEM	仿真 32.4 → 84.0，真机最高 98.7，20% 数据即可竞争
UniMem	仿真 93.4 vs 固定间隔采样 68.2；真机 80.0 vs 分层基线 43.5
Chronos（单 RGB 双臂）	4 任务 78%，记忆依赖子集 72%，$\pi_{0.5}$ 为 7% / 0%

三个观察：第一，RMBench 上”只用初始帧 + 近期帧”已经能到 67.8%，说明这套基准的记忆需求偏浅，Chronos 的 73.6% 也是靠全历史 SSM 而非显式事件记忆拿到的；第二，RoboMME 和 RoboMemArena 上领先的都是双系统 / 智能体方案（PonderPounce、HyMeS、BATON），说明当任务需要**计数和过程记忆**时，符号化的进度状态目前比端到端 latent 更可靠；第三，真机瞬态证据任务上端到端事件记忆（EventVLA、KEMO、UniMem）的绝对成功率最高。**没有一种记忆在所有基准上领先**——RoboMME 论文的结论”记忆表征的效果高度依赖任务”被后续一整年的结果反复印证。

4. 十个洞见

洞见 1：前沿从”存多少”转到了”何时写”

2025 年的记忆 VLA 在比谁的窗口长、压缩得好（CronusVLA、ContextVLA、MemoryVLA）。2026 年中的论文几乎同时转向**写入时机**：EventVLA 学未来关键帧概率，KEMO 用运动学 + 视觉变化检测事件，UniMem 训练事件分类器，WeaveLA 在子目标完成时触发，MemoryWAM 用事件边界锚帧，TFP 的机制分析发现记忆写入增益在操作事件附近比非事件阶段大约 6 倍，OnEvoMemory 从 rollout 结果学该留什么。这些工作从不同架构出发，落到同一个结论：**历史里绝大多数帧是当前帧的冗余，值得写的只有状态转移的瞬间。**

洞见 2：”Present but Not Remembered”给出了机理解释

Liao & Cao 用逐层线性探针和因果交换干预审计三个冻结 VLA，发现：过去帧内容在全网络都可线性解码，但**只属于历史、当前帧没有的信息几乎不存在**——存下来的历史基本是现在的副本；历史只有在当前帧严重退化时才被因果地使用。这解释了为什么帧堆叠收益不稳定（ContextVLA 的出发点），也解释了为什么稀疏事件帧有效：事件帧携带的正是当前帧没有的信息。它给记忆增强定了一条设计准则：**注入只属于过去的信息，而不是更多的历史。** TRACE 的签名增量 `δ_t` 、CAMP 的动作历史压缩、PTP 的过去 token 预测都可以读成这条准则的实例。

洞见 3：原图还是 latent，取决于要恢复多少比特

EventVLA 的隐式记忆库消融（24.9% vs 75.2%）和 TRACE 的 512 维槽（69.23%）看起来矛盾，其实测的是两类信息。EventVLA 的任务要同时记 3–5 个物体的颜色、位置、顺序——高熵、需要空间细节；TRACE 的任务要记的是"来自哪一侧""是哪本书"——一两个比特的分支变量。RoboMME 的结论（表征效果依任务而异）在这里有了可操作的版本：**先估计下游决策需要从过去恢复多少信息，再选表征。** 需要细节的用原图或密集 token（EventVLA、NativeMEM、UniMem 的密集空间记忆），需要分支变量的用槽或信念（TRACE、TFP、RB-VLA），需要计数的用符号（AGM 的进度指针、Notes-to-Self 的草稿板）。

洞见 4：寻址是被低估的一轴

多数工作默认"按时间拼接、让注意力自己找"。TRACE 是少数把寻址当一等问题的：轨迹签名做键，保序变换下路由相似度 ~90、倒序 37.8。Keyframe-Chaining 的进度感知查询、MAP-VLA 的轨迹相似度检索、ELMUR 的 LRU 槽是另外三种寻址。寻址的价值在于**写入时刻和读取时刻之间有一个稳定的对应关系**——TRACE 用的是"走过的路"，进度感知方法用的是"做到第几步"。当任务的分支结构和机器人运动不对应时（线索只是颜色差异、动作完全相同），轨迹寻址会退化，这是下一步要解决的边界。

洞见 5：长上下文没那么脆，但虚假相关是真的

Agarwal 等（长上下文扩散策略）系统扫了上下文长度，结论是"naive 加长没文献说的那么脆"，用 UNet + 交叉注意力在常规数据量下多数任务能拿高成功率。但另一条证据链同样扎实：PTP 指出扩散策略会**忽略**过去-未来动作依赖（copycat 的反面），GMP 发现直接加长历史会因分布偏移和过拟合显著掉点（MemMimic 上门控记忆比长历史基线 +30.1），HALO 需要 VLM 先验蒸馏和稀疏注意力来压虚假相关。两者的调和：长上下文能不能用，取决于监督是否把注意力引向任务相关的历史。EventVLA 的软标签 + 师生课程、TRACE 的平衡 / 熵 / 一致性损失、μVLA 的 TBPTT 都是在做这件事。

洞见 6：记忆的可靠性来自"有纪律的状态更新"，不是容量

AGM 的核心论断是：外部记忆如果把"尝试过"当成"完成了"，局部执行错误会变成持久的任务状态错误，所以进度指针只在物理证据验证子目标后才推进。同样的纪律出现在 EventVLA 的 NMS + 冷却（防止同一事件灌满缓冲区）、TRACE 的门控写入（未被路由到的槽几乎不动）、BATON 的验证器智能体（腕部相机确认场景就位才调用 VLA）、HELM 的状态验证器（执行前预测失败）。**记忆写入应当是一个带验证的事务**，这一原则在端到端和智能体两条路线上同时被独立发现。

洞见 7：双系统在计数与过程记忆上仍然领先

RoboMME 和 RoboMemArena 的榜首（PonderPounce、HyMeS、BATON）都是"高层管状态、低层 VLA 执行"。HyMeS 的表述最直接："技能在权重里，记忆在代码里"——低层技能靠模仿学，高层记忆管理靠编码智能体从 rollout 反馈里迭代启发式。EventVLA 对双系统的批评（延迟高、误差传播）在瞬态视觉证据任务上成立，但在"按 3 次、跳过已完成的子任务"这类离散过程记忆上，符号状态目前更稳。PonderPounce 用 MLLM 的原生因果上下文当记忆、异步只传最新的认知 token（p50 78 ms 刷新、25 ms 动作），是在缩小双系统的延迟劣势。**端到端与双系统的分界线正在从"哪个更好"变成"哪类信息该放在哪一层"。**

洞见 8：记忆的成本正在趋近于零

EventVLA 的多帧输入把吞吐从 2.91 Hz 压到 0.94 Hz，是这批论文里最贵的；同期的工作在往另一头走：TRACE 每步 +2.1 ms；NativeMEM 用 VLA 自己的视觉编码器把每帧每视角压成 1 个 token；TempoFit 完全不训练、不加

token，只复用中间层的前缀 K/V；StreamPI 不加任何参数，靠指令锚定的因果注意力和长度外推做流式多帧；DySta 只保留一份静态 token 并复用其 KV 缓存，推理快 2 倍还涨点。**“记忆几乎免费”已经成为可以达到的工程目标，** EventVLA 的三倍延迟会在后续版本里成为首要优化对象。

洞见 9：多机器人是同一个问题的另一张脸

SAI 把”搭档看不全”处理成数据课程问题，但论文自己的消融显示历史 token 是执行可靠性的关键（提前松手 64.5% → 16.1%）。把搭档相位当成一个从历史推断的隐变量，SAI 与 TRACE 的延迟证据设定是同构的——只是”早期线索”换成了”对方刚才做了什么”。目前多机器人工作几乎都在解数据采集（HATS 单人 + 智能体控四臂，Duet 人-人演示预训练，Tri-Manual 离线重排时序），**还没有人把显式记忆模块用在搭档状态估计上**。这是一个明确的空白。

洞见 10：评测在分裂，诊断在成熟

一年之内出现了 RMBench、RoboMME、RoboMemArena、RoboTwin-MeM、ReMemBench、LIBERO-Mem、MemMimic、MemoryRTBench、RuleSafe、Memory-T-Bench、MemoryBench、LongBench 至少 12 套记忆相关基准，各自的任务集和口径不同，跨论文的数字无法比较（第 3 节的表只能按基准分块）。同时，**诊断方法**在成熟：TRACE 的路由相似度 / 分支一致性 + 顺序反转负对照，Present-but-Not-Remembered 的探针 + 因果干预，TFP 的隐状态干预，LongBench 把长时程失败拆成执行鲁棒性和上下文依赖两个机制。下一阶段的评测应当同时报告”成功率”和”记忆到底用了什么”。

5. 开放问题

1. **缓冲区饱和**：EventVLA 的 5 帧 FIFO 在 >10 分钟、事件密集的任务上会挤掉早期证据；TRACE 的固定槽在 2× 历史下一致性降到 0.928。分层（近期 / 事件 / gist，如 MemoryWAM）是目前的折中，还没有在同一基准上和平面记忆正面比过。
2. **寻址的可区分性**：轨迹签名要求分支对应不同的运动；签名维度随状态维度立方增长（17 维 → 5219，深度 4 → 88740）。高维人形 / 灵巧手需要 log-signature 或学习的低维轨迹键。
3. **写入触发的监督来源**：EventVLA 靠 235B 模型离线标注，KEMO 靠规则，UniMem 靠分类器，OnEvoMemory 靠 rollout 结果。哪一种能跨任务泛化，尚无对照。
4. **记忆与动作块长度的耦合**：EventVLA 的前瞻能力随动作块从 50 缩到 15 而崩塌（75.2 → 13.6）；WhyChunking 指出动作块的好处之一正是非马尔可夫表达力。记忆机制与块长度应当联合设计。
5. **错误记忆的安全性**：AGM 指出未经验证的记忆会把局部错误固化；EventVLA 若把错误帧写入缓冲区如何恢复，论文没有讨论。
6. **多机器人的显式记忆**：见洞见 9。
7. **基准统一**：RoboMME 的四类记忆（时间 / 空间 / 物体 / 过程）+ RoboTwin-MeM 的 n 参数化 + TRACE 的阶段进度指标，可以拼成一个共同的报告模板。

6. 三篇论文教给我们的方法论

- **把一个默认做法拆开单独测**。EventVLA 只换记忆表征（原图 vs latent）；TRACE 只换寻址（有无签名路由、有无增量）；SAI 只换数据课程（架构固定）。三篇论文的说服力都来自这种单变量对照。
- **报告代价**。EventVLA 报延迟（0.94 Hz），TRACE 报每步毫秒和参数增量，SAI 报干预数据量占比（30%）。这让读者能判断收益是否值得。
- **用诊断替代猜测**。TRACE 的顺序反转负对照是最值得推广的一个设计：它把”记忆在用历史”从假设变成可测量。

7. 我们的下一步（研究方向建议）

1. **前瞻写入 × 轨迹寻址**：把 EventVLA 的 KEM 头作为写入触发、TRACE 的签名作为槽地址，检验能否在 RoboTwin-MeM（高熵视觉证据）和 TRACE 任务（低熵分支变量）上同时保持性能。假设：稀疏写入解决容量，轨迹寻址解决读写对齐。
2. **搭档相位记忆**：在 SAI 的两机器人设定上，把 A 的 30 步 GRU 换成 TRACE 式签名槽记忆（键 = A 自己的轨迹，内容 = 对 B 的观测），测 Yield/Wait 是否随 B 的延迟长度保持稳定。
3. **记忆诊断基准**：给 RoboMME 四类任务各配一组保序 / 破序历史变换，把路由相似度 / 分支一致性变成所有记忆 VLA 的标准报告项。
4. **写入触发的跨任务迁移**：在同一骨干上对照 EventVLA（学习触发）、KEMO（规则触发）、UniMem（分类器触发）三种写入策略在未见任务上的迁移，回答"写入时机是任务特定的还是通用的"。

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2. EventVLA 深读

> **论文**：EventVLA: Event-Driven Visual Evidence Memory for Long-Horizon Vision-Language-Action Policies
> **作者**：Ganlin Yang, Zhangzheng Tu, Yuqiang Yang, Sitong Mao, Junyi Dong, Tianxing Chen, Jiaqi Peng, Jing Xiong, Jiafei Cao, Jifeng Dai, Wengang Zhou, Yao Mu, Tai Wang（中科大 / 上海 AI Lab / 华为 / 港大 / 清华 / 北大 / 上交）
> **arXiv**：2606.20092（2026-06-18）· 代码与数据：InternRobotics/EventVLA
> **仓库内文件**：英文 PDF · 中文翻译 PDF · 英文版解读

0. 一句话

EventVLA 让 VLA 策略自己预测"接下来这 50 步里哪几帧将来会用得上"，只把这几帧原图存进一个容量为 5 的缓冲区，再和初始帧、近期帧一起送回视觉编码器。这样做在作者新建的 RoboTwin-MeM 基准上把平均成功率从 18.0%（只用固定锚帧）提到 75.2%，在 RMBench 上只用固定锚帧就拿到 67.8%（此前最好的 Mem-0 是 42.0%），真机四个任务成功率 90 / 60 / 90 / 75%。

1. 要解决的问题

标准 VLA 是马尔可夫的：`a_t = π(o_t, l)`，默认所有任务相关信息一直看得见。真实操作里信息经常只出现一瞬间：掀开盖子看一眼颜色再盖上、读一张随机数字卡、看别人用木棍点一遍顺序。这些证据出现后就消失，而动作要在几百步之后才依赖它。作者把这类任务叫"非马尔可夫操作"，并把已有的记忆增强路线分成三类，各有一个具体的失败方式：

路线	代表	失败方式
双系统（高层 VLM 管记忆 + 低层策略执行）	MemER、Mem-0、MEM（π_MEM）	延迟高、误差沿层级传播
循环 / 隐状态压缩	AVA-VLA、Recurrent Memory Transformer	信息瓶颈，细粒度视觉细节被丢掉
帧缓冲（保留历史原图）	MemoryVLA、CronusVLA、ContextVLA、LoLA	无选择地堆帧，冗余淹没稀疏关键证据，计算开销大

论文提出的核心问题只有一句：**VLA 究竟应该在什么时候、保留什么视觉证据，才能既完成任务又不撑爆算力？**

2. 方法

2.1 记忆的结构：锚帧 + 事件帧

把策略改写成 $a_t = \pi(o_t, M_{t-1}, l)$，其中记忆 $M_t = A_t \cup E_t$：

- **基础视觉锚帧 $A_t = \{o_0\} \cup \{o_{t-K}, \dots, o_{t-1}\}$**：初始帧提供不变的全局布局（东西一开始在哪），短期滑窗提供运动与任务进度线索。这部分是规则化的、不需学习。仿真里用 o_0, o_{t-30}, o_{t-15}；真机用 $o_0, o_{t-60}, o_{t-40}, o_{t-20}$。
- **事件关键帧 E_t**：由 KEM 模块动态写入，容量上限 $N_{max} = 5$，FIFO 淘汰。

推理时三部分按时间顺序拼成一个图像序列 $I_{input} = \mathrm{concat}([A_t, E_{t-1}, o_t])$，直接进 VLM 的视觉编码器，让自注意力自己在稀疏历史帧之间建立时间关联。没有独立的记忆读取模块——“读”就是普通的多帧注意力。

2.2 KEM：预测未来关键帧的概率

KEM 是一个挂在动作头旁边的轻量 MLP 头。它的输入是 VLA 自回归 Transformer 最后一层的隐状态 $h_t \in R^{H\times d}$（$H = 50$ 是动作块长度），输出一个长度为 H 的概率向量：

```
p̂_t = σ(KEM_mlp(h_t)) = [p̂_t^1, …, p̂_t^H] ∈ [0,1]^H
```

$\hat{p}_t^i$ 表示”未来第 i 步会是任务关键帧”的概率。之所以按整块预测而不是逐步做二分类，是因为关键事件常常在一个动作块执行到一半时出现又消失——逐步分类器根本来不及。共享隐状态是关键设计：h_t 同时编码了观测和动作查询，所以 KEM 天然知道策略接下来要做什么，能”预约”一次记忆写入。

写入规则：$\hat{p}_t^i \ge \tau_{commit}$（0.55）时，把 $t+i$ 时刻的原图写进 E_t。为了防止同一事件的相邻帧把缓冲区灌满，在线还有两步后处理：

1. **一维 NMS**：在半径 $w = 8$ 的滑窗内只保留局部峰值；
2. **冷却期**：距离上一次写入不足 $C = 10$ 步的候选被丢弃。

2.3 训练：自动标注 + 软标签 + 师生课程

- **标签来源**：用 Qwen3-VL-235B-A22B（8 卡 A800，vLLM）离线看演示视频，每回合均匀采 128 帧、拼多视角，few-shot 提示后直接输出 JSON 格式的关键帧步数。仿真里对照物理引擎的真值，平均时间误差小于 10 步；真机对照人工标注误差在 50 步以内（回合长度 1500–2000 步）。
- **软标签**：物理事件在时间上是模糊的，硬 0/1 标签让预测头不稳定。作者用升余弦核把标签平滑到半径 R 内：$y_t^i = 0.5(1 + \cos(\pi|t+i-t*|/R))$。
- **损失**：$L = L_{action} + \lambda L_{kem}$，$L_{kem}$ 是序列平均的 BCE，$\lambda = 0.1$。
- **课程**：训练早期缓冲区用真值关键帧构造（teacher forcing），概率 α 从 1 线性衰减到 0，逐步换成模型自己的阈值预测。这一步是为了消除”训练用真值、测试用预测”的分布偏移。

2.4 骨干与超参

设置	RMBench / RoboTwin-MeM	真机
基座	QwenOFT（Qwen3-VL-4B-Instruct，StarVLA 代码库）	$\pi_{0.5}$（PaliGemma）

设置	RMBench / RoboTwin-MeM	真机
动作头	OFT，`H = 50`，动作维 14	flow 头，`H = 50`，动作维 32
训练步数	80k，VLM 学习率 1e-5，新模块 1e-4	60k，5e-5 余弦衰减，全局 batch 32
图像分辨率	224×224	224×224
KEM	τ=0.55，N_max=5，w=8，C=10，λ=0.1	同左

3. RoboTwin-MeM 基准

作者认为 RMBench 和 RoboMME 这类基准大多能靠"初始帧 + 近期帧"解决，真正只在交互中间短暂出现的证据没有被单独测过。RoboTwin-MeM 建在 RoboTwin 2.0（SAPIEN）上，8 个双臂任务，每个任务用一个参数 `n ∈ [1, 5]` 标明"必须记住几个中间关键帧"：

任务	n	平均步数	考什么
Rearrange Blocks Hard	1	879	记住搬走的是哪块
Put Back Block Hard	2	1468	记住两块分别去过哪个外侧垫
Pick Objects in Order / Pick the Unhidden Block	3	1124 / 699	掀盖看内容，盖回后按顺序/排除法取
Cover Blocks Hard / Find Seal Stamp / Reproduce Route	4	1544 / 1338 / 1417	多次掀盖记颜色；找印章并放回原盖下；看一遍演示后复现路线（上下文学习）
Press Button Keyframe	2–5	430	读两张数字卡，按对应次数按按钮（计数）

三类能力被拆开测：瞬态证据记忆（掀盖类）、序列/计数（按钮类）、上下文模仿（Reproduce Route）。每个任务 50 条演示，附带与每个动作-观测对对齐的细粒度语言标注。

4. 实验结果

4.1 RMBench：只用锚帧就够了

方法	平均成功率
$\pi_{0.5}$ / X-VLA / QwenOFT（无记忆）	10.4 / 9.8 / 5.6
MemER / Mem-0（双系统）	8.7 / 42.0
MemoryVLA (OpenVLA) / MemoryVLA (QwenOFT)	19.4 / 41.7
EventVLA 去掉初始帧 / 去掉短期窗	33.7 / 23.8
EventVLA（仅视觉锚帧）	**67.8**

RMBench 的 9 个任务依赖持久的空间布局和固定的动作风格，所以作者只部署了锚帧版本。两个消融说明初始帧和短期窗都不可少：去掉任一个就分别跌到 33.7% 和 23.8%。逐任务看，Rearrange Blocks 96%、Put Back Block 95%、Swap Blocks 96%、Cover Blocks 97%，但 Observe and Pick Up 只有 21%，Press Button 只有 3%——计数类任务锚帧解决不了。

4.2 RoboTwin-MeM：KEM 带来的是质变

方法	n=1	n=2	n=3	n=3	n=4	n=4	n=4	n=5	平均
$\pi_{0.5}$	20	19	1	14	0	8	0	0	7.8
MemER	32	4	12	2	0	26	3	5	10.5
MemoryVLA (QwenOFT)	39	0	1	9	1	11	0	25	10.8
EventVLA 仅锚帧	62	13	5	20	0	26	0	18	18.0
EventVLA 锚帧 + KEM	**62**	**93**	**90**	**54**	**94**	**63**	**98**	**48**	**75.2**

（列顺序：Rearrange Blocks Hard、Put Back Block Hard、Pick Objects in Order、Pick the Unhidden Block、Cover Blocks Hard、Find Seal Stamp、Reproduce Route、Press Button Keyframe。）

从 18.0% 到 75.2% 的跳变全部来自 KEM。n=1 的任务锚帧已经够用（62% 不变），n≥2 的任务几乎全靠事件帧。Mem-0 在这套基准上是 0%。

4.3 消融：每个设计选择值多少

变体	平均	说明
完整 EventVLA	75.2	—
隐式记忆库（把关键帧压成一个 latent 向量）	24.9	多事件挤进一个向量，信息瓶颈
硬标签	48.8	预测头不稳定，漏写
去掉 NMS	53.4	同一事件的相邻帧灌满缓冲区，早期证据被 FIFO 挤出
N_max = 2	32.0	容量不够
动作块 30 / 15	31.1 / 13.6	前瞻窗口缩短，KEM 来不及预约写入

“隐式记忆库 24.9 vs 原图 75.2”是全文最有分量的一个数字：它直接反驳了”把历史压成 latent 就够了”的默认做法——至少在需要同时记住 3 到 5 个视觉细节的任务上不够。

4.4 不损伤马尔可夫任务

RoboTwin 2.0 标准任务上，EventVLA 相对 QwenOFT 基座 Easy 80.0→83.8%、Hard 78.0→81.6%，比 $\pi_{0.5}$（82.7 / 76.8）也略高。

4.5 真机：ARX ACONE 双臂

四个任务各 20 次：Find Block Easy / Hard（藏块位置只短暂可见）、Pick-X-Times（读随机数字后计数）、Pick in Order（木棍点一遍顺序）。

方法	Find Block Easy	Find Block Hard	Pick-X-Times	Pick in Order
$\pi_{0.5}$	0–10%	0–10%	0–10%	0–10%
π_MEM（复现）	50	—	30	40
EventVLA	**90**	**60**	**90**	**75**

4.6 代价：推理速度降到三分之一

方法	平均延迟 (s/chunk)	吞吐 (chunks/s)
QwenOFT	0.36	2.91
EventVLA 仅锚帧	0.96	1.07
EventVLA 锚帧 + KEM	1.09	0.94

多帧输入让视觉 token 数量翻了几倍，吞吐从 2.91 Hz 降到 0.94 Hz。作者的辩护是 VLA 通常作为高层规划器搭配高频底层控制器，1 Hz 够用。

5. 判读

5.1 真正的贡献在哪

1. **把”何时写入”从规则变成预测。** 之前的方法要么每帧都写（帧缓冲），要么按固定间隔采样，要么交给一个外挂 VLM 决定。KEM 用策略自己的隐状态预测未来 50 步的写入概率，相当于让策略”预约”记忆。这和动作块预测共享同一个表征，几乎不增加参数。
2. **用一个干净的对照证明了原图优于 latent。** 75.2 vs 24.9 的差距是在同一框架、同一数据、只换记忆表征下测出来的，比跨论文比较可信得多。
3. **用 n 参数化记忆需求。** RoboTwin-MeM 把”要记住几件事”变成任务的显式属性，可以画出成功率随 n 的衰减曲线，这是以前的记忆基准做不到的诊断能力。

5.2 需要留意的地方

1. **KEM 没有在 RMBench 上测。** RMBench 的 67.8% 是纯锚帧版本，KEM 的收益只在作者自建的 RoboTwin-MeM 上体现。两个基准都出自 RoboTwin 系（RMBench 第一作者 Tianxing Chen 也是本文作者），设计取向天然贴合”锚帧 + 关键帧”的假设。RoboMME、MIKASA-Robo 上的表现没有报告。
2. **KEM 与长动作块强耦合。** 动作块从 50 缩到 15 时成功率从 75.2% 掉到 13.6%，比去掉 KEM（18.0%）还低。这意味着 KEM 的前瞻能力是动作块长度的函数；对需要短块高频响应的任务（接触丰富、动态场景），这套机制未必能直接搬过去。
3. **容量 5 + FIFO 的天花板。** 作者在局限里明说超过 10 分钟、事件密集的任务会撑爆缓冲区并挤掉早期证据。RoboTwin-MeM 最难的 n=5 任务（Press Button Keyframe）也只有 48%，n=4 的 Find Seal Stamp 63%，成功率随 n 明显下降。
4. **标注依赖 235B 模型。** 8 卡 A800 跑 Qwen3-VL-235B 做离线标注，这个成本在论文里没有量化。真机上关键帧误差在 50 步以内，靠软标签的时间容忍度吃掉。
5. **三倍延迟。** 0.94 Hz 对桌面双臂任务够用，对移动操作或动态场景是实打实的代价。NativeMEM、TempoFit 这类工作正是在这个维度上做文章（每帧压成 1 个 token / 不加 token 只复用 KV）。
6. **真机每任务 20 次**，没有置信区间；π_MEM 是作者复现版。

5.3 与 TRACE 的对照

两篇论文解决的是同一个问题（决策时刻证据已经消失），但在三个轴上取了相反的设计：

维度	EventVLA	TRACE
记忆内容	原始图像帧	512 维 latent 槽

维度	EventVLA	TRACE
写入触发	学到的"未来关键帧"概率 + NMS + 冷却	每步都写，但由轨迹签名路由到不同槽、门控更新
寻址方式	时间顺序拼接，靠注意力自己找	机器人状态轨迹的路径签名作为键
容量	5 帧 + 锚帧	K = 4 或 6 槽，固定
与骨干的关系	端到端，进 VLM 视觉编码器	适配器外挂，骨干、动作头、损失都不改
推理开销	吞吐 2.91→0.94 Hz	每步 +2.1 ms

两种设计对应两类信息：需要保留的是**高维视觉细节**（颜色、位置、多个物体）还是**低维分支变量**（是哪个来源、走哪条路）。EventVLA 的隐式记忆库消融（24.9%）说明前者压不进一个向量；TRACE 的成功说明后者放进几个槽就够。选哪种记忆，取决于下游要恢复的信息有多少比特。

6. 关联阅读

- **KEMO**（2606.23589，同月）：也是事件驱动关键帧记忆，但事件检测用机器人运动学 + 视觉过滤的规则而不是学习的预测头；关键帧编码为 token 后经交叉注意力和门控残差融合进 VLA。真机相对 $\pi_{0.5}$ 任务成功率 +23.6%。两篇几乎同时出现，说明"事件驱动的稀疏写入"是 2026 年中的共识方向。
- **UniMem**（2608.22869）：单骨干里放一个事件分类器决定何时更新记忆、一个关键帧编码器做密集空间记忆，仿真 93.4% vs 固定间隔采样 68.2%。
- **MemoryWAM**（2606.20562）：世界动作模型里用"近期帧 + 事件边界锚帧 + gist token"三层记忆，与 EventVLA 的锚帧 + 事件帧结构同构。
- **MemER**（2510.20328）：双系统里由高层 VLM 选关键帧；本文把它当作双系统基线，RoboTwin-MeM 上 10.5%。
- **RMBench**（2603.01229）与 **RoboMME**（2603.04639）：本文认为二者可被锚帧解决，RoboTwin-MeM 正是为了补"瞬态证据"这一空白。
- **Present but Not Remembered**（2607.03372）：探针实验发现冻结 VLA 里的历史帧信息基本是当前帧的冗余副本，建议记忆增强应注入"只属于过去"的信息——这正是稀疏事件帧在做的事。

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3. TRACE 深读

> **论文**：TRACE: Trajectory-Routed Causal Memory for Delayed-Evidence Visuomotor Imitation
> **作者**：Zihao Li, Ranpeng Qiu, Yincong Chen, Guoqiang Ren, Weiming Zhi（浙江大学 / 浙江工业大学 / 悉尼大学）
> **arXiv**：2606.14551（2026-06-12，v3）· 项目页：jeong-zju.github.io/trace
> **仓库内文件**：英文 PDF · 中文翻译 PDF · 英文版解读

0. 一句话

TRACE 给 ACT 和 Diffusion Policy 外挂一个 K 个槽（4 或 6 个、每槽 512 维）的固定容量 latent 记忆，用机器人状态轨迹的**路径签名**（path signature）作为读写地址：线索可见时把视觉-状态证据写进由"到这里为止怎么走过来的"决定的槽，线索消失后到了视觉上分不清的分岔点，再用同样的轨迹地址把它读出来。五个真机延迟证据任务上，ACT 从 25.50 提到 69.23 平均阶段进度，Diffusion Policy 从 25.00 提到 59.53，超过用同样数据微调的 $\pi_{0.5}$（50.47）；顺序反转的负对照把路由相似度打到 37.8%，证明它读的是有序历史而不是当前帧。

1. 要解决的问题：延迟证据

作者定义了一类任务：早期出现一个线索（物体身份、来源货架、衣物来自哪一侧），中间一段共享的执行段，之后到一个**分岔点**，此时不同历史下的观测几乎一样，但正确动作不同。形式化地，短窗口长度为 `w` 时存在两条历史使得

```
x_{t-w+1:t} ≈ x'_{t-w+1:t}，但 a*_t(H_t) ≠ a*_t(H'_t)
```

于是 `π(a_t | x_t)` 只能给这两个几乎相同的观测分配同一个动作分布，注定至少错一半。需要的是一个紧凑的因果历史摘要 `m_t = m(H_t)`，使 `π(a_t | x_t, m_t)` 能恢复分支决策，并且接口大小固定。

作者对三种常见补救的批评很具体：加长窗口——线索一旦掉出窗口就失效，且长窗口逼策略自己去猜哪些早期帧还有用；循环网络——分支相关线索会被覆盖、稀释，或者和任务进度信号纠缠在一起；通用外部记忆——写入和读取没有和”到分岔点时的执行历史”绑定。

2. 方法

2.2 记忆内容与记忆地址分离

这是 TRACE 最核心的设计决定：**当前图像和状态提供”记什么”，执行过的轨迹决定”存到哪、从哪取”。**

- 内容：`e_t = φ_x(x_t)`，多视角视觉特征池化 + 本体感受嵌入。
- 地址：对归一化后的 17 维机器人状态路径 `S_t`（底盘平面速度 2 + 偏航角速度 1 + 左右臂各 7 关节）计算截断到深度 `p = 3` 的路径签名

```
Sig_{≤p}(S_t) = (1, ∫dS, ∫∫_{u1<u2} dS⊗dS, ∫∫∫ dS⊗dS⊗dS)
```

去掉标量项后维度 `17 + 17² + 17³ = 5219`，再加一个签名空间的增量 `δ_t = ξ_t − ξ_{t−1}`。两者经学习的投影得到 `g_t = φ_g(ξ_t)`、`Δg_t = φ_Δ(δ_t)`，拼成轨迹键 `q_t = φ_q([g_t, Δg_t])`。

为什么是路径签名而不是时间步或另一个 RNN 隐状态：签名对单调时间重参数化不变（走得快慢不影响地址）、对常数偏移不变（基点签名），但对**顺序**敏感——把历史倒放地址就变了。它还支持流式更新，每步 0.52 ms。签名本身不存视觉线索，只是钥匙。

2.2 签名路由的槽记忆

`K` 个槽 `M_t = {m_{t,1}, …, m_{t,K}}`，每回合开始清零。每一步：

1. **路由**：`ρ_t = φ_ρ(q_t)`，与上一步的槽内容（可加固定槽身份 `η_k` 打破对称）做点积，softmax 得到写入权重 `ω_{t,k}`（公式 6）。路由同时看轨迹键和当前槽内容，所以不是一个固定的哈希。
2. **门控写入**：写入提案 `u_t = φ_w(e_t, q_t)`，候选内容 `m̃_{t,k} = tanh(φ_m(m̃_{t−1,k}, u_t, ρ_t))`，门 `β_{t,k} = ω_{t,k} · σ(φ_β(·))`，更新 `m_{t,k} = (1−β)m_{t−1,k} + β m̃_{t,k}`（公式 7–8）。权重小的槽几乎不动，被选中的槽吸收当前证据。
3. **读出**：用当前证据和路由状态构造查询，对更新后的槽做注意力，`z_t^mem = Σ_k α_{t,k} W_V^r m_{t,k}`（公式 9）。

共享状态 `R_t = (M_t, z_t^mem, g_t, Δg_t)` 交给适配器。训练和部署走同一条因果扫描：只用到当前时刻为止的观测和状态，不用线索标签、分支标签、未来帧，也不做测试时的演示检索。

2.3 适配器：只翻译，不改策略

更新器在两个策略族之间共享，只有面向策略的适配器不同：

- **回归（ACT）适配器**：把每个槽映射成 512 维记忆 token，把 `[z_t^mem, g_t, Δg_t]` 映射成一个摘要 token，拼进 ACT 的注意力 memory；动作头照旧回归动作块，损失仍是 chunk L1（+ 可选 KL）。参数 0.79 M。
- **扩散适配器**：按读出权重池化槽，与 `z_t^mem, g_t, Δg_t` 拼接，过一个零初始化 MLP 得到加性全局条件向量；去噪过程和损失不变。参数 16.03 M。

共享更新器 11.91 M。相对 ACT（51.62 M）增加 24.6%，相对 Diffusion Policy（270.96 M）增加 10.3%。

2.4 三个稳定器

有限槽记忆训练时有三种典型失败：**槽塌缩**（大部分证据写进少数槽）、**弥散写入**（一个观测摊到所有槽，地址失效）、**读写失配**（写进去的形式读不出来）。TRACE 各加一项辅助损失：平衡损失 `L_bal` 惩罚平均路由不均、熵损失 `L_ent` 惩罚每步高熵路由、一致性损失 `L_cons` 让读出与写入提案对齐（公式 41–43）。这些只作用于记忆模块，下游模仿损失不变。去掉它们平均进度从 69.23 掉到 66.10。

3. 实验设置

- **任务**（TriPilot-FF 全身遥操作采集，30 Hz，1 顶视 + 2 腕部相机，17 维状态）：

任务	早期证据 → 分岔	演示数	回合长度	子任务/阶段
Tool	初始物体身份决定后续工具顺序	100	2910 帧（97 s）	2/4
Book	来源货架决定路线与放置位置	150	1392 帧（46 s）	3/4
Laundry	来源一侧决定”刷+篓”还是”叠+收”	60	2220 帧（74 s）	2/4
Cable	来源一侧决定匹配哪台设备	30	1770 帧（59 s）	1/4
Medicine	托盘来源决定取哪个药盒并放回	60	1759 帧（59 s）	2/4

- **指标**：阶段级进度（%），每个任务 25 次真机 rollout，任务平衡平均，bootstrap 标准误。
- **基线**：ACT、Diffusion Policy（TRACE 的两个基座）；SmolVLA、$\pi_{0\cdot5}$、X-VLA、GR00T N1.6，全部在同样的单任务演示上各微调 180k 步，语言模型只给通用任务提示（不泄露线索）。

4. 结果

4.1 主表

方法	Tool	Book	Laundry	Cable	Medicine	平均
Diffusion Policy	18.00	12.33	22.00	34.00	38.67	25.00
ACT	31.00	13.67	11.50	44.00	27.33	25.50
$\pi_{0\cdot5}$	45.50	51.00	47.50	49.00	59.33	50.47
GR00T N1.6	39.00	12.67	24.00	38.00	39.33	30.60
SmolVLA	25.00	20.00	12.00	50.00	34.67	28.33
X-VLA	5.50	47.00	41.00	36.00	40.00	33.90

方法	Tool	Book	Laundry	Cable	Medicine	平均
TRACE（回归）	54.50 ± 17.96	**83.00**	**81.00**	**51.00**	**76.67**	**69.23**
TRACE（扩散）	**58.50**	68.33	70.50	51.00	49.33	59.53

收益最大的是 Book、Laundry、Medicine——恰好是”来源线索决定后续分支”的任务；Cable 只有 +7，因为设备附近的局部几何本身就还带着信息。Tool 标准误高达 17.96，作者专门做了去掉 Tool 的稳健性检验：TRACE 回归仍是 72.92，领先最强非 TRACE 基线 21.21 个点。

4.2 与通用记忆模块的对照（同一 ACT 基座）

记忆模块	在线	固定容量	签名	平均
无记忆	—	—	—	25.50
GRU 循环记忆	✓	✓	—	45.83
Transformer 历史上下文	—	—	—	52.10
LRU 外部记忆	✓	✓	—	53.93
检索提示记忆（MAP-VLA 式）	—	—	—	56.30
TRACE 签名路由槽	✓	✓	✓	**69.23**

任何记忆都比没有好（+20 到 +30），但四个对照彼此之间只差 10 个点，TRACE 再往上 13 个点。作者的解释：这些对照能保留历史，却没有把延迟线索绑定到”分岔点时会被读到的那个轨迹地址”上。

4.3 消融与诊断

组件消融	平均
只用当前观测	25.50
只有签名（不存视觉证据）	45.50
无路由的槽记忆	52.17
去掉增量路由 `Δg_t`	61.43
平均读出（不用注意力）	62.80
去掉辅助损失	66.10
完整 TRACE	69.23

“只有签名”到 45.50 说明轨迹形状本身就带一部分分支信息（毕竟从不同货架过来路径不同），但 TRACE 剩下的 24 个点来自把视觉证据存进去。

历史变换诊断是这篇论文最有说服力的部分。对同一条在线 rollout 的状态路径施加变换后重跑记忆模块，测两个量：路由相似度（分岔点之前写入的槽序列是否一致）和分支一致性（分岔点读出与分支动作是否不变）。

变换	测什么	路由相似度	分支一致性
时间重采样	时间重参数化不变性	92.4	96.0
速度抖动	同上	89.8	94.7

变换	测什么	路由相似度	分支一致性
状态偏移	偏移不变性	91.2	95.1
稀疏采样	采样鲁棒性	86.5	92.3
顺序反转（负对照）	破坏因果顺序	**37.8**	**41.6**

保序变换都在 90 上下，倒放历史掉到 40 以下。同样的分岔点画面、同样的策略权重，只是把历史倒过来，记忆读出就变了——这是”记忆确实依赖有序历史”的直接证据，而不是靠成功率间接推断。

4.4 长历史稳定性

把在线历史延长到 1.25× / 1.5× / 2.0×，或插入重复干扰段，槽切换率保持在 0.05 以下，分支决策一致性最低 0.872（重复干扰段）。作者明确写这是”在评测过的延长范围内的诊断，不是任意长度的保证”。

4.5 部署开销

路径	基座前向	签名	槽更新	读出	适配器	TRACE 总开销	占比
回归	5.90 ms	0.52	0.90	0.58	0.10	2.11 ms	35.8%
扩散（100 步去噪）	118.00 ms	0.52	0.90	0.58	0.14	2.15 ms	1.8%

RTX 5090、PyTorch 2.9.1。记忆是一次前向里的固定小开销，不是第二次策略调用，也不是离线检索。

5. 判读

5.1 值得记住的三点

1. **地址和内容分离，是对”记忆该如何寻址”的一个新答案。** 帧堆叠按时间寻址，注意力检索按内容寻址，TRACE 按”执行过的轨迹几何”寻址。对延迟证据任务这是合理的：线索出现时和分岔点到达时，机器人走过的路是对应的，而画面不是。
2. **诊断而非只报成功率。** 路由相似度 / 分支一致性 + 顺序反转负对照，这套方法论可以直接搬到其他记忆工作上，用来回答”你的记忆到底在用什么”。它和 *Present but Not Remembered* 的探针实验是互补的两种审计手段。
3. **不改骨干的外挂设计。** 更新器共享、适配器只做翻译，同一套记忆状态同时提升回归和扩散两个策略族，是模块化主张的实证。

5.2 需要留意的地方

1. **签名维度的指数增长。** 17 维状态深度 3 已经是 5219 维，深度 4 是 88740 维；作者承认这是主要局限。状态维数更高（人形、灵巧手）时要么降深度要么换 log-signature（1785 维，但流式实现和归一化更复杂，论文没做）。
2. **地址依赖轨迹的可区分性。** 路由靠”从不同来源走过来路径不同”。如果两条分支在分岔点之前的机器人运动完全一致（比如线索只是颜色不同，动作序列一模一样），签名地址就分不开，只能退化为靠内容路由。论文的五个任务里来源都对应不同的空间位置，正好落在 TRACE 的舒适区。
3. **写入是每步进行的，没有”该不该写”的判断。** 与 EventVLA 相反，TRACE 每步都写，靠路由和门控决定写多少、写到哪。稳定性诊断显示 2× 历史时读出一致性降到 0.928，重复干扰段 0.907——有限槽在长历史下确实会被慢慢冲淡。

4. **基线的公平性。** VLA 基线各微调 180k 步且只给通用提示，但它们没有加任何记忆模块；TRACE 只和 ACT/DP 加记忆做了同基座对照。"TRACE + $\pi_{0.5}$"会怎样，论文没有回答。
5. **Tool 任务方差极大**（± 17.96），作者用去 Tool 分析和失败分类表处理得比较诚实，但也说明真机 25 次 rollout 对接触密集任务不够。
6. **单任务训练。** 每个任务单独训练一套策略，没有多任务或跨任务的记忆迁移结果。

5.3 与 EventVLA 的对照

见 EventVLA 深读 §5.3。一句话：EventVLA 存**多比特的视觉细节**（原图），靠学到的写入时机控制稀疏度；TRACE 存**少比特的分支变量**（latent 槽），靠轨迹地址控制读写对齐。两者在"记忆必须有界"上一致，在"存什么、怎么找"上互补。一个自然的后续问题是：用 EventVLA 的前瞻写入触发决定何时写，用 TRACE 的签名决定写到哪，能否同时拿到稀疏性和可寻址性。

6. 关联阅读

- **同组工作**：TriPilot-FF（2602.09888，数据采集系统）、SAI（2606.16490，本仓库第三篇）、AutoIntervene（2608.07065，用视觉-动作支持记忆决定何时交还人类接管）、Tri-Manual（2607.25731）。
- **同样是槽 / 固定容量 latent 记忆**：ELMUR（2510.07151，层内外部记忆 + LRU 更新，MIKASA-Robo 上大幅领先）、MANN（1605.06065，内容寻址外部记忆的源头）、VPWEM（2603.04910，把窗外观测递归压成固定数量的情景记忆嵌入）。
- **同样强调"记忆应是策略的状态"**：Chronos（2606.30318，选择性 SSM 传播全历史状态，RMBench 73.6%）、TFP（2607.08283，液态时间常数信念，事件附近写入增益放大约 6 倍）、RB-VLA（2602.20659）。
- **路径签名**：Signatory（2001.00706）、CILO（2407.04856，模仿学习中用签名编码轨迹约束）。
- **检索式对照的来源**：MAP-VLA（2511.09516）。

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4. SAI 深读

> **论文**：Robots that Collaborate: Sequential Asymmetric Imitation for Learning Coupled Robot Policies
> **作者**：Yincong Chen, Ranpeng Qiu, Zihao Li, Yanan Zhou, Guoqiang Ren, Weiming Zhi（浙江大学 / 浙江工业大学 / 悉尼大学）
> **arXiv**：2606.16490（2026-06-15，v2）· 项目页：cyc0429.github.io/sai-project-page
> **仓库内文件**：英文 PDF · 中文翻译 PDF · 英文版解读

0. 一句话

两台双臂移动操作机器人抬画、铺床罩、铺桌布、共同收衣，彼此不通信、看不全对方。SAI 用一个遥操作者分三阶段采数据：先让 A 跟着顺从的人类搭档学会任务，再冻结 A、让人遥操 B 去适应 A 的节奏，最后把 A、B 同时部署，只在协调快失败的地方对 A 做稀疏干预。四个真机任务上，成功率从独立模仿的 23–50% 提到 53–70%，"该等就等、该让就让"的 Yield/Wait 指标从 27–47% 提到 68–72%。策略本身只是带 30 步历史 token 和级联底盘-手臂头的 ACT；协调行为来自数据课程，不来自架构。

1. 为什么这篇进了"记忆 VLA"的清单

SAI 表面上是多机器人模仿学习，骨子里处理的是同一个非马尔可夫问题：**搭档的状态是部分可观的**，A 只能从自己的局部相机和共享物体上的力反馈去推断 B 处于哪个阶段、有没有延迟、是不是卡住了。这个推断必须靠历史。论文的两处证据直接说明这一点：

- 策略里的 30 步历史 token（对数采样 12 帧 → GRU）把 Laundry 任务里"提前松手"的失败率从 64.5% 压到 16.1%，因为策略要靠历史区分"正在搬运的中间状态"和"该松手的终止状态"——单帧看两者一样。
- 第三阶段的干预专门教 A 在"B 还没到位"时减速、等待、恢复；B 是否到位是一个需要从时间序列里读出来的隐变量。

所以 SAI 是"记忆 VLA"的多智能体版本：要记住的不是被盖住的物体，而是看不全的搭档走到了哪一步。它和 TRACE 是同一个组的工作，两篇论文的策略骨干（带历史编码的 ACT）和数据采集系统（TriPilot-FF）也相同。

2. 问题设定

- 去中心化部分可观控制：每台机器人 `i ∈ {A, B}` 只拿自己的局部观测 `o_i^t = O_i(s_t)`，各自策略 `a_i^t ~ π_i(·|o_i^t)`，不交换消息、状态、动作或隐向量。协调只能通过共享工作空间的局部观测和共享物体的物理耦合产生。
- 三类失败：**时间相位耦合**（抓、起、放、松要对齐）、**搭档相关的让步**（对方延迟或卡住时自己要慢下来）、**交互冲突**（内力过大、物体变形、反向拉扯、碰撞）。
- 单遥操作者约束：一次只能直接遥操一台机器人。监督被拆成三个非对称数据集：`D_A^solo`（A 与顺从人类搭档）、`D_B^A`（B 在 A 的学习策略部署下）、`D_A^int`（联合部署时对 A 的干预）。

核心问题：这三个顺序、非对称的数据集，够不够学出接近同步双人演示效果的耦合策略？

3. 方法：三阶段课程

3.1 阶段一：用人类搭档引导 A

人扶着共享物体另一端，遥操 A 完成任务。A 学到抓、拉、对齐这些基础技能。因为训练时搭档是人、部署时搭档是机器人，直接训练会让 A 学到人类外观的捷径（衣服、皮肤、体形）。作者用离线的**搭档区域随机化**：实例分割出人体区域，膨胀后在训练加载时随机换成 RGB 随机噪声、强高斯模糊或 Stable Diffusion 修补（附录 B）。目标不是抹掉交互证据，而是让人类外观线索不可靠，逼策略看物体几何、形变、接触进度。

3.2 阶段二：让 B 在部署的 A 面前学

冻结 `π_A^(1)` 部署，人遥操 B。采数据时给 A 加有界的部署扰动：速度缩放、动作噪声、时序偏移、初始物体状态扰动。B 由此见到真实的搭档偏差，学到等待、让步、恢复、续接。这一阶段是非对称的——B 不是对着理想搭档或人工同步的人练，而是对着 A 实际学出来的那套行为练，所以能补偿 A 的时序、运动风格和残余错误。

3.3 阶段三：DAgger 式干预微调 A

此时最大的分布错配在 A 身上：它只见过顺从的人类搭档，没见过自主的 B。联合部署两台机器人，操作者只在协调开始失败的状态附近对 A 给出纠正动作（拉太早、不让步、B 延迟时继续、相位错配后恢复差）。干预数据与原始 A 演示聚合微调：

```
π_A^(3) = BC(D_A^solo ∪ D_A^int)
```

干预样本相对名义回放加权；只纠正部分动作维度时用二值动作损失掩码（比如只改底盘、保持手臂），避免稀疏纠正标签覆盖掉未受影响维度的名义行为。干预集只有基线数据量的约 30%。

3.4 策略实例化

每台机器人一个 ACT 风格策略：冻结 DINOv3-80M（ViT-B）视觉骨干、顶视 + 腕部相机、17 维本体状态（3 底盘 + 14 臂/夹爪）、臂侧力矩反馈。历史窗 `H = 30` 步，对数时间采样 12 帧，多视角特征池化后与 64 维状态嵌入融合，单层 GRU（隐 512）末隐状态作为历史 token。动作头是级联的：先用 MLP 预测底盘 `a_base ∈ R³ = (v_x, v_y, ω)`，再把它与解码器查询拼起来预测 14 维臂/夹爪动作，动作块长度 100，时序集成系数 0.01。

4. 实验

4.1 四个真机任务，三个训练流程

任务：Bed-throw Spreading、Tablecloth Spreading（柔性物体对齐）、Laundry Collection（共享空间异步协调）、Painting Transport（刚体接触运输）。三个流程共用同一架构：独立模仿（各自从单边演示学）、搭档条件模仿（有阶段二、无阶段三）、完整 SAI。每个任务-方法 30 次 rollout；Phase Sync 按预定义检查点计（Painting 5 个 × 30 = 150），Yield/Wait 按搭档延迟机会计。

任务	方法	Success	Phase Sync	Yield/Wait
Bed-throw	独立 / 搭档条件 / **SAI**	23.3 / 36.7 / **53.3**	30.8 / 60.8 / **62.5**	26.7 / 35.0 / **68.3**
Tablecloth	独立 / 搭档条件 / **SAI**	43.3 / 60.0 / **66.7**	54.4 / 67.8 / **68.9**	46.7 / 61.7 / **70.0**
Laundry	独立 / 搭档条件 / **SAI**	50.0 / 63.3 / **70.0**	51.7 / 68.3 / **73.3**	31.7 / 48.3 / **71.7**
Painting	独立 / 搭档条件 / **SAI**	33.3 / 46.7 / **56.7**	42.7 / 67.3 / **74.7**	34.4 / 48.9 / **68.9**

读法：阶段二主要抬的是 Phase Sync（Bed-throw 30.8→60.8），因为 B 学会了跟 A 的节奏；阶段三主要抬的是 Yield/Wait（Bed-throw 35.0→68.3），因为 A 终于学会在 B 掉队时等。成功率是两者叠加。

4.2 受控延迟测试

Tablecloth 任务里人为暂停 B。独立模仿的 A 继续按名义轨迹拉，桌布错位、B 失去抓握；SAI 的 A 观察到 B 没前进就减速等待，B 恢复后再续。这是 Yield/Wait 指标背后的具体行为。

4.3 架构消融：历史与级联头

失败模式	变体	失败率
提前松手（Laundry 交付阶段）	无历史 / 有历史	64.5% / **16.1%**
提前下放（搬运→放置过渡）	无级联 / 有级联	51.6% / **9.7%**

历史 token 让策略分得清中间搬运态和终止松手态；级联头让手臂命令以底盘意图为条件，避免底盘还没到位手臂就开始放。作者强调这两个组件提升的是局部执行可靠性，**不能**解释协作收益——表 1 里三个流程用的是同一架构，独立模仿照样在搭档延迟时失败。

4.4 骨干无关性

Painting Transport 上换 Diffusion Policy 做动作头：独立模仿 23.5 / 35.2 / 30.7 → SAI 52.9 / 62.2 / 76.9（Success / Phase Sync / Yield/Wait，34 次 rollout）。ACT 上是 33.3 / 42.7 / 34.4 → 56.7 / 74.7 / 68.9。课程的效果不依赖动作解码器。

5. 判读

5.1 值得记住的

1. **协调是"搭档分布"的函数，不是架构的函数。** SAI 的实验设计把架构固定、只换数据课程，因此收益能干净地归因到"策略在训练时见过什么样的搭档"。这与 TRACE 把记忆模块外挂、骨干不动的思路一致：先证明是这一个变量在起作用。
2. **非对称是特性不是妥协。** B 对着真实的 A 学、A 再对着真实的 B 补——每一步都只修正当前最大的分布错配。同步双操作者数据既贵又不一定更好（那是对着一个完美搭档学）。
3. **历史在多机器人里的角色被量化了。** 64.5% → 16.1% 的提前松手是一个很直接的"历史区分视觉相同状态"的例子，与 TRACE 的延迟证据设定同构。

5.2 需要留意的

1. **干预覆盖不足。** 阶段三数据只有基线的约 30%，作者自己说恢复行为"尚未饱和"；干预过多又可能让策略学会不必要的等待。多少干预、往哪里打，目前靠人判断。同组后续的 AutoIntervene（2608.07065）正是在做这件事：用成功执行构建视觉-动作支持记忆，按分位数校准阈值决定何时交给人、何时收回。
2. **A 中心的设计。** 只对 A 做干预。需要同时纠正两台机器人的任务不适用；B 在阶段二之后再没更新过，A 变了之后 B 面对的分布其实也变了，论文没做第四阶段。
3. **历史只有 30 步（1 秒量级）。** GRU 历史 token 足够分辨相邻阶段，但覆盖不了"B 两分钟前做过什么"。搭档相位这类隐变量在更长任务里可能需要 TRACE 式的显式记忆槽——这是两篇论文最自然的结合点。
4. **评测指标是描述性的。** Phase Sync 和 Yield/Wait 的分母是事件数不是独立试验数（Painting 150 个检查点来自 30 次 rollout），不能当独立样本做显著性检验；论文附录明说了这一点。
5. **两台机器人是同型号**（双臂移动操作器 + TriPilot-FF），异构搭档（人形 + 轮式，如 Duet）下人类外观随机化那一套是否够用未知。
6. **没有和同步双操作者演示训练的策略直接比**——那是 SAI 想替代的目标，但成本正是不采它的理由，所以"接近同步演示效果"这句只是假设。

5.3 与另两篇的关系

	EventVLA	TRACE	SAI
看不见的是什么	被盖住/消失的物体证据	早期出现的分支线索	搭档的内部相位
靠什么补	稀疏原图缓冲	轨迹寻址的 latent 槽	30 步 GRU 历史 + 训练分布覆盖
干预点	架构（KEM 头）	模块（外挂适配器）	数据（课程 + DAgger）
共同点	都拒绝"加长窗口"这一默认答案，都用有界的历史表征		

6. 关联阅读

- **同组**：TRACE、TriPilot-FF、AutoIntervene、Tri-Manual（2607.25731：单人为三臂系统演示，离线按依赖关系重排时序）。
- **多机器人数据采集**：HATS（2606.16491，与 SAI 相邻编号，单人 + MLLM 智能体控制四臂）、Duet（2606.20990，人-人演示预训练 + 少量双机器人遥操微调，Unitree G1 + Vega1 异构搭档）、Mobile ALOHA（2401.02117）。
- **干预式模仿**：DAgger（1011.0686）、IntervenGen（2405.01472，从少量人类干预自动生成大量纠正数据——可直接补 SAI 阶段三的覆盖不足）。
- **双臂解耦**：Decoupled Interaction Framework（2503.09186，每臂独立模型 + 选择性交互模块，RoboTwin 上 +23.5%）——单机双臂版本的"去中心化 + 局部交互"。
- **两个动作头如何一致**：Making two action heads agree（2608.15748），流匹配策略双分支协调机制与运行时塌缩证书，可作为 SAI 里"两台机器人选到不同模态"这一失败的形式化参考。

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5. 设计空间矩阵

每个记忆 VLA 都要回答六个问题：**存什么**（内容）、**何时写**（触发）、**写到哪 / 从哪读**（寻址）、**存多少**（容量）、**接在哪**（集成点）、**靠什么学**（监督）。下表把三篇核心论文和 30 个代表性工作填进这六列，条目内容只取自原文或摘要。方法名后的括号是 arXiv 编号。横向解读见趋势与洞见 §2。

方法	存什么	何时写	寻址	容量	集成点	监督	主要数字
EventVLA (2606.20092)	原始图像帧	学习的未来关键帧概率 ≥0.55 + NMS + 冷却	时间顺序拼接，注意力自找	5 事件帧 + 初始帧 + 2–3 近期帧	VLM 视觉编码器输入	Qwen3-VL 自动标签 + 升余弦软标签 + 师生课程	RoboTwin-MeM 18.0→75.2；RMBench 67.8（仅锚帧）
TRACE (2606.14551)	视觉 + 状态的 512 维 latent	每步，门控决定写多少	机器人轨迹路径签名（深度 3，5219 维）+ 增量	K = 4/6 槽	动作头前适配器（ACT 记忆 token / DP 全局条件）	无标签；平衡 + 熵 + 一致性稳定器	ACT 25.5→69.2；顺序反转路由相似度 37.8
SAI (2606.16490)	GRU 隐状态（12 帧对数采样）	每步	无	30 步	ACT 解码器输入	三阶段数据课程 + DAgger 干预	提前松手 64.5→16.1%；成功 23–50→53–70%
KEMO (2606.23589)	关键帧 token	运动学 + 视觉过滤检测事件	时序 token，交叉注意力	事件数	门控残差融合进 VLA	事件附近样本加权	真机 +23.6 成功 / +34.1 阶段
UniMem (2608.22869)	关键帧密集空间记忆	事件分类器	注意力	关键帧缓存	单骨干	分类器监督	仿真 93.4 vs 68.2
WeaveLA (2606.17463)	完成段的 latent token	子目标完成事件	直接路由进下一子任务	每子任务若干 token	动作生成路径（冻结 VLA）	模仿损失	SwingXtimes N=3：0→47.8

方法	存什么	何时写	寻址	容量	集成点	监督	主要数字
Keyframe-Chaining (2603.01465)	关键帧作交错视觉 token	判别式嵌入选帧	进度感知查询	若干关键帧	VLA 输入	对比嵌入	ManiSkill 4 任务
MemER (2510.20328)	关键帧 + 文本指令	高层 VLM 选帧	VLM 推理	选中帧 + 近期帧	双系统（Qwen2.5-VL-7B → $\pi_{0.5}$）	少量语言标注	分钟级真机任务
MEM / π_MEM (2603.03596)	视频编码短时 + 文本长时	持续	文本检索 / 视频窗口	文本无界	双系统	演示	15 分钟级任务
MemoryVLA (2508.19236)	感知 + 认知 token 库	每步，冗余合并	内容相似检索	库大小	记忆条件扩散专家	模仿损失	MIKASA-Robo 41.2
ContextVLA (2510.04246)	单个上下文 token	每步	—	1 token	VLM 输入	模仿损失	多帧收益、低成本
HAMLET (2510.00695)	moment token	每步	轻量记忆模块	历史长度	GR00T N1.5	时间对比初始化	真机 76.4（+47.2）
NativeMEM (2607.06678)	每帧每视角 1 token	每帧	序列位置	线性增长（token 极小）	追加到 VLA 输入	冻结 VLA 监督的 tokenizer → 解冻微调	仿真 32.4→84.0
TempoFit (2603.07647)	中间层前缀 K/V	每步 FIFO	K-to-K 检索 + 帧间隙偏置	层内 FIFO	注意力前残差	免训练	LIBERO-LONG +4.0
StreamPI (2608.26067)	（观测，指令）对的 KV	每帧	因果注意力	随流增长	骨干注意力掩码	随机间隔流式训练	零新增参数
DySta (2602.03983)	静态 / 动态 token	静态 token 仅在需要时重缓存	KV 复用	一份静态 + 动态	VLM 输入	模仿损失	2 倍加速 +2.3
PRISM (2606.16178)	分层压缩 token	每步	门控注意力	两分钟	Transformer 策略	模仿损失	ReMemBench +5–12
HALO (2606.25136)	长上下文 token	每步	稀疏注意力检索	八分钟	Transformer 策略	VLM 生成的记忆问答蒸馏	—
AtlasVLA (2608.06729)	4D 体素哈希空间状态 + 自我状态	每帧更新	空间哈希	场景体素	DiT 条件	模仿损失	LIBERO-Long +9.4
LaMem-VLA (2607.07608)	短期 / 长期 latent token	curator 整理	seeker 多模态查询	有界上下文	编织进嵌入序列	模仿损失	SimplerEnv / LIBERO
μVLA (2606.12497)	m 个记忆 token	每步自注意力更新	隐式	m 个 token	骨干内	TBPTT，无辅助损失	MIKASA 0.42→0.84
VPWEM (2603.04910)	情景记忆嵌入	窗外观测递归压缩	交叉注意力	固定数量	扩散策略条件	模仿损失	MIKASA +20%

方法	存什么	何时写	寻址	容量	集成点	监督	主要数字
VQ-Memory (2603.09513)	离散 VQ token（本体）	每步	序列	码本	VLA / DP 条件	VQ-VAE	RuleSafe
MemoAct (2603.18494)	感觉 / 短期无损 / 长期压缩	分层	分层	三层	策略条件	模仿损失	MemoryRTBench
Chronos (2606.30318)	每控制步一个状态 token	每步	选择性 SSM	全历史（压缩）	策略 latent 状态	IMLE + 薛定谔桥	RMBench 73.6
TFP (2607.08283)	进度信念（LTC 动力学）	每步，事件附近增益 ~6×	—	单信念	流匹配解码器调制	模仿损失	MIKASA ShellGameTouch 75.0
CAMP (2606.21188)	压缩的动作历史	每步	—	固定	策略条件	自监督（动作历史）	Memory-T-Bench
GMP (2604.18933)	latent 记忆	记忆门决定何时回忆	交叉注意力	固定	视觉运动策略	历史动作加扩散噪声	MemMimic +30.1
ELMUR (2510.07151)	层内外部记忆嵌入	每段	LRU 替换 / 凸混合	每层固定	Transformer 各层	RL 目标	MIKASA 21/23
RB-VLA (2602.20659)	紧凑信念状态	每步	—	固定	扩散策略条件	自监督世界模型	32.5→77.5
FM-VLA (2607.18231)	力记忆 token	每步	—	短历史	动作专家条件	VAE 力重建	80%+
AGM (2608.29537)	子目标序列 + 进度指针	物理证据验证后才推进	指针	子目标数	冻结 VLA 外部	2.43M 验证头	RoboMME Counting 领先
HyMeS (2608.09410)	代码里的记忆结构	阶段完成验证	代码逻辑	无界	编码智能体 → VLA	rollout 反馈迭代启发式	RoboMemArena 52.5→66.2
PonderPounce (2608.24115)	MLLM 原生因果上下文	持续	注意力	上下文长度	异步认知 token → VLA	端到端联合训练	RoboMME 60.83
BATON (2608.16889)	子任务解 + 转移记忆	探索后写入	子任务索引	子任务数	LLM 智能体 + 冻结 VLA	无参数更新	RoboMemArena +11.6
Notes-to-Self (2602.21013)	语言草稿板	模型自写	文本	文本	VLA 提示	演示	ClevrSkills / MemoryBench
MemoryWAM (2606.20562)	近期帧 + 事件边界锚帧 + gist token	事件边界	定制注意力	三层有界	世界动作模型	视频 + 动作	长时程记忆任务
MemoryVAM (2606.20679)	Perceiver 压缩的记忆 token	每帧	交叉注意力	固定	视频骨干 + 动作解码器	视频预测 + 回合边界监督	LIBERO-Mem 5→42.5

三个空白（读表可见）

1. **「何时写」和「寻址」同时学的方法还没有出现**：EventVLA 学写入时机但按时间寻址，TRACE 按轨迹寻址但每步都写。
2. **多机器人一列几乎为空**：SAI 只有 30 步 GRU，没有显式记忆模块用于搭档状态估计。
3. **监督来源五花八门**（235B 标注 / 规则 / 分类器 / rollout 结果 / 自监督），没有跨任务对照说明哪种写入策略能迁移。

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6. 研究机会清单

每条按「为什么重要 / 最小可行实验（MVE）/ 相关解读」组织。MVE 的标准是：一个人、一块 GPU、两周内能拿到第一个能证伪的数字。编号按 README 的家族顺序排列；只做一件事就做第 1 条。所有数字只取自原文，见数字口径账本。

A. 写入与寻址（EventVLA × TRACE 的中间地带）

1. 前瞻写入 × 轨迹寻址

为什么重要：EventVLA 学"何时写"但按时间拼接、靠注意力自己找（RoboTwin-MeM 18.0 → 75.2）；TRACE 按轨迹签名寻址但每步都写（ACT 25.5 → 69.2）。两者各占一个极端，"既学写入时机、又学寻址"的组合没人做过。稀疏写入解决容量（EventVLA 的 5 帧 FIFO 在 >10 分钟任务上会饱和），轨迹寻址解决读写对齐（TRACE 的每步写入在 2× 历史下一致性降到 0.928）。

MVE：在 TRACE 的 ACT 基座上加一个 KEM 式的写入门（用 EventVLA 的软标签监督，标签可用 Qwen3-VL 或 KEMO 的运动学规则生成），只在门打开时执行签名路由写入。在 TRACE 的 Book / Laundry 两个任务上比较：完整 TRACE、门控 TRACE 的阶段进度，以及 2× / 4× 延长历史下的分支一致性。预期：成功率持平、长历史一致性提升；若门控反而掉点，说明分支线索出现的时刻并不稀疏。

相关解读：EventVLA §5.3、TRACE §5.3、KEMO。

2. 写入触发的跨任务迁移

为什么重要：现有触发器各有监督来源——EventVLA 用 235B 模型离线标注，KEMO 用运动学 + DINOv2 去重的规则并逐任务手调窗口（w 10–40、冷却 8–60），UniMem 训练事件分类器，OnEvoMemory 从 rollout 结果学。没有任何对照告诉我们哪种触发在未见任务上还能工作。

MVE：在同一个 $\pi_{0.5}$ 骨干上实现三种触发（学习 / 规则 / 分类器），在 RoboTwin-MeM 的 4 个任务上训练、在另外 4 个任务上零样本测触发质量：以物理引擎真值关键帧为参照报告触发的精确率 / 召回率和时间误差，再报告下游成功率。三种触发的排序若在训练任务和留出任务上颠倒，就说明写入时机是任务特定的。

相关解读：KEMO、UniMem、EventVLA §2.3。

3. 轨迹地址的可区分性边界

为什么重要：TRACE 的寻址靠"不同来源走过来的路不同"。当两条分支在分岔点之前动作完全一致（线索只是颜色差异），签名地址重合，只能退化为内容路由。TRACE 的五个任务里来源都对应不同空间位置，正好落在舒适区，边界

没被测过。

MVE：构造一个"同轨迹、异线索"任务（掀同一个盖子，颜色决定后续分支），对照 TRACE、去掉签名的无路由槽记忆（TRACE 表 3 中为 52.17）、EventVLA 式原图缓冲。预期 TRACE 掉到无路由水平，原图缓冲保持——这会把"何时用轨迹寻址、何时用内容寻址"变成可查表的结论。

相关解读：TRACE §5.2、设计空间矩阵。

4. 高维状态下的签名

为什么重要：17 维状态、深度 3 的签名已是 5219 维，深度 4 为 88,740 维；人形或灵巧手的状态维数翻倍后标准签名不可用。log-signature（深度 3 为 1785 维）是论文自己点出但没做的替代。

MVE：在 TRACE 的公开任务数据上替换为 log-signature 与"学习的低维轨迹键"（对状态路径做 1D 卷积 + 池化），比较路由相似度、分支一致性和每步延迟（当前签名 0.52 ms）。目标是在 40 维以上状态下把地址维度压到 2000 以内且保持顺序反转负对照低于 50。

相关解读：TRACE §5.2、Signatory（2001.00706）。

B. 容量、纪律与安全

5. 缓冲区饱和后的记忆分层

为什么重要：EventVLA 的 N_max = 5 + FIFO 会在事件密集的长任务上挤掉早期证据（n=5 的任务只有 48%）；MemoryWAM 的"近期帧 + 事件边界锚帧 + gist token"三层是目前的折中，但没有在同一基准上和平面记忆正面比过。

MVE：把 RoboTwin-MeM 的 Press Button Keyframe 扩到 n = 8、12（更多数字卡），对照 EventVLA 原版、N_max = 12 的平面缓冲、以及"事件帧满时压成一个 gist token 而非丢弃"的分层版本。报告成功率随 n 的曲线和吞吐。

相关解读：EventVLA §5.2、MemoryWAM 笔记（notes）。

6. 把"验证后才推进"搬进端到端记忆

为什么重要：AGM 的结论是可靠记忆靠有纪律的状态更新而非容量——进度指针只在物理证据验证子目标后推进，否则"尝试过"会被当成"完成了"。端到端方法里对应的机制只有 EventVLA 的 NMS + 冷却和 TRACE 的门控，都不检查写入内容是否被后续观测证实。

MVE：给 EventVLA 的事件缓冲加一个"回读验证"：写入后 k 步内若当前观测与写入帧的预测状态矛盾（用 KEM 头的下一块预测做代理），撤销写入。在 Pick-X-Times 类计数任务上对照写入撤销率与成功率。

相关解读：AGM、HyMeS、趋势与洞见 洞见 6。

7. 错误记忆的恢复

为什么重要：AGM 指出未经验证的记忆会把局部错误固化为持久的任务状态错误。EventVLA 若把一帧错误观测（遮挡、误触发）写进缓冲，论文没有讨论如何恢复；TRACE 的槽会被后续写入慢慢冲淡但没有主动纠错。

MVE：在 RoboTwin-MeM 上做注入实验——在推理时强制写入一帧随机历史帧，测成功率下降幅度与随后能否自恢复；对照"允许覆盖同槽"与"只追加"两种策略。这给出记忆机制的"错误容忍度"这一新指标。

相关解读：AGM、EventVLA §5.2。

8. 记忆与动作块长度的联合设计

为什么重要：EventVLA 的前瞻能力是动作块长度的函数：块从 50 缩到 15，成功率 75.2 → 13.6，低于不用 KEM 的 18.0。WhyChunking 又指出动作块的收益之一正是非马尔可夫表达力。需要短块高频响应的接触任务目前无法享受前瞻写入。

MVE：把 KEM 的预测窗口与执行窗口解耦——预测未来 50 步的关键帧概率，但每次只执行 10 步再重预测。对照原版（预测 = 执行 = 50）和短块原版（15/15），在 RoboTwin-MeM 与一个接触丰富任务上同时报告成功率与反应延迟。

相关解读：EventVLA §4.3、WhyChunking（2608.02547）。

C. 评测与诊断

9. 标准化的记忆诊断

为什么重要：成功率不能区分"记忆在用历史"和"碰巧对了"。TRACE 的路由相似度 / 分支一致性 + 顺序反转负对照，Present-but-Not-Remembered 的探针 + 因果干预，TFP 的隐状态干预，是三套独立的诊断，但都只在各自论文里用过一次。

MVE：给 RoboMME 的四类任务（时间 / 空间 / 物体 / 过程）各配保序 / 破序两组历史变换，对 $\pi_{0.5}$ + 帧堆叠、MemoryVLA、EventVLA、μVLA 四种记忆报告"破序后成功率下降幅度"。下降为零的记忆就是没用历史的记忆。

相关解读：TRACE §4.3、Present but Not Remembered、RoboMME。

10. 跨基准的统一报告模板

为什么重要：一年内出现至少 12 套记忆基准，口径互不相通：RMBench 上锚帧就能到 67.8，RoboMemArena 68.9% 子任务依赖记忆，RoboTwin-MeM 用 n 参数化瞬态证据数，TRACE 用阶段进度而非成功率。跨论文的数字目前只能分块列表，不能比较。

MVE：定义一个最小报告模板——任务的记忆类型（RoboMME 四类）、必须记住的中间事件数 n、证据出现与使用的时间间隔、指标类型（成功 / 阶段进度）、试验数——并为本仓库 17 篇有译文的论文回填这张表（见数字口径账本的雏形）。

相关解读：RoboMME、RoboMemArena、RMBench。

11. RMBench 到底测了什么

为什么重要：EventVLA 只用初始帧 + 近期帧就在 RMBench 拿到 67.8%，Chronos 用全历史 SSM 拿到 73.6%——两者都不需要事件记忆。如果 RMBench 的 9 个任务大多能被静态锚帧解决，那它测的是"能否保留初始布局"而不是"能否记住中间事件"。

MVE：在 RMBench 上跑三个退化基线：只加初始帧、只加近 2 帧、两者都加，逐任务报告。哪些任务只靠初始帧就能过 80%，就把它们标为"布局记忆"而非"事件记忆"；剩下的（Observe and Pick Up 21%、Press Button 3%）才是记忆基准的硬核。

相关解读：RMBench、EventVLA §4.1、Chronos。

D. 路线之争与成本

12. 计数与过程记忆能否端到端

为什么重要：RoboMME 与 RoboMemArena 的榜首（PonderPounce、HyMeS、BATON）都是高层管符号状态的方案；端到端事件记忆在瞬态视觉证据上最强，但在"按 3 次、跳过已完成子任务"上还没有正面胜过符号进度指针。FM-VLA 用力历史记计数（80%+）提示了另一条端到端路径。

MVE：在 RoboMME 的 Counting 子集上对照 AGM（符号指针）、EventVLA（事件帧计数）、FM-VLA 式力记忆、WeaveLA（子目标完成事件），同时报告成功率与每步延迟。回答"符号状态的优势是表征上的还是只是验证机制上的"。

相关解读：AGM、HyMeS、MEM、MemER。

13. 免训练记忆的失效边界

为什么重要：TempoFit 不训练、不加 token，只复用中间层 K/V 就在 LIBERO-LONG 上 +4.0；StreamPI 零新增参数；NativeMEM 每帧 1 token。如果免训练记忆够用，EventVLA 三倍的延迟和 235B 标注就没有必要。但它们没有在 RoboTwin-MeM 这类瞬态证据任务上测过。

MVE：把 TempoFit 与 StreamPI 直接接到 EventVLA 的 QwenOFT 基座，在 RoboTwin-MeM 的 n=1 与 n=4 任务上对照 EventVLA。若免训练方法在 n=1 持平、n=4 崩塌，就给出"何时必须学习写入"的边界。

相关解读：趋势与洞见 洞见 8、TempoFit / StreamPI / NativeMEM 笔记（notes）。

E. 多机器人

14. 搭档相位的显式记忆

为什么重要：SAI 只用 30 步 GRU 历史（约 1 秒）区分"搬运中"与"该松手"（提前松手 64.5% → 16.1%），但搭档两分钟前做过什么它记不住。搭档相位是一个只能从历史推断的隐变量，与 TRACE 的延迟证据设定同构，却没有人用显式记忆模块做搭档状态估计。

MVE：在 SAI 的 Tablecloth 受控延迟实验上，把 A 的 GRU 换成 TRACE 式签名槽记忆（键 = A 自己的轨迹，内容 = 对 B 的观测），把 B 的暂停时长从 2 秒逐步拉到 30 秒，报告 Yield/Wait 随延迟时长的曲线。GRU 版应随时长下降，槽记忆版应保持。

相关解读：SAI §5.2、TRACE、AutoIntervene。

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7. 数字口径账本

并排任何两个数字之前先查这张表。每个头条数字标注：它在什么任务集上、是什么类型的指标、试验规模多大、由谁测的（作者自测 / 作者复现别人 / 基准方测）、以及并排比较时最容易踩的口径坑。全部数字取自原文或本仓库基于全文的笔记；找不到的字段写"未报告"。

一、三篇核心论文

#	数字	出处	任务集	指标类型	试验规模	独立性	口径提醒
1	**75.2%**（vs 仅锚帧 18.0%）	EventVLA 表 2	RoboTwin-MeM，8 个双臂仿真任务，n = 1–5 个中间关键帧，430–1544 步/回合	成功率，任务平均	表 2 为整数百分比，每任务试验数正文未写；50 条演示/任务	作者自建基准、自测；基线 MemER / Mem-0 / MemoryVLA 为作者复现	基准与方法同出 RoboTwin 系；n=1 任务锚帧已够（62% 不变），收益集中在 n≥2
2	**67.8%**（vs Mem-0 42.0、MemoryVLA 41.7、$\pi_{0.5}$ 10.4）	EventVLA 表 1 / 表 8	RMBench，9 任务	成功率，任务平均	RMBench 协议每任务 100 次	作者自测；MemoryVLA 数字为作者在 QwenOFT 上复现（OpenVLA 版 19.4）	这是**仅锚帧**版本，KEM 未在 RMBench 上评测；Press Button 只有 3%、Observe and Pick Up 21%
3	**24.9%**（隐式记忆库） vs 75.2%	EventVLA 表 2 消融	RoboTwin-MeM	成功率	同 #1	同框架单变量消融	全文最干净的对照：只换记忆表征（原图 vs latent）
4	**13.6%**（动作块 15）	EventVLA 表 2 消融	RoboTwin-MeM	成功率	同 #1	同框架消融	低于不用 KEM 的 18.0；说明 KEM 前瞻依赖块长度
5	**90 / 60 / 90 / 75%**（vs π_MEM 50 / – / 30 / 40）	EventVLA 图 4	ARX ACONE 双臂真机 4 任务	成功率	每任务 20 次	作者自测；π_MEM 为作者复现，基座 $\pi_{0.5}$	真机基座换成 $\pi_{0.5}$（仿真为 QwenOFT）；无置信区间
6	**2.91 → 0.94 Hz**	EventVLA 表 9	RoboTwin-MeM 8 任务	吞吐（chunks/s），延迟 s/chunk	表 9 逐任务平均	作者自测	仅锚帧版已降到 1.07 Hz，KEM 只再降 0.13
7	**69.23**（ACT 25.50 →） / **59.53**（DP 25.00 →）	TRACE 表 1	5 个真机延迟证据任务	**阶段进度 %**（非成功率），任务平衡平均 ± bootstrap SE	每任务 25 次真机 rollout	作者自测	Tool 任务 SE ± 17.96；去 Tool 后回归版 72.92
8	**50.47**（$\pi_{0.5}$）	TRACE 表 1	同 #7	阶段进度 %	25 次/任务	作者在同数据上微调 180k 步，只给通用提示	VLA 基线**无记忆模块**；”TRACE + $\pi_{0.5}$” 未测
9	**37.8 / 41.6**（顺序反转） vs 保序 86.5–92.4 / 92.3–96.0	TRACE 表 3	同 #7	归一化诊断量（路由相似度 / 分支一致性，完整 TRACE = 100）	留出 rollout	作者自测	不是成功率；只说明记忆依赖有序历史
10	**+2.11 ms/步**（占回归路径 35.8%；扩散路径 1.8%）	TRACE 表 14	—	每步延迟 mean / p95	1900 步	RTX 5090, PyTorch 2.9.1	扩散基座前向 118 ms（100 步去噪）稀释了占比

#	数字	出处	任务集	指标类型	试验规模	独立性	口径提醒
11	**53.3 / 66.7 / 70.0 / 56.7%**（独立模仿 23.3 / 43.3 / 50.0 / 33.3）	SAI 表 1	Bed-throw / Tablecloth / Laundry / Painting 真机	成功率	每任务-方法 30 次	作者自测	Phase Sync / Yield 的分母是事件数（120 / 90 / 60 / 150 检查点），非独立试验
12	**64.5% → 16.1%**（提前松手）	SAI 表 2	Laundry 交付阶段	失败率	31 个从高风险阶段起始的 rollout 片段	作者自测	不是完整任务 rollout；架构消融，与协作收益无关

二、14 篇邻居

#	数字	出处	任务集	指标类型	试验规模	独立性	口径提醒
13	**27.8% → 51.4%**（聚合 TSR），SCR 42.3 → 76.4	KEMO	6 个 YAM 双臂真机任务	任务成功率 / 阶段完成率	每任务 12 次	作者自测；MemoryVLA 基线由作者训练（单相机、不同骨干）	Cover Blocks 9/12 vs 0/12；MemoryVLA 聚合 TSR 1.4% 受基线设置影响
14	**93.4 vs 68.2**（仿真）/ **80.0 vs 43.5**（真机）	UniMem	robosuite 5 任务 / xArm6 4 任务	成功率（部分任务为子任务平均或位置精度分）	仿真 N = 25，真机 N = 15	作者自测；68.2 为 6 s 定间隔采样基线，43.5 为 MemER	UpDown3Times 报子任务平均而非整任务成功
15	**71.9%**（SimplerEnv-Bridge，+14.6）/ **41.2%**（MIKASA-Robo）	MemoryVLA	SimplerEnv / MIKASA-Robo	成功率	基准协议	作者自测	在 RoboMME 附录中总分低于 10%，在 RoboMemArena 中 TSR 15.0 ——同一方法跨基准差异极大
16	**错舀 1 次**（vs 无历史 61、32 帧长历史 12）	MemER	Counting 真机任务	计数错误次数（越低越好）	20 trials	作者自测	加文本模态反而变差（MemER + Text 错舀 13）；API 模型在线部署因 10–15 s 延迟全部失败

#	数字	出处	任务集	指标类型	试验规模	独立性	口径提醒
17	**15 分钟级任务**；上下文自适应 +11 / +62 个百分点	MEM（π_MEM）	Recipe Setup / Clean Kitchen；捡筷子 / 开冰箱	进度分 / 成功率	每任务 10 次 rollout，柱状图	作者自测（$\pi_{0\cdot6}$ 基座）	主结果**未给精确数值**，只有图；EventVLA 真机基线 π_MEM 为 EventVLA 作者复现
18	**42.0%**（Mem-0 总平均；M(1) 52.8 / M(n) 28.5）	RMBench	9 任务	成功率	每任务 100 次	基准方自测	Press Button 所有方法 0%；真机 3 任务各 40 次 Mem-0 22.50%
19	**17.93**（$\pi_{0\cdot5}$）/ **44.51**（FrameSamp+Modul）/ **42.38**（MemER）/ **84.08**（oracle 上界）	RoboMME	16 任务，Counting / Permanence / Reference / Imitation	成功率，类别平均	基准协议	基准方对 14 种记忆变体统一实现	动作 token 对记忆 token 平均注意力只有 0.7–4.8%；真机 4 任务各 10 次 $\pi_{0\cdot5}$ 4/40
20	**60.83**（PonderPounce 9B）/ **75.54**（9× 数据）	PonderPounce	RoboMME	成功率	基准协议	作者自测，基线 FrameSamp+Modul 44.51 / 57.88 沿用 RoboMME 数字	不同数据量档位不可混比
21	**38.5 TSR / 55.2 CSR**（PrediMem，Qwen3-VL-8B）vs $\pi_{0\cdot5}$ 21.5 / 38.7	RoboMemArena	26 任务，Transfer / Occlusion / Counting / Sequence	任务成功率 / 累计成功率	基准协议	基准方自测	S2 规模 1.7B / 4B / 8B 对应 19.9 / 31.9 / 38.5；真机 $\pi_{0\cdot5}$ 20%、PrediMem 52%
22	**41.3 → 60.1**（任务成功）/ 52.5 → 66.2（累计）	HyMeS	RoboMemArena **修正后的 12 任务协议**，160 回合	任务成功率 / 累计成功率	160 回合	作者自测；比 PrediMem 高 14.5 / 4.5	12 任务子集由作者选定；开发期程序编辑依赖基准阶段判定（真机无此特权反馈）
23	**55.96%**（Counting 四任务均值；PickXTimes 100、BinFill 84）	AGM	RoboMME Counting 官方 50 回合测试集	成功率	50 回合，两次独立运行均值	作者自测，基线含 MemER 48.83	SwingXTimes / StopCube 上 AGM = 冻结 $\pi_{0\cdot5}$（子目标不锚定夹爪，按设计退回）
24	**73.6%**（RMBench 7 任务；Mem-0 50.8、$\pi_{0\cdot5}$ 11.2）	Chronos	RMBench **7 任务**子集	成功率	每任务 100 次	作者自测；基线为论文报告值	只有 7 任务，不能与 EventVLA 的 9 任务 67.8 直接比；Press Button 2%、Battery Try 29%

#	数字	出处	任务集	指标类型	试验规模	独立性	口径提醒
25	**0.42 → 0.84**（5 训练任务）/ 0.07 → 0.23（留出）	μVLA	MIKASA-Robo	成功率	基准协议	作者自测	受控隔离研究，只加记忆 token 不加辅助损失；LIBERO 96.2 说明全可观下无退化
26	**30.9% → 80.0%**（两轮自动干预；人判切换 68.6%）	AutoIntervene	7 个真机双臂任务	成功率均值	每格 25 次无辅助 rollout	作者自测	累计操作者时间 122.9 s vs 179.9 s；对照 LazyDAgger / RND-DAgger 的切换指标为 1.00 vs 0.00 / 0.90 误触发
27	**A_4 ≈ +0.019 / +0.015 / −0.011**（历史独有信息）	Present but Not Remembered	Octo-Small / Octo-Base / CronusVLA-0.5B	线性探针可解码性差值	逐层探针 + 因果交换干预，5 种子	独立审计（非方法论文）	不是成功率；结论是历史 ≈ 当前帧副本；注入历史能否驱动动作因架构而异（Octo 不能，CronusVLA +0.167）
28	**32.4 → 84.0**（仿真）/ **98.7**（真机）/ 20% 数据	NativeMEM	论文自选任务	成功率	摘要口径	作者自测（仅据摘要）	本仓库未译全文，数字仅据摘要

三、并排时最常踩的四个坑

1. **RMBench 的 67.8 与 73.6 不是同一个任务集**：EventVLA 报 9 任务，Chronos 报 7 任务；Chronos 表里的 Mem-0 是 50.8（7 任务），EventVLA 表里是 42.0（9 任务）。
2. **成功率 vs 阶段进度**：TRACE 的 69.23 是阶段进度百分比，MemER 的 Counting 用错舀次数，RoboMemArena 同时报 TSR 与 CSR；三者与”成功率”不可互换。
3. **基线是谁跑的**：EventVLA 表里的 MemoryVLA 41.7 是 EventVLA 作者复现，KEMO 表里的 MemoryVLA 是单相机、换骨干的复现，RoboMME / RoboMemArena 里的 MemoryVLA 是基准方统一实现——同一方法三个数字，三种口径。
4. **图内数字与文字数字**：MEM 的主结果只有柱状图，EventVLA 真机结果只有图 4；账本里这类条目标”图”或”未给精确数值”，引用时不要补数。

8. 分类文献解读

分组与 README 一致，每组先给一段定位。三篇核心论文的完整解读见第 2–4 章，这里只保留元数据与一句话定位；14 篇邻居论文为基于全文的手写笔记（一句话定位 / 解决什么问题 / 方法 / 关键数字 / 局限与注意 / 与核心论文的关系 / 关联阅读）；其余 86 篇为一句话定位与原文摘要。

核心论文（逐篇深读） / Core Papers (Deep Dives)

本仓库围绕这三篇 2026 年 6 月的论文展开。它们回答同一个问题——决策时刻证据已经消失，策略靠什么补——但分别把答案放在架构（EventVLA 的关键帧证据记忆）、模块（TRACE 的轨迹寻址槽记忆）和数据（SAI 的搭档分布课程）三个层面。每篇都有中英文详细解读、英文原文 PDF 与保版式中文翻译 PDF。

1. EventVLA: Event-Driven Visual Evidence Memory for Long-Horizon Vision-Language-Action Policies

Ganlin Yang, Zhangzheng Tu, Yuqiang Yang, Sitong Mao, Junyi Dong, Tianxing Chen, Jiaqi Peng, Jing Xiong et al. — arXiv 2026, arXiv 2606.20092

核心论文之一：稀疏视觉证据记忆 = 初始/近期锚帧 + 学习的关键帧证据记忆（KEM），KEM 从 VLA 隐状态预测未来 50 步的关键帧概率，写入容量 5 的原图缓冲；RoboTwin-MeM 18.0→75.2%，RMBench 67.8%（仅锚帧）。

Abstract. Memory remains a critical bottleneck for long-horizon robotic manipulation, as standard Vision-Language-Action (VLA) policies often fail when task-relevant cues become occluded or unobservable over time. While existing memory-augmented methods utilize historical context, they either suffer from severe information bottlenecks, incur high latency via decoupled dual systems, or rely on unselective buffers that accumulate massive visual redundancies. To address these limitations, we introduce EventVLA, an end-to-end framework founded on the concept of sparse visual evidence memory that comprises two core components: foundational visual anchors to retain initial and short-term contexts, and a dynamic Keyframe Evidence Memory (KEM) module. Specifically, KEM directly predicts future keyframe probabilities from the VLA's latent embeddings to autonomously capture and store sparse, task-critical visual events. This foresight-driven mechanism empowers the policy to dynamically evaluate the future causal utility of current observations, preserving transient visual evidence before it becomes unobservable. Furthermore, we propose RoboTwin-MeM, a diagnostic benchmark specifically designed to evaluate non-Markovian manipulation tasks with interactive visual evidence. Extensive evaluations show that across 17 memory-requiring simulation tasks and 4 real-world bimanual tasks, EventVLA achieves an average success rate improvement of +40% over state-of-the-art memory-augmented VLAs.

2. TRACE: Trajectory-Routed Causal Memory for Delayed-Evidence Visuomotor Imitation

Zihao Li, Ranpeng Qiu, Yincong Chen, Guoqiang Ren, Weiming Zhi — arXiv 2026, arXiv 2606.14551

核心论文之一：固定 K 槽 latent 记忆，用机器人状态轨迹的路径签名做读写地址，适配器外挂到 ACT/DP；五个真机延迟证据任务 ACT 25.5→69.2，顺序反转负对照把路由相似度打到 37.8。

Abstract. Robots under autonomous operation may require decisions based on evidence that is no longer visible. We study delayed-evidence tasks, where an early cue disappears before a later decision point, so visually similar observations can require different actions. In these settings, the current observation is not a sufficient state for control. We introduce TRAjectory-routed Causal Evidence (TRACE), a memory framework for visuomotor imitation policies. TRACE stores task-relevant visual and robot-state evidence, such as object identity, target choice, or route-dependent state, in a fixed-size latent memory that remains bounded over long episodes. Instead of indexing memory by raw time or manually provided task labels, TRACE uses path signatures: compact, order-sensitive features of the executed robot-state trajectory. These signatures do not store the visual cue itself; rather, they provide trajectory-conditioned keys for writing and retrieving the evidence stored when the cue was visible. When the robot later reaches an ambiguous observation, the policy conditions on TRACE memory to recover the missing context and choose the correct branch. TRACE attaches through lightweight adapters to policies, without changing

the policy backbone, action head, or imitation objective. Across real-world long-horizon manipulation tasks with visually ambiguous branch points, TRACE improves branch selection and task success over alternative baselines, including short-history and recurrent memory. Project page: https://jeong-zju.github.io/trace

3. Robots that Collaborate: Sequential Asymmetric Imitation for Learning Coupled Robot Policies

Yincong Chen, Ranpeng Qiu, Zihao Li, Yanan Zhou, Guoqiang Ren, Weiming Zhi — arXiv 2026, arXiv 2606.16490

核心论文之一：单遥操作者三阶段课程（A 与人类搭档→B 对着部署的 A→对 A 做稀疏干预）学出无通信的双机器人耦合策略；历史 token 把提前松手从 64.5% 降到 16.1%。

Abstract. Collaborative mobile manipulation requires robots to coordinate with a partially observed partner while physically interacting through shared objects. This is difficult because failures often arise not from poor local skills, but from mistimed waiting, yielding, pulling, releasing, or repositioning. We study this problem with two bimanual mobile manipulators coupled through rigid and deformable objects. We propose Sequential Asymmetric Imitation (SAI), a single-teleoperator curriculum for learning coupled multi-robot behaviors without synchronized dual-operator demonstrations or explicit inter-robot communication. SAI trains Robot A from unilateral demonstrations with a compliant human partner, trains Robot B against the deployed Robot A policy, and then refines Robot A using sparse interventions near coordination failures. This staged process exposes the policies to increasingly realistic partner behaviors, including delay, phase mismatch,insufficient yielding, and interaction conflict. Across real-world dual-robot manipulation tasks, SAI improves task success, phase synchronization, and partner-contingent yielding over independent imitation and curriculum-ablation baselines. These results suggest that physically coupled collaboration can be learned through the structure of the imitation curriculum, rather than through synchronized multi-operator demonstrations or explicit coordination mechanisms. More videos on project page:http://cyc0429.github.io/sai-project-page/

综述与分析 / Surveys and Analyses

先读这里的三篇分析再读方法：《Present but Not Remembered》用探针和因果干预说明冻结 VLA 里的历史基本是当前帧的冗余副本；长上下文扩散策略研究说明 naive 加长窗口没有传说中那么脆；WhyChunking 说明动作块的价值之一正是非马尔可夫表达力。

1. Weights or Skills? A Survey of Robot-Learning Techniques: from Action-Predicting Weights to Robots that Write their Own Skills

Gaytri Jena, Kapil Wanaskar, Vinija Jain, Aman Chadha, Vasu Sharma, Amitava Das — arXiv 2026, arXiv 2608.01851

「权重 vs 技能（代码）」的机器人学习综述，梳理持久技能记忆与自我改进闭环；与 HyMeS「技能在权重、记忆在代码」同一坐标。

Abstract. Robot learning is splitting into two bets: policies that bake competence into frozen weights (vision-language-action, or VLA, models), and agents that write and refine their own executable skills as code. This survey organises the field around that axis of weights versus skills. Its central analytical contribution is a deep-dive that arranges code-as-policy methods by their degree of self-improvement, from zero-shot program synthesis, through closed-loop self-repair and persistent skill memory, to the sparsely populated cell in which execution feedback, skill memory, and evolutionary search combine into one open-ended loop; only a few very recent systems (for example ASPIRE, ENPIRE, and RoboClaw) occupy that cell. We map the complementary “skills” pole, from unsupervised

reinforcement-learning skill discovery to large-language-model skill libraries, and show that the word “skill” is used in at least five distinct senses, of which only the code sense self-improves without gradient updates. We then connect the taxonomy to the emerging skill economy: commercial robot-skill marketplaces now distribute one-tap skills across robots but ship only static playback, which surfaces open problems of adaptation, cross-embodiment portability, provenance, safety verification, composition, and standardisation. This is a deliberately focused survey. Rather than cataloguing the field exhaustively, it examines 77 representative systems across six technique families through one taxonomy and a set of contrast tables, and it supplies operational definitions of the self-improvement mechanisms together with a statement of what each family cannot do.

2. Why Does Action Chunking Improve Behavioral Cloning Performance in Robotic Control?

Filippo Lazzati, Kyle Stachowicz, William Chen, Alberto Maria Metelli, Andrew Wagenmaker, Sergey Levine — arXiv 2026, arXiv 2608.02547

动作块之所以有效，是因为更强的非马尔可夫表达力、更少的复合误差与隐式集成，而非时间一致性等旧假说；解释了 EventVLA 的 KEM 为何依赖长动作块。

Abstract. Action chunking—predicting and executing multiple actions instead of a single action—has proven to be a critical component for learning effective robotic control policies. However, our precise understanding of why action chunking improves performance has remained limited. In this work we seek to close this gap. Through rigorous experimental evaluations in both simulated and real-world settings, we show that existing hypotheses for the success of action chunking—temporal consistency, horizon reduction, and representation learning—fail to explain the success of action chunking. Instead, we find that action chunking benefits from greater non-Markovian expressivity and reduced compounding error compared to Markovian policies, but, in many settings of interest, these effects can be fully captured by delayed policies, which at each step predict a single action based on the observation MATH1 steps in the past. We then show that there exists an additional benefit of action chunking that we refer to as implicit ensembling. In particular, by learning a diversity of temporal relationships (that is, MATH2), action-chunked policies exhibit behavior matching that of a model ensemble, increasing their robustness and generalization ability over policies that only learn a single temporal relationship. Building on these insights, we show that in simulated and real-world robotic control settings, we can match the performance of action chunking without action chunking—by deploying an action chunking policy as an ensemble of policies with randomized delays. Furthermore, we propose a policy class that amplifies the benefits of action chunking by explicitly instantiating an ensemble, and which we show significantly improves over the performance of action chunking in many domains.

3. Present but Not Remembered: Auditing How Frozen VLAs Encode, Deploy, and Steer Visual History

Chih-Ting Liao, Xin Cao — arXiv 2026, arXiv 2607.03372

> **arXiv 2607.03372** · arXiv 2026 · 提交 2026-07-03 · 分类 `survey` / 综述与分析
> *Chih-Ting Liao, Xin Cao*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

对三个冻结 VLA（Octo-Small 27M、Octo-Base 93M、CronusVLA-0.5B）做逐层线性探针与激活置换干预，结论是「在场、冗余、仅兜底」：上一帧内容各层可解码（Octo-Small 峰值 R^2=0.398），但扣除当前帧已能解释的部分后，历史独有信息上界只有 +0.019（81 种探针配置，A_4/A_1≈0.05）；历史只在当前帧被完全遮黑时才进入动作，第 6 层后依赖消失。

解决什么问题

记忆增强工作默认「原生 VLA 不会用历史」，却没人先诊断：多帧策略里的过去是独立存储还是当前帧的副本，是否曾被读入动作。与 Grant 等人只看单帧内模态融合的机理研究互补，作者用无需训练、开环、单卡可跑的审计回答：历史是否在场（A_1）、是否独有（A_4）、何时何处被使用（$\Delta\ell$ 与截断层 ℓ^*）。

方法

刺激集与模型。 BridgeData V2 中 250 对匹配的 (o_{t−1}, o_t, 指令)，213 条指令，哈希在分析前冻结；开环读取 7 维动作块的扩散均值。Octo 窗口 2 帧；CronusVLA 的历史是每帧一个 896 维认知特征。

解码腿。 岭回归从第 ℓ 层历史 token 激活预测上一帧场景得 A_1；投影掉当前帧探针已能解释的部分、只解码残差得 A_4。扫 81 种（层 × 池化 × 目标 × 正则）配置取上界，打乱标签作对照。

部署腿。 在第 ℓ 层把历史 token 激活换成供体的，读动作变化，减去自我置换噪声底（约 0.15）得校正贡献 Cℓ；部署间隙 $\Delta\ell$ = C(全黑遮挡) − C(干净)，配对 bootstrap CI、5 种子符号一致、5,000 次置换检验三者同时成立才算「被使用」。退化梯度含部分遮挡、重遮挡、模糊、全黑，帧序打乱作顺序盲对照。第二种干预是注意力敲除：屏蔽读出 token 对历史 key 的注意力。

注入门。 在截断层之前把样本自己的干净历史注回遮挡状态，看能否修复动作或消除状态别名；在 CronusVLA 上看能否把动作推向供体。四项合成可复用的 Temporal-Deployment Audit。

关键数字

量	Octo-Small	Octo-Base	CronusVLA-0.5B
A_1（历史可解码）	0.389	0.362	0.142
A_4（历史独有）	+0.019	+0.015	−0.011
全黑遮挡下历史贡献	上升（兜底）	上升，部署速率约为 Small 的 31%	下降 0.0498→0.0357（常驻）
注入能否驱动动作	不能（修复 −0.010，CI 含 0）	未测	能（+0.167，CI [0.106, 0.227]）

Octo-Small 逐层：L4 部署间隙 +0.054（CI [0.028, 0.079]，置换 p=0.0004，5/5 种子），L6 为 −0.005；注意力敲除 L0 +0.323、L4 +0.137、L6 +0.045，同样在 L6 断开。退化梯度里只有全黑显著（+0.043），部分/重遮挡与模糊都不显著；帧序打乱把两种架构的贡献都打到约 0。夹爪只占历史效应的 9.3%；自然遮挡在 250 对里只有 3 对。

局限与注意

- 审计的「历史」只是 Bridge 帧率下的前一帧，演示又近似马尔可夫，A_4≈0 在此设定下并不意外；外推到长视距记忆任务是推论而非实验，论文没有测任何真正的记忆依赖任务。
- A_4 以上一帧像素为目标残差化，排除不了模型存有抽象历史（任务进度、意图）；作者承认这点，以部署腿作旁证。

- 开环、250 对、两大架构族、消费级单卡；满足「冻结、开源、多帧、单卡」的第三个家族没找到（$\pi_0/\pi_{0\cdot5}$、X-VLA、ACT 单帧，OpenVLA-OFT 多视角非多帧），「兜底 vs 常驻」只有两个端点。「在中层截断前注入」的位置结论来自 12 层的 Octo，能否迁移到 3B 级 VLA 未验证。
- 对本仓库的含义：「多给几帧」的设计（ContextVLA、CronusVLA）按本文预测主要在加冗余；作者的可检验假设：在近马尔可夫数据上训练的记忆模块会继承冗余，除非目标显式奖励「当前帧不可还原」的信息。

与核心论文的关系

- **EventVLA**（2606.20092）：KEM 只存「以后会看不见」的帧（掀开又盖回的方块、短暂出现的数字卡），正是本文定义的历史独有信息而非更多历史；RoboTwin-MeM 上锚帧 18.0% 到加 KEM 75.2% 与「加独有信息才有收益」同向。张力在短期锚帧：按本文逻辑近期帧几乎是当前帧副本，但 EventVLA 在 RMBench 上去掉短期窗从 67.8% 掉到 23.8%；可能的调和是近期帧携带运动与阶段信息而非像素内容，本文以 RGB 场景为目标的探针测不到。
- **TRACE**（2606.14551）：路径签名增量是机器人状态轨迹的有序特征，从单张当前图像无法还原，按定义是当前帧不可还原的信息；本文发现两种冻结架构都对帧序盲（打乱后贡献归零），TRACE 的顺序反转负对照（37.8 对比保序变换约 90）恰好证明其记忆用到了顺序。经适配器外挂到 ACT/DP，也符合本文对兜底型模型「走工程化通路注入」的建议。
- **SAI**（2606.16490）：历史 token 让策略分清单帧看起来一样的「搬运中」与「该松手」两态，过早释放 64.5%→16.1%；这类状态别名在训练数据里本就存在，模仿目标因此天然奖励历史独有信息，正是本文假设中记忆模块不继承冗余的条件。三篇核心论文都在注入独有信息而非更多历史。

关联阅读

- Grant et al.（2603.19233）：单帧内模态融合轴的跨架构机理研究，本文的正交对照。
- CronusVLA（2506.19816）：被审计的第二架构族，$A_4 \approx 0$ 但常驻使用、可被注入引导。
- ContextVLA（2510.04246）：多帧历史 token 直接拼入输入，本文预测主要增加冗余。
- Past-Token Prediction（2505.09561）：显式预测过去 token 的训练目标，即「目标函数奖励历史信息」的一种实现。
- Buurmeijer et al.（2603.05487）：观测与控制 VLA 内部特征，同属 VLA 机理可解释性线。

4. World Action Models: A Survey

Qiuhong Shen, Shihua Zhang, Yue Liao, Qi Li, Zhenxiong Tan, Shizun Wang, Shuicheng Yan, Xinchao Wang — arXiv 2026, arXiv 2606.20781

世界动作模型综述，讨论持久性（persistence）与记忆在预测-动作耦合中的位置，指出趋势是「生成更少的未来、保留控制所需的部分」。

Abstract. World Action Models (WAMs) are embodied predictive-action models that make a forecast of the future available to action. Recent WAMs repurpose large video generation models, and a parallel line relies on language or vision-language backbones without a video-generation core. This rapid expansion has blurred the boundary among broad world models, video generation models, action-grounded video world models, Vision-Language-Action policies, and WAMs. This survey gives the field a common account. It first clarifies these boundaries, then organizes existing works through two complementary views. The first view asks what each method is required to generate, spanning rendered futures, latent futures, and video-generation-free action reasoning. The second view

decomposes each method by predictive substrate, backbone, action coupling, and deployment regime. This anatomy supports a unified discussion of interactability, causality, persistence, physical plausibility, and generalization, followed by data, evaluation, and open challenges. Across these axes, a consistent design pattern emerges: WAMs are not simply video generators with action heads, but predictive-action methods whose design choices trade representational richness against compute, memory, latency, and action-label cost. The field is moving toward methods that generate less of the future while preserving what control requires. The survey homepage is available at https://world-action-models.github.io/.

5. Training and Evaluating Diffusion Policies with Long Context Lengths

Abhinav Agarwal, Adam Wei, Taylan Kargin, Michael Zeng, Cole Becker, Arif Kerem Dayi, Pablo Parrilo, Asuman Ozdaglar et al. — arXiv 2026, arXiv 2606.16447

系统扫描扩散策略的上下文长度：naive 加长没有文献说的那么脆（UNet+交叉注意力下多任务可行），并提出多上下文长度联合训练；是「长窗口是否可用」争论的关键数据点。

Abstract. Imitation learning has enabled highly-dexterous robotic manipulation from RGB observations. Policies trained with these methods, however, typically condition robot actions on only a short history of observations. These policies cannot solve tasks that require memory and can get stuck repeatedly executing the same failing motions. In this work, we first benchmark policy performance as context length is incrementally increased from short to long, across a spectrum of tasks with varying local stability and memory requirements, and in multiple data regimes. To our knowledge, this is the first study to investigate context length for Diffusion Policies at this level of detail. Our results challenge prior claims: naively scaling context length is not as brittle as advertised in literature. With an appropriate conditioning method and denoising backbone (UNet+Cross-Attention), single-task policies achieve high success rates on many tasks in the usual data regime even with naive scaling. Next, we propose a training algorithm to jointly train policies at multiple context lengths, further reducing the sample complexity of long-context learning. Finally, we apply our findings to re-evaluate some previously proposed solutions to long-context imitation learning.

6. Large VLM-based Vision-Language-Action Models for Robotic Manipulation: A Survey

Rui Shao, Wei Li, Lingsen Zhang, Renshan Zhang, Zhiyang Liu, Ran Chen, Liqiang Nie — arXiv 2025, arXiv 2508.13073

大 VLM 基 VLA 的系统综述，把记忆机制列为有前景方向之一；本仓库的分类沿用其单体/双系统/分层划分。

Abstract. Robotic manipulation, a key frontier in robotics and embodied AI, requires precise motor control and multimodal understanding, yet traditional rule-based methods fail to scale or generalize in unstructured, novel environments. In recent years, Vision-Language-Action (VLA) models, built upon Large Vision-Language Models (VLMs) pretrained on vast image-text datasets, have emerged as a transformative paradigm. This survey provides the first systematic, taxonomy-oriented review of large VLM-based VLA models for robotic manipulation. We begin by clearly defining large VLM-based VLA models and delineating two principal architectural paradigms: (1) monolithic models, encompassing single-system and dual-system designs with differing levels of integration; and (2) hierarchical models, which explicitly decouple planning from execution via interpretable intermediate representations. Building on this foundation, we present an in-depth examination of large VLM-based VLA models: (1) integration with advanced domains, including reinforcement learning, training-free optimization, learning from

human videos, and world model integration; (2) synthesis of distinctive characteristics, consolidating architectural traits, operational strengths, and the datasets and benchmarks that support their development; (3) identification of promising directions, including memory mechanisms, 4D perception, efficient adaptation, multi-agent cooperation, and other emerging capabilities. This survey consolidates recent advances to resolve inconsistencies in existing taxonomies, mitigate research fragmentation, and fill a critical gap through the systematic integration of studies at the intersection of large VLMs and robotic manipulation. We provide a regularly updated project page to document ongoing progress: https://github.com/JiuTian-VL/Large-VLM-based-VLA-for-Robotic-Manipulation

记忆依赖操作的基准 / Benchmarks for Memory-Dependent Manipulation

2026 年春天落地的 RMBench、RoboMME、RoboMemArena 是夏天方法论文的共同靶子。RoboMME 的结论（记忆表征的效果高度依赖任务）被之后一年的结果反复印证。EventVLA 自带的 RoboTwin-MeM 用 n 参数化「必须记住几个中间关键帧」，是目前唯一专门隔离瞬态证据的基准；更多嵌在方法论文里的基准（ReMemBench、LIBERO-Mem、MemMimic、MemoryRTBench、RuleSafe、Memory-T-Bench、MemoryBench）见对应条目。

1. RoboDojo: A Unified Sim-and-Real Benchmark for Comprehensive Evaluation of Generalist Robot Manipulation Policies

Tianxing Chen, Yue Chen, Zixuan Li, Junyuan Tang, Kailun Su, Haoran Lu, Weijie Wan, Baijun Chen et al. — arXiv 2026, arXiv 2607.04434

统一仿真+真机基准（42 仿真 + 18 真机任务），记忆是五个评测维度之一；提供公共榜单与 XPolicyLab 集成。

Abstract. Generalist robot manipulation policies have advanced rapidly, yet existing benchmarks remain limited in systematically evaluating their capabilities. Many rely on simple, short-horizon, or skill-narrow tasks with limited capability coverage, and are often conducted only in simulation or only in the real world. Simulation enables scalable feedback but misses physical deployment challenges, while real-world evaluation is costly, time-consuming, and difficult to reproduce. We introduce RoboDojo, a unified sim-and-real benchmark for comprehensive evaluation of generalist robot manipulation policies. RoboDojo includes 42 simulation tasks and 18 real-world tasks covering diverse and complementary manipulation capabilities. The simulation benchmark evaluates five dimensions: generalization, memory, precision, long-horizon execution, and open-vocabulary instruction following, while the real-world benchmark exposes policies to challenging physical-world deployment conditions. RoboDojo supports scalable evaluation through heterogeneous parallel simulation in Isaac Sim and provides RoboDojo-RealEval, a reproducible real-world evaluation system with remote cloud access, standardized hardware, scene reset, evaluation protocol, and deployment interface. Together with XPolicyLab, policies can be integrated once and evaluated across simulation and real-world settings with minimal adaptation. We integrate 30 policies into XPolicyLab and evaluate them on RoboDojo, establishing a public leaderboard and systematic analysis of current policy performance. The website is available at http://robodojo-benchmark.com/.

2. RoboMemArena: A Comprehensive and Challenging Robotic Memory Benchmark

Huashuo Lei, Wenxuan Song, Huarui Zhang, Jieyuan Pei, Jiayi Chen, Haodong Yan, Han Zhao, Pengxiang Ding et al. — arXiv 2026, arXiv 2605.10921

> **arXiv 2605.10921** · arXiv 2026 · 提交 2026-05-11 · 分类 `bench` / 记忆依赖操作的基准
> *Huashuo Lei, Wenxuan Song, Huarui Zhang, Jieyuan Pei, Jiayi Chen, Haodong Yan, Han Zhao, Pengxiang Ding, Zhipeng*

> *Zhang, Lida Huang, Donglin Wang, Yan Wang, Haoang Li*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

RoboMemArena 用 VLM 分解子任务 + AnyGrasp 自动执行生成 26 个任务、2,600 条平均 1,076 步的轨迹，151 个子任务中 104 个（68.9%）依赖历史，并原生提供子任务指令与关键帧标注；配套双系统 PrediMem 以 38.5% TSR 领先 MemER 27.3%、$\pi_{0.5}$ 21.5%。要记的对照：拿掉关键帧库只剩 5 帧近期缓冲，TSR 跌到 17.7%，低于无记忆的 $\pi_{0.5}$。

解决什么问题

三个缺口：已有记忆基准缺少能直接监督双系统规划器的多模态标注（关键帧图像 + 子任务语言）；任务短、结构简单、有些并不真需要记忆（点名 MemoryBench、MIKASA、RMBench、RoboMME）；没有配对的真机评测。要回答的是：在平均超过 1,000 步、多数子任务不可由当前帧判定的任务里，反应式 VLA 差多少，关键帧该按什么规则写入。

方法

任务。 四类：Transferring 4 个（相同容器间搬运，记源-目标映射）、Occlusion 11 个（开抽屉或放微波炉后物体被遮住）、Counting 7 个（倒酱两次等重复动作，前后场景几乎一样）、Sequence 4 个（后续动作依赖前一步结果）。单任务平均步长 472（Pour Wine into Mug Twice）到 1,835（Put Cookies Butter into Drawer Respectively）；双相机 256×256。

生成管线。 VLM 从指令和当前图像生成有序子任务并指派到 {Move, Place, Pour, Open, Close} 五个原语，不合理的分解人工修正；AnyGrasp 估 6-DoF 抓取位姿驱动原语，失败则重试；关键帧取夹爪开合切换帧与末端速度极小/方向突变帧的并集。每任务 100 条演示，共 15,100 段关键帧对齐片段。

评测。 TSR 要求全部阶段谓词同时成立；CSR 为满足的阶段比例，每任务 3–9 个验证阶段。真机 5 个任务在 AgileX Cobot Mobile Aloha 双臂平台各 10 次，IHMB（看人类演示后做早餐）最长超过 3 分钟。

PrediMem。 S2 为 Qwen3-VL-8B-Instruct（冻结视觉塔，其余全参微调 2 轮），输入 5 帧近期缓冲 + 不设上限的关键帧缓冲，输出当前子任务和近期窗内哪几帧写入关键帧库；训练时加预测编码头，从当前隐状态预测下一帧冻结 ViT 特征（MSE + 余弦，权重 0.1），推理时移除。S1 沿用 $\pi_{0.5}$ 的 flow-matching 目标。异步运行：S2 1.06 Hz，S1 3.40 Hz。

关键数字

方法（TSR %）	Transfer	Occlusion	Counting	Sequence	平均 TSR	平均 CSR
$\pi_{0.5}$	20.0	12.7	14.3	60.0	21.5	38.7
HiF-VLA	17.5	12.7	8.6	42.5	16.9	39.8
MemoryVLA	15.0	7.3	14.3	37.5	15.0	35.3
MemER	20.0	16.4	27.1	65.0	27.3	49.1
GPT-5.4 作 S2（冻结）	13.8	1.8	12.9	15.0	8.7	30.5
PrediMem 去预测编码头	25.0	19.5	38.6	63.8	32.3	49.0

方法（TSR %）	Transfer	Occlusion	Counting	Sequence	平均 TSR	平均 CSR
PrediMem 去关键帧库	17.5	6.4	20.0	45.0	17.7	41.6
PrediMem（Qwen3-VL-8B）	22.5	27.3	45.7	72.5	38.5	55.2
Ground Truth（oracle 参考）	32.5	33.6	51.4	85.0	46.1	64.8

S2 缩放：1.7B / 4B / 8B 的 TSR 为 19.9 / 31.9 / 38.5。预测编码损失权重 0 / 0.1 / 0.5 / 1 对应 TSR 32.3 / 38.5 / 31.0 / 29.8。关键帧库存 2 帧时 CSR 很低，4–8 帧改善，不设上限最好。真机平均成功率：$\pi_{0.5}$ 20%，MemER 40%，PrediMem 52%；IHMB 只有 PrediMem 成功过（10%）。

局限与注意

- oracle 上界只有 46.1% TSR，一半以上失败来自 S1 低层执行而非记忆，TSR 对记忆能力的分辨率受限，CSR 部分弥补。
- 「初始帧 + 近期窗」能否解题：去掉关键帧库即只剩 5 帧近期窗，TSR 17.7% 低于反应式 $\pi_{0.5}$；单独的初始帧锚点未测。Occlusion 类的证据（抽屉里有什么）只在开抽屉中途可见，Counting 类靶向的正是前后场景相同，都不在初始帧里；Transferring 的源-目标映射在初始帧中部分可见。三个基准里它的瞬态证据比例最高。
- 「26 个任务」高估了多样性：物体只有饼干、黄油、巧克力、酱等几样，11 个 Occlusion 任务多为抽屉/微波炉模板换物体。
- 关键帧标注由夹爪切换和运动学拐点自动给出，标的是操作阶段边界而非「证据出现」时刻，拿来监督记忆写入有语义偏差；VLM 分解也需人工修正。
- 68.9% 是作者用 LLM 辅助评分自算并与其他基准对比；Sequence 类上 $\pi_{0.5}$ 已有 60% TSR，该类不少子任务由局部视觉规律决定。真机每任务仅 10 次；HiF-VLA、MemoryVLA 低于 $\pi_{0.5}$，基线调优存疑。

与核心论文的关系

- **EventVLA**（2606.20092）与 PrediMem 的记忆结构几乎同构：近期锚帧 + 存原图的关键帧缓冲。差别在写入决策：PrediMem 由约 1 Hz 的 S2 VLM 回看近期窗做出，监督来自启发式关键帧并靠预测编码增敏；KEM 则从动作块隐状态前瞻预测「这帧将来是否关键」，端到端、无双系统延迟。去关键帧库后跌破反应式基线，独立佐证了 EventVLA「仅近期窗不够」的判断；EventVLA 未在此基准上报告。
- **TRACE**（2606.14551）的延迟证据任务与 Occlusion 类同型（看到抽屉内容、关闭、稍后据此决策）；但 TRACE 用固定 K 槽有界记忆，而这里关键帧库不设上限最好，千步以上任务对有界记忆是压力测试。Counting 类的重复倒酱在状态轨迹上是重复片段，路径签名能否区分第一次与第二次值得验证。
- **SAI**（2606.16490）关注双臂耦合下的阶段判断；真机任务同在双臂 Aloha 平台，IHMB「看人演示后复现」与 SAI 的历史 token 都把阶段信息显式化。四类任务对应 RoboMME 分类的时间（Counting、Sequence）与空间/物体（Occlusion、Transferring），仿真里没有过程模仿类。

关联阅读

- MemER（2510.20328）：最强基线，同为双系统 + 关键帧，差在关键帧选择对任务动态不敏感。
- Keyframe Chaining（2603.01465）：关键帧链处理非马尔可夫长视距操作，思路与关键帧库相近。
- HiF-VLA（2512.09928）：基线之一，运动表征强但不存事件级记忆。
- Mem（2603.03596）：多尺度具身记忆，被引用来论证记忆的多模态性。
- RoboCerebra（2506.06677）：另一长视距基准，但 RMBench 指出其任务信息全程可见。

3. LongBench: Evaluating Robotic Manipulation Policies on Real-World Long-Horizon Tasks

Xueyao Chen, Jingkai Jia, Tong Yang, Yibo Fu, Wei Li, Wenqiang Zhang — arXiv 2026, arXiv 2604.16788

真机 1000+ 回合长时程基准，把任务分成全可观（考执行鲁棒性）与上下文依赖（考歧义推理）两类，发现记忆方法并不稳定地改善上下文难度。

Abstract. Robotic manipulation policies often degrade over extended horizons, yet existing benchmarks provide limited insight into why such failures occur. Most prior benchmarks are either simulation-based or report aggregate success, making it difficult to disentangle the distinct sources of temporal difficulty in real-world execution. We introduce LongBench, a real-world benchmark for evaluating long-horizon manipulation. LongBench consists of over 1,000 real-world episodes, covering two complementary regimes: Context-Independent (fully observable) and Context-Dependent (ambiguity-driven). By organizing tasks into capability- and ambiguity-specific subsets, LongBench enables mechanism-aware evaluation of execution robustness, temporal consistency, and context-dependent reasoning. Evaluating six state-of-the-art policies reveals that long-horizon performance is not governed by a single factor. We observe that performance in fully observable settings is more strongly associated with execution robustness, while contextual difficulty varies across tasks and is not consistently improved by memory-based methods. We hope that LongBench serves as a useful benchmark for studying long-horizon manipulation and for developing policies with stronger robustness across both execution and contextual challenges.

4. RoboMME: Benchmarking and Understanding Memory for Robotic Generalist Policies

Yinpei Dai, Hongze Fu, Jayjun Lee, Yuejiang Liu, Haoran Zhang, Jianing Yang, Chelsea Finn, Nima Fazeli et al. — arXiv 2026, arXiv 2603.04639

> **arXiv 2603.04639** · arXiv 2026 · 提交 2026-03-04 · 分类 `bench` / 记忆依赖操作的基准
> *Yinpei Dai, Hongze Fu, Jayjun Lee, Yuejiang Liu, Haoran Zhang, Jianing Yang, Chelsea Finn, Nima Fazeli, Joyce Chai*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

RoboMME 把记忆拆成时间/空间/物体/过程四类，各 4 个任务共 16 个、1,600 条演示、77 万步，并在同一 $\pi_{0.5}$ 骨干上受控比较 14 种记忆变体与 4 个已有方法。要记的结论：没有一种表征通吃，感知记忆 + 调制器注入（FrameSamp+Modul）总分最高 44.51%，符号记忆在计数类领先；真值子目标的 GroundSG+Oracle 84.08%，人类 90.5%。

解决什么问题

已有记忆型 VLA（MemoryVLA、MemER、ContextVLA 等）各用不同骨干、各评自设任务，回答不了「哪种记忆表征对哪类任务有效」；已有基准也不够：MemoryBench 只有 3 个近乎解决的任务，MIKASA-Robo 面向 RL、平均仅 72 步、演示不足。RoboMME 同时提供显式非马尔可夫、覆盖四类记忆的测试床，和固定骨干、固定记忆预算的模型族，把表征与注入方式拆开比较。

方法

任务。 ManiSkill 中 7 自由度 Franka Panda 桌面，前视 + 腕部 256×256。Counting（时间）：PickXTimes、BinFill、SwingXTimes、StopCube；Permanence（空间）：VideoUnmask、ButtonUnmask 及各自的 Swap 版；Reference（物体）：PickHighlight、VideoRepick、VideoPlaceButton、VideoPlaceOrder；Imitation（过程）：MoveCube、InsertPeg、PatternLock、RouteStick。带 Video 前缀的任务和 Imitation 套件在初始步给一段视频，其余任务的证据在执行中出现（按按钮期间方块被短暂高亮或盖住）。平均 481 步（PatternLock 208 到 VideoPlaceOrder 1,134），评测上限 1,300 步。演示由关键路点回放生成并加 5% 噪声再恢复以注入纠错，分易/中/难，每任务 100 条，附子目标与关键帧标注。

模型族 MME-VLA。 三种表征：符号（SimpleSG / GroundSG 子目标，由 Gemini-2.5-Pro 提示、微调 Qwen3-VL-4B 或仿真真值 Oracle 生成）、感知（TokenDrop 按 RGB 差异保留变化区块并完整保留首帧；FrameSamp 均匀采 32 帧各池化到 4×4）、循环（TTT 快权重、RMT 记忆槽）。三种注入：Context（拼进 VLM 输入）、Modul（动作专家每层交叉注意记忆后做 AdaLN 调制，约 80M 参数）、Expert（独立 18 层记忆专家，约 190M）。记忆预算统一 512 token。

协议。 16 任务联合训练 80k 步，动作块 20 执行 16，冻结 SigLIP；每任务 50 个固定种子回合，取最后 3 个 checkpoint × 3 个随机种子共 9 次平均。对比 $\pi_{0.5}$、$\pi_{0.5}$+过去动作、SAM2Act+、MemER。人类研究把任务改为在线 VideoQA 加 oracle 规划器，18 人 800 回合。

关键数字

方法（成功率 %）	Counting	Permanence	Reference	Imitation	平均
$\pi_{0.5}$（无记忆）	28.78	17.00	17.17	8.78	17.93
$\pi_{0.5}$ + 过去动作	—	—	—	—	19.73
SAM2Act+	35.33	26.00	16.83	7.33	21.37
MemER	48.83	53.17	38.00	29.50	42.38
GroundSG+QwenVL（符号最佳）	38.00	39.34	31.56	21.89	32.70
FrameSamp+Modul（感知最佳）	65.22	25.11	36.33	51.39	44.51
TTT+Expert（循环最佳）	36.00	22.95	19.67	10.78	22.35
GroundSG+Oracle（上界）	83.86	93.31	95.17	63.98	84.08
人类（oracle 规划器）	—	—	—	—	90.50

其他：记忆预算从 64 增到 1024，FrameSamp+Modul 从 30.42 升到 45.87，512 之后增益变小；动作 token 对记忆 token 的平均注意力只有 0.7–4.8%，峰值可到 21.8%，记忆使用稀疏；GroundSG+QwenVL 计算量约为 $\pi_{0.5}$ 的 3 倍，MemER 约 5 倍。真机 4 任务各 10 次：$\pi_{0.5}$ 4/40，GroundSG+QwenVL 19/40，FrameSamp+Modul 25/40。附录中 DP 与 MemoryVLA 总分均低于 10%。

局限与注意

- 论文自述：桌面单臂、固定资产、几乎只测 $\pi_{0.5}$ 一个骨干；同样的记忆设计搬到 OpenVLA-OFT 收益很小（最高 21.6%）。
- 「初始帧 + 近期窗」能否解题：论文没有这一基线，最接近的是完整保留首帧的 TokenDrop，它在三种注入下都低于均匀采样全程的 FrameSamp。按任务结构，9 个任务的证据在初始视频这段静态前缀里，但 VideoPlaceOrder、PatternLock 需要前缀内部的顺序而非单帧；ButtonUnmask、ButtonUnmaskSwap、

PickHighlight 的证据只在按按钮期间出现，是瞬态证据；4 个计数任务需要事件累积。EventVLA「RoboMME 可被静态锚帧解决」的判断在本文内部找不到直接支持，也未被检验。

- 真值子目标能到 84.08%，而人类在 StopCube、SwingXTimes 只有 78 / 80，部分任务的困难来自时序精度与低层控制，不全是记忆。
- 符号记忆变体依赖子目标标注（仿真自动生成，真机需人工标注接地子目标），这项成本应计入比较。
- 多任务联合训练、512 token 预算、冻结视觉编码器，与 RMBench 单任务从零训练的设定不同，数字不能横比。

与核心论文的关系

- **EventVLA**（2606.20092）未在 RoboMME 上报告结果，却认为其可由锚帧解决。RoboMME 的 Button 系列与 PickHighlight 恰是 KEM 针对的瞬态证据：在高亮出现时判断「这帧将来有用」并存原图。RoboMME 发现保留关键帧原图的 MemER 在动态场景变化类任务上最强（54.67），支持存原图关键帧的路线；Counting 套件对应 EventVLA 的 Press Button Keyframe（n=2–5）。
- **TRACE**（2606.14551）的路径签名是机器人状态轨迹的有序特征，天然适合 Imitation 套件里 PatternLock、RouteStick 这类复现轨迹与绕行方向的过程记忆；RoboMME 在这两项上符号记忆几乎为零（6.67 / 6.00），FrameSamp+Modul 为 53.56 / 66.67，过程记忆需要低层运动信息。其顺序反转负对照可搬到 VideoPlaceOrder 这类要求顺序的任务上。
- **SAI**（2606.16490）是双移动机械臂的协同问题，RoboMME 把移动操作留作未来工作；SAI 靶向的过早释放属于时间记忆里的阶段判断，与 Counting 套件「何时按停止按钮」同类。三篇核心论文合起来覆盖时间（EventVLA 计数、SAI 阶段）、空间/物体（EventVLA 遮挡证据、TRACE 目标选择）与过程（EventVLA Reproduce Route）记忆，但都没有 RoboMME 数字。

关联阅读

- MemER（2510.20328）：本文最强的外部基线，关键帧图像 + 语言子目标的混合记忆。
- SAM2Act（2501.18564）：MemoryBench 的提出者，本文以其记忆库 + 离散路点动作为基线。
- ContextVLA（2510.04246）：历史视觉 token 直接拼入输入，本文 Perceptual+Context 一族的代表。
- Mem（2603.03596）：多尺度具身记忆，本文的符号记忆刻意去掉了其推理链以便标准化。
- MIKASA-Robo（2502.10550）：被本文批评为步长短、演示不足的 RL 取向记忆基准。

5. RMBench: Memory-Dependent Robotic Manipulation Benchmark with Insights into Policy Design

Tianxing Chen, Yuran Wang, Mingleyang Li, Yan Qin, Hao Shi, Zixuan Li, Yifan Hu, Yingsheng Zhang et al. — arXiv 2026, arXiv 2603.01229

> **arXiv 2603.01229** · arXiv 2026 · 提交 2026-03-01 · 分类 `bench` / 记忆依赖操作的基准
> *Tianxing Chen, Yuran Wang, Mingleyang Li, Yan Qin, Hao Shi, Zixuan Li, Yifan Hu, Yingsheng Zhang, Kaixuan Wang, Yue Chen, Hongcheng Wang, Junjie Wang, Tianhang Yang, Renjing Xu, Ruihai Wu, Yao Mu, Yaodong Yang, Hao Dong, Ping Luo*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

RMBench 用「任务记忆复杂度」M(n) 把 9 个 RoboTwin 2.0 双臂任务分成 5 个 M(1) 和 4 个 M(n)，并配套模块化记忆策略 Mem-0 做受控消融。要记的数字：Mem-0 平均 42.0%，对比 $\pi_{0.5}$ 10.4%、X-VLA 9.8%；消融里去掉锚帧记忆使 M(1) 平均从 52.8% 掉到 26.8%，去掉子任务关键记忆使 M(n) 从 28.5% 掉到 4.8%。

解决什么问题

现有策略默认马尔可夫假设，只看固定长度的近期窗；已有记忆基准要么可复现任务少（MemoryBench 7 个任务仅 3 个可稳定复现）、要么面向 RL（MIKASA）、要么信息全程可见（LIBERO-Long）。RMBench 给出任务侧的记忆需求刻度 TMC：最优策略所需保留的任务相关历史观测的最小数量 m，M(0) 无需记忆，M(1) 需一帧，M(n) 需多帧。另一目标是拆出哪些架构组件真正贡献记忆能力。

方法

任务。 M(1)：Observe and Pick Up、Rearrange Blocks、Put Back Block、Swap Blocks、Swap T，依赖一帧或固定少量历史帧；M(n)：Battery Try、Blocks Ranking Try、Cover Blocks、Press Button，需要试错、按外部反馈重复、按数字卡计数。基于 SAPIEN，另有 Isaac Lab-Arena 实现；每个动作-观测对附细粒度语言标注。论文未报告各任务的平均步长。

协议。 每任务 50 条合成演示训练，100 次 rollout；基线 DP、ACT（无预训练）、$\pi_{0.5}$、X-VLA（预训练），均不做子任务分解。Mem-0 在 M(1) 任务只用执行模块，在 M(n) 任务启用子任务分解。

Mem-0。 双系统：规划模块（Qwen3-VL-8B-Instruct，LoRA）以初始帧 o_0、任务指令和「关键记忆窗」（已完成子任务文本 + 各子任务结束帧）预测下一个子任务；执行模块把当前帧与子任务经 VLM 编码后均值池化，分别与锚帧记忆（子任务起始帧 latent）和滑动窗记忆（最近 K 帧 latent）做交叉注意力，送入 H=30 的 DiT；子任务结束分类器（MLP）连续 8 步判定结束才切换并清空两个缓冲。规划只在子任务边界触发。执行模块每任务从零训练，无预训练。

关键数字

设置（成功率 %）	DP	ACT	$\pi_{0.5}$	X-VLA	Mem-0
M(1) 平均	6.4	6.8	14.4	11.8	52.8
M(n) 平均	5.0	4.8	5.5	7.3	28.5
总平均	5.8	5.9	10.4	9.8	42.0
Press Button	0	0	0	0	0
Observe and Pick Up	1	1	9	9	4
真机平均（3 任务，各 40 次）	—	0.0	5.83	—	22.50

Mem-0 消融：M(1) 去锚帧 26.8、去滑动窗 40.4；M(n) 去关键记忆 4.8、去锚帧 26.8、去滑动窗 25.3、用仿真真值替代结束分类器 45.3。Swap T 去滑动窗反而从 14 升到 20；Cover Blocks 去锚帧 92、去滑动窗 84，都高于完整版 68。Press Button 用真值分类器也只有 14。

局限与注意

- 主结果全部在仿真；真机只有 3 个任务、Mem-0 22.5%，失败多为低层抓放精度而非规划。

- 「初始帧 + 近期窗」能否解题：Mem-0 的锚帧就是子任务起始帧、滑动窗就是近期帧，M(1) 任务不分解时锚帧即整局初始帧。消融说明这两项是 M(1) 成绩的主体（52.8 对比无锚帧 26.8），与 EventVLA 仅用锚帧在 RMBench 拿到 67.8% 方向一致（骨干与训练配置不同，数字不能直接对齐）。真正抵抗静态锚帧的是计数与试错类：Press Button 所有方法为 0，Blocks Ranking Try 最高 18。
- 协议不对称：Mem-0 在 M(n) 任务额外使用子任务标注和结束分类器，基线则端到端无分解；真值分类器带来 28.5→45.3 的提升说明结果里相当一部分来自分解本身。
- TMC 标注为人工给定，M(1) 定义里「固定少量历史帧」与「一帧」并未严格区分。
- 标注成本：逐帧语言标注、子任务边界与结束信号都来自合成管线；真机每任务需 100 条人工演示。

与核心论文的关系

- **EventVLA**（2606.20092）在 RMBench 上只部署锚帧版本（初始帧 + 短期窗）得 67.8%，并据此认为 RMBench 大多依赖持久空间布局、可被静态锚帧解决；RMBench 自己的锚帧消融支持这一判断。EventVLA 报告的两个低分任务 Observe and Pick Up、Press Button 也正是 Mem-0 的短板，说明 RMBench 的剩余难度在语义辨识与计数，而非瞬态证据。EventVLA 因此在同一平台另建 RoboTwin-MeM，用 n 参数化「中途出现又消失」的关键帧数量。RMBench 一作也在 EventVLA 作者列表中。
- **TRACE**（2606.14551）针对的延迟证据任务属于 M(1) 型，但它用机器人状态轨迹的路径签名寻址 latent 记忆而非存原图；RMBench 指出 Swap T 的目标朝向无法用语言可靠表达，正是符号化子任务记忆失效、需要 latent 记忆的场景。M(n) 的试错任务要求有序历史，TRACE 的顺序反转负对照可借来检验记忆是否真的用到了顺序。
- **SAI**（2606.16490）的历史 token 解决的是过早释放；RMBench 附录里 Swap Blocks 的典型失败同样是子任务过早或过晚终止，两者都指向「阶段判断」这一时间记忆问题。按 RoboMME 的分类，RMBench 主要覆盖空间记忆（M(1)）和时间/计数记忆（M(n)），不含物体指代与过程模仿。

关联阅读

- SAM2Act / MemoryBench（2501.18564）：RMBench 批评其 7 个任务仅 3 个可稳定复现。
- MIKASA-Robo（2502.10550）：32 个记忆任务但面向 RL 形式化。
- MemER（2510.20328）：逐帧推理子任务的双系统，RMBench 用结束分类器把规划降到边界触发。
- RoboMME（2603.04639）：同期基准，按时间/空间/物体/过程四类记忆设计 16 任务。
- RoboMemArena（2605.10921）：后续基准，认为 RMBench 任务覆盖偏少，把平均步长推到 1,076。

6. RoboCerebra: A Large-scale Benchmark for Long-horizon Robotic Manipulation Evaluation

Songhao Han, Boxiang Qiu, Yue Liao, Siyuan Huang, Chen Gao, Shuicheng Yan, Si Liu — NeurIPS 2025, arXiv 2506.06677

长时程操作基准，评测 System 2 的规划/反思/记忆，GPT 生成任务并由人执行子任务；偏高层推理而非瞬态视觉证据。

Abstract. Recent advances in vision-language models (VLMs) have enabled instruction-conditioned robotic systems with improved generalization. However, most existing work focuses on reactive System 1 policies, underutilizing VLMs' strengths in semantic reasoning and long-horizon planning. These System 2 capabilities-characterized by deliberative, goal-directed thinking-remain under explored due to the limited temporal scale and structural complexity of current benchmarks. To address this gap, we introduce RoboCerebra, a benchmark for

evaluating high-level reasoning in long-horizon robotic manipulation. RoboCerebra includes: (1) a large-scale simulation dataset with extended task horizons and diverse subtask sequences in household environments; (2) a hierarchical framework combining a high-level VLM planner with a low-level vision-language-action (VLA) controller; and (3) an evaluation protocol targeting planning, reflection, and memory through structured System 1-System 2 interaction. The dataset is constructed via a top-down pipeline, where GPT generates task instructions and decomposes them into subtask sequences. Human operators execute the subtasks in simulation, yielding high-quality trajectories with dynamic object variations. Compared to prior benchmarks, RoboCerebra features significantly longer action sequences and denser annotations. We further benchmark state-of-the-art VLMs as System 2 modules and analyze their performance across key cognitive dimensions, advancing the development of more capable and generalizable robotic planners.

7. Memory, Benchmark & Robots: A Benchmark for Solving Complex Tasks with Reinforcement Learning

Egor Cherepanov, Nikita Kachaev, Alexey K. Kovalev, Aleksandr I. Panov — arXiv 2025, arXiv 2502.10550

32 个记忆密集型桌面操作任务（RL 出身）+ 记忆任务分类框架；MemoryVLA、VPWEM、μVLA、ELMUR 均在其上报告。

Abstract. Memory is crucial for enabling agents to tackle complex tasks with temporal and spatial dependencies. While many reinforcement learning (RL) algorithms incorporate memory, the field lacks a universal benchmark to assess an agent's memory capabilities across diverse scenarios. This gap is particularly evident in tabletop robotic manipulation, where memory is essential for solving tasks with partial observability and ensuring robust performance, yet no standardized benchmarks exist. To address this, we introduce MIKASA (Memory-Intensive Skills Assessment Suite for Agents), a comprehensive benchmark for memory RL, with three key contributions: (1) we propose a comprehensive classification framework for memory-intensive RL tasks, (2) we collect MIKASA-Base – a unified benchmark that enables systematic evaluation of memory-enhanced agents across diverse scenarios, and (3) we develop MIKASA-Robo (pip install mikasa-robo-suite) – a novel benchmark of 32 carefully designed memory-intensive tasks that assess memory capabilities in tabletop robotic manipulation. Our work introduces a unified framework to advance memory RL research, enabling more robust systems for real-world use. MIKASA is available at https://tinyurl.com/membenchrobots.

事件与关键帧记忆（EventVLA 一族） / Event and Keyframe Memory

只在「发生了什么」的时刻写入，存原图或原图 token。2026 年中的共识方向：EventVLA 学未来关键帧概率，KEMO 用运动学 + 视觉规则，UniMem 训练事件分类器，WeaveLA 在子目标完成时触发，Keyframe-Chaining 用进度感知查询检索。

1. UniMem: Unifying Multimodal Memory and Control for Vision-Language-Action Models

Lars Osterberg, Maggie Wang, Mac Schwager — arXiv 2026, arXiv 2608.22869

> **arXiv 2608.22869** · arXiv 2026 · 提交 2026-08-24 · 分类 `event` / 事件与关键帧记忆（EventVLA 一族）
> *Lars Osterberg, Maggie Wang, Mac Schwager*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

UniMem 把「何时记」和「记什么」都放进 $\pi_{0.5}$ 单一骨干：最后一层 latent 上的 MLP 事件头一旦预测出非 null 事件，就把事件名追加进文本历史、把当前多视角原图存入关键帧集合（3 张历史 + 当前帧），再经加了时间注意力的 SigLIP 读回。仿真五任务平均 93.4%（固定 6 s 间隔采帧的 $\pi_{0.5}$+V.E. 为 68.2%），真机四任务 80.0%（MemER 43.5%）；靠缓存关键帧 hidden state，推理保持约 90 ms 的单帧速度。

解决什么问题

Perceptual aliasing：观测几乎相同、正确动作却取决于历史（拿起再放回、数第几勺）。作者批评两条路线：分层方案（MemER、MEM、Hi Robot）让额外的 VLM 管记忆、只给 VLA 发文本子任务，执行器拿不到过去画面里的空间细节，流水线割裂、延迟高；多帧输入若按固定间隔取帧则引入与动作无关的伪相关，反而拉低成绩。UniMem 要事件驱动、文本 + 视觉双模态、单模型端到端。

方法

存什么：文本记忆 M_t 是事件词表 E 中的离散事件名，以自然语言追加到指令上（"History: human tap, grabbed spoon, …"），起始为空。视觉记忆 H_t 是事件时刻的多视角原图 {I_wrist, I_ext}，起始只含初始场景帧；受训练算力限制容量为 3 个历史里程碑 + 当前帧（$|K| \le 4$，仿真用 3、真机用 4），超出丢最旧。

何时写：事件头 f_φ（MLP）读 backbone 最后一层本应生成文本 token 的 latent z_t，输出 E ∪ {null} 上的分布，argmax 非 null 即触发写入（Eq. 1），文本与视觉同时写。z_t 已注意过 M_t 与 H_t，分类器因此条件于全部历史，构成递归回路。记忆只增不删：漏检下一步还能补，误检则永久污染两种记忆，因此 null 类不屏蔽而是降权 w_null = 0.02。

如何读：改 $\pi_{0.5}$ 的 SigLIP，每隔几层在空间注意力之间插入因果时间自注意力，复用该层自身的 QKV 与 LayerNorm 权重（配置取自 MEM 与 TimeSformer）。缓存：历史关键帧在时间注意力之前的 hidden state 被缓存，新帧到来只平移位置嵌入、不重算空间注意力，当前帧独自作 query。

监督：$L = L_a + \lambda L_e$，$\lambda = 0.1$；L_a 是不变的 flow-matching 损失，L_e 是类别加权交叉熵，梯度回传到共享 LM trunk。标签由 Claude Sonnet 5.0 生成的脚本按本体感知动作签名（gripper 闭合、roll/pitch 模式）自动打，事件窗口取检测点前 5 帧到后 20 帧，人工抽查后改脚本；M_t 只在事件窗口结束后才更新以防泄漏。训练时从每个过去事件窗口均匀采一帧组成 H_t。全部模型 LoRA 微调；BeanScoop、TapScoopPour 上把决策时刻的采样概率上调约 4×。

关键数字

仿真 robosuite Franka Panda，每任务 N = 25；真机 xArm6，每任务 N = 15，人工判定子任务二值成功，real-time chunking 约 10 Hz。UpDown3Times 报子任务平均成功率，UpDownSpatial 报放回位置精度分。

仿真任务	$\pi_{0.5}$+V.E.（6 s 定间隔，同编码器）	No Memory	Text Only	Keyframe Only	UniMem
UpDown	84	52	96	92	100
UpDown3Times	16	22	93	23	96
OccludedTap	96	60	88	100	96
UpDownSpatial	49	6	30	52	79
PlateRecall	96	8	20	96	96
平均	68.2	29.6	65.4	72.6	93.4

真机任务	MemER	No Memory	Text Only	Keyframe Only	UniMem
HammerMeasure	87	13	53	53	87
BeanScoop	67	0	27	20	93
TableClean	13	0	0	47	80
TapScoopPour	7	7	7	27	60
平均	43.5	5.0	21.8	36.8	80.0

延迟（RTX 4090）：四路相机、16 个关键帧的上下文只比双相机单帧基座多约 25 ms；正文称比分层记忆基线快 6×。

局限与注意

论文自述：未在数十分钟到小时级任务上验证；没有记忆编辑（剪枝、整合）；关键帧标注依赖离线自动脚本和预定义事件词表；只有一条几分钟的时间上下文，不分短期长期。

我的读法： - 事件词表逐任务预定义、标签来自人工核对的规则脚本，TapScoopPour 的人类敲击还是采集时手动打标：这是子任务级监督，「统一」指架构而非去掉标注。 - 容量只有 3 张历史帧：Keyframe Only 在 UpDown3Times 只有 23%，部分是容量装不下三次循环，靠文本记忆补。 - 所有消融都带事件头辅助损失训练，No Memory 不是原版 $\pi_{0.5}$，辅助损失本身对骨干的影响没被拆出来。 - 真机 MemER 基线由作者自己 LoRA 微调低层 $\pi_{0.5}$（附录 B），高层如何训练、用多少演示未交代；N = 15，一次 trial 值 6.7 个点。 - 仿真基线 $\pi_{0.5}$+V.E. 用 UniMem 自己的编码器、只换取帧规则，是选帧规则的消融，不是与 MemoryVLA、EventVLA 类方法的比较；九个任务全是自建，没用 RMBench、MIKASA-Robo 等公开基准。 - 93.4 的平均混合了二值成功、子任务平均、空间精度三种度量；正文里 Keyframe Only 在 TableClean 的 53% 与表 II 的 47% 不一致，前者应是阶段成功率。 - 决策时刻 4× 过采样是与 KEMO 损失加权同类的训练技巧，是否也用于基线未说明。

与核心论文的关系

与 EventVLA 最近：都在 VLA 最后一层 hidden state 上接头决定关键帧，都存原图（EventVLA 5 槽 FIFO，UniMem 3 张历史 + 当前帧、丢最旧），都以初始帧作锚，都把帧送回 VLM 的视觉编码器。四点不同。预测目标：EventVLA 的 KEM 头对未来 50 步中每一步输出「将成为关键帧」的概率（前瞻式，阈值 0.55 + NMS + 冷却），标签是 Qwen3-VL-235B 的软标签；UniMem 的头对当前步做离散语义事件分类（回顾式，argmax），标签是 Claude 写的本体感知规则脚本。记忆模态：EventVLA 纯视觉；UniMem 把事件名写进指令，且发现计数任务离不开它（UpDown3Times 文本 93 vs 关键帧 23）。读出成本：EventVLA 把帧当额外视觉 token 拼接，吞吐 2.91→0.94 Hz；UniMem 改 SigLIP 加时间注意力并缓存 hidden state，保持约 90 ms，补的正是 EventVLA 付出的代价。短期窗：EventVLA 保留，UniMem 没有。

与 TRACE：TRACE 每步门控写入 4/6 个 512 维 latent 槽、以路径签名寻址、无需标签；UniMem 只在事件时写显式图像 + 文本、按时间顺序寻址、需要事件标签。与 SAI 的 30 步 GRU 历史 token 没有机制重叠，只有 TapScoopPour「等人敲杯再动」与 SAI 的伙伴条件时序问题相似。UniMem 未被任何核心论文用作基线（2026-08，晚于 EventVLA）；它和 EventVLA 用了同一个双系统基线 MemER，数字分别是真机 43.5% vs 80.0% 与 RoboTwin-MeM 10.5% vs 75.2%，协议不同。

关联阅读

- EventVLA（2606.20092）：设计最接近的对照，学习式未来关键帧头 vs 事件分类头。

- MemER（2510.20328）：被 UniMem 在真机上击败的分层基线（43.5 vs 80.0）。
- MEM（2603.03596）：VLM 文本摘要 + 固定间隔帧，UniMem 的时间注意力 SigLIP 配置取自它。
- KEMO（2606.23589）：规则式事件检测的同族方案，没有文本记忆。
- Past-Token Prediction（2505.09561）：长上下文带来伪相关的论证来源，是 UniMem 事件驱动选帧的动机。

2. KEMO: Event-Driven Keyframe Memory for Long-Horizon Robot Manipulation with VLA Policies

Yihan Zeng, Minghao Ye, Yiyuan Chen, Yide Shentu, Philipp Wu, Zike Yan, Zhongyu Li — arXiv 2026, arXiv 2606.23589

> **arXiv 2606.23589** · arXiv 2026 · 提交 2026-06-22 · 分类 `event` / 事件与关键帧记忆（EventVLA 一族）
> *Yihan Zeng, Minghao Ye, Yiyuan Chen, Yide Shentu, Philipp Wu, Zike Yan, Zhongyu Li*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

KEMO 是挂在 $\pi_{0.5}$ 上的插件式关键帧记忆：用「关节运动放缓 + DINOv2 视觉去重」两条规则在线检测事件，把事件帧压成 16 个 SigLIP token 存进容量 K 的时序记忆库，再经门控交叉注意力注回当前视觉 token。六个真机双臂任务（每任务 12 trials）上聚合 TSR 从 $\pi_{0.5}$ 的 27.8% 升到 51.4%（+23.6 点），SCR 从 42.3% 升到 76.4%（+34.1 点）。最值得记的是 Cover Blocks：KEMO 9/12，$\pi_{0.5}$ 与 MemoryVLA 都是 0/12。

解决什么问题

阶段歧义：长程任务中视觉相似的观测出现在不同阶段，正确动作取决于已完成的操作，Cover Blocks 里盖第二个杯子和揭第二个杯子的画面几乎一样。作者批评两类现有记忆：稠密记忆（MemoryVLA 的 token 合并、RoboMME 的递归状态、CronusVLA 的固定 latent）把稀疏的状态转移和大量冗余观测一起压缩，关键事件反而难以分辨；任务分解方法（MemER、Mem-0）要子任务级标注、在线多跑一个 VLM，高层摘要的错误向下传播。KEMO 只保留稀疏事件帧，不压缩、不分解、不加在线高层推理。

方法

存什么：事件帧的 SigLIP patch token（256 个）经 4×4 平均池化压到 16 个，加可学习的槽位时间嵌入 PE(i)。存的是 token 而非原图。

何时写：两级规则。(1) 事件显著性 $S_t = 1/(1+\delta_t)$，δ_t 为长度 w 的滑窗内关节位移均值，机器人减速时趋近 1；对 S_t 做因果峰值检测（在线需等一个短窗确认）得候选帧。(2) 候选帧与上一个已接受关键帧用冻结 DINOv2 全局嵌入算余弦距离 v_i，$v_i > \varepsilon = 0.05$ 才接受，否则丢弃并保留原参考帧；再加 r 帧冷却期。w、峰值窗、r 逐任务手调（Table 6）：w 取 10–40，峰值窗 20 或 60，r 为 8（Drawer 任务 60）。

容量与读出：记忆库保留最近 K 个关键帧，K 按任务的子任务转移数手动设定，不足时用最近关键帧填充并以 padding mask 屏蔽。读出发生在视觉编码阶段、进入语言模型之前：每个相机的当前帧 token X_t 作 query，K×16 个记忆 token 作 key/value 做 masked cross-attention 得 X'_t，再经残差门 $\hat{X}_t = X_t + \sigma(g([X_t; X'_t]))\cdot X'_t$ 融合。g 用负偏置初始化，训练初期门几乎关闭以保住基座行为；替换原图像 token 后 prefix 长度与 LM 结构不变。

监督：没有记忆专属损失。检测出的关键帧索引同时用于损失加权：距任一关键帧 ≤ $\delta = 3$ 步的样本，其 flow-matching 损失权重 $\lambda = 8.0$，其余为 1（Eq. 4），写进记忆的时刻与训练最看重的时刻是同一批索引。基座 $\pi_{0.5}$，30k

步，batch 32，2×A100，顶视 + 双腕三相机，joint-space，horizon 50。

关键数字

协议：YAM 双臂真机，每任务 100–200 条遥操作演示，12 次随机初始状态 trial。TSR = 全部阶段按序完成的 trial 数；SCR = 从头连续完成的阶段数均值，某阶段失败后即停计。

任务（阶段数）	$\pi_{0\cdot5}$ TSR / SCR	MemoryVLA TSR / SCR	KEMO TSR / SCR
Swap Foods (2)	6/12 / 1.500	1/12 / 0.750	8/12 / 1.580
Find Block (4)	0/12 / 0.353	0/12 / 1.000	2/12 / 3.167
Cover Blocks (6)	0/12 / 1.333	0/12 / 0.333	9/12 / 5.250
Box Refill (4)	4/12 / 2.580	0/12 / 0.000	7/12 / 3.330
Make Sandwich (4)	10/12 / 3.330	0/12 / 1.000	11/12 / 3.750
Drawer Items Replacement (4)	0/12 / 0.000	0/12 / 0.000	0/12 / 1.417
聚合 TSR	27.8%	1.4%	51.4%
聚合 SCR	42.3%	比 KEMO 低 60.9 点	76.4%

Cover Blocks 消融（Table 1，12 trials，6 阶段）：

变体	TSR	SCR
均匀采样（关闭损失加权）	0/12	1.833/6
最近 K 帧（关闭损失加权）	1/12	0.700/6
事件关键帧（关闭损失加权）= w/o Loss Weighting	3/12	2.500/6
w/o Gated Fusion（直接相加）	0/12	0.000/6
完整 KEMO	9/12	5.250/6

局限与注意

论文自述：检测器超参（w、峰值窗、r）需逐任务调；K 按预期阶段转移数手设；未来想学事件选择和动态容量。

我的读法： - 「task-agnostic」只是流程通用，参数不通用：Table 6 里 w、峰值窗、r 和 K 都随任务变，等于把任务阶段数以先验喂了进去。 - 损失加权贡献了一半以上的增益：Cover Blocks 上纯记忆机制 3/12，加权后 9/12；缺「$\pi_{0\cdot5}$ + 损失加权、无记忆」这一对照，记忆本身值多少没被隔离。 - MemoryVLA 基线不公平：Prismatic 骨干、仅单个顶视相机、40k 步 batch 128，对三相机 $\pi_{0\cdot5}$；聚合 TSR 1.4% 远低于无记忆 $\pi_{0\cdot5}$，比的主要是骨干与输入。 - 消融只做了一个任务、12 trials；Swap Foods 8 vs 6、Make Sandwich 11 vs 10 在噪声内。 - 最长的 Drawer Items Replacement（95 s，2846 步）所有方法 TSR 0/12。 - 没有延迟或吞吐数字，DINOv2 与在线峰值检测的开销未报告；去重用哪路相机未说明。 - 依赖「转移前减速」假设，推、擦等连续动作的阶段变化检测不到；无仿真、无公开基准。

与核心论文的关系

最近的是 EventVLA：都给 $\pi_{0\cdot5}$ 类 VLA 加稀疏事件帧、都在视觉侧注入、容量都很小（KEMO 按任务设 K，EventVLA 5 槽 FIFO）。差别有四处。何时写：EventVLA 用 KEM 头从最后一层 hidden state 预测未来 50 步中每步是

关键帧的概率（阈值 0.55 + NMS + 冷却），标签来自 Qwen3-VL-235B 离线标注；KEMO 用手写的减速 + DINOv2 差异规则，不学、不用 VLM 标注，代价是逐任务调参。存什么：EventVLA 存原图并拼进视觉编码器输入，每帧占满 token；KEMO 每帧 16 个池化 token 走 cross-attention，prefix 不变长，理论上便宜得多但没测（EventVLA 吞吐 2.91→0.94 Hz）。锚帧：EventVLA 有初始帧 + 短期窗，KEMO 没有锚帧，只有事件帧；它的「最近 K 帧」基线 1/12 说明短期窗单独不够。训练信号：KEMO 的关键帧对齐损失加权（λ = 8）是三篇核心论文都没有的部件。

与 TRACE：TRACE 每步门控写入 K = 4/6 个 512 维 latent 槽，用机器人状态轨迹的路径签名寻址；KEMO 也用机器人状态，但只当写入触发器，寻址靠槽位时间顺序，槽内容是半显式的视觉 token 而非 latent。SAI 的 30 步 GRU 历史 token 正是 KEMO 消融里输掉的「最近帧」式记忆。KEMO 未被任何核心论文用作基线（与 EventVLA 同月）；它与 EventVLA 都把 MemoryVLA 当记忆基线，各自复现的数字是聚合 TSR 1.4% 与 RMBench 41.7%（QwenOFT 骨干），协议不同，不能横比。

关联阅读

- EventVLA（2606.20092）：同月的学习式关键帧头，是 KEMO 规则检测器最直接的对照。
- MemoryVLA（2508.19236）：KEMO 的记忆基线，聚合 TSR 1.4%。
- BPP（2602.15010）：用现成 VLM 标关键帧再喂扩散策略，KEMO 反对的「VLM 在线检测」路线。
- FrameSkip（2605.13757）：由动作变化与 gripper 切换算连续重要性分数，KEMO 运动学线索的前身。
- Keyframe-Chaining VLA（2603.01465）：学出来的关键帧选择模块 + FiLM 调制，介于规则与 VLM 之间。

3. WeaveLA: Event Driven Cross-Subtask Latent Memory Weaving for Repetitive Robot Manipulation

Shoujing Zhu, Zhenyang Liu, Fungmiu Wang, Jiafeng Wang, Bo Yue, Guiliang Liu, Simo Wu, Xiangyang Xue et al. — arXiv 2026, arXiv 2606.17463

以「子目标完成事件」为跨子任务记忆交接的时间单位：把完成段压成 latent token 直接路由进下一子任务的动作生成；RoboMME 最难重复切片 0→47.8%，单次执行任务不变。

Abstract. Vision-Language-Action (VLA) policies have achieved remarkable single-step manipulation, yet they remain brittle precisely where each stage depends on what was just completed. The core issue is structural: short-window VLAs lack an explicit channel for rouxting information across sub-task boundaries, and existing memory-augmented variants either write at every frame, retrieve from demonstration-time stages, or fire at sub-goal events without performing an explicit sub-task-to-sub-task hand-off into the action expert. We identify the sub-goal completion event as the natural temporal unit for cross-subtask memory hand-off, and present WeaveLA (Weave Latent memory for Vision-Language-Action policies), a cross-subtask memory interface that, on top of a frozen VLA backbone, compresses each completed segment into latent tokens via query-driven attention pooling and routes them directly into the action-generation path of the next sub-task. This event-triggered, action-side design preserves the base policy's short-window interface while adding a lightweight cross-subtask channel. Through stratified evaluation on RoboMME with a MATH3 backbone, WeaveLA's gains land exactly where the channel is needed: on the hardest repetition slice (SwingXtimes, MATH4), success rises from MATH5 to MATH6, while single-execution episodes remain unchanged. Per-episode paired analysis confirms the gains are confined to tasks whose causal structure requires cross-subtask information.

4. Bi-HIL: Bilateral Control-Based Multimodal Hierarchical Imitation Learning via Subtask-Level Progress Rate and Keyframe Memory for Long-Horizon Contact-Rich Robotic Manipulation

Thanpimon Buamanee, Masato Kobayashi, Yuki Uranishi — arXiv 2026, arXiv 2603.13315

双边控制 + 分层模仿：关键帧记忆与子任务级进度率同时条件高低层策略，面向接触丰富的长时程任务。

Abstract. Long-horizon contact-rich robotic manipulation remains challenging due to partial observability and unstable subtask transitions under contact uncertainty. While hierarchical architectures improve temporal reasoning and bilateral imitation learning enables force-aware control, existing approaches often rely on flat policies that struggle with long-horizon coordination. We propose Bi-HIL, a bilateral control-based multimodal hierarchical imitation learning framework for long-horizon manipulation. Bi-HIL stabilizes hierarchical coordination by integrating keyframe memory with subtask-level progress rate that models phase progression within the active subtask and conditions both high- and low-level policies. We evaluate Bi-HIL on unimanual and bimanual real-robot tasks, demonstrating consistent improvements over flat and ablated variants. The results highlight the importance of explicitly modeling subtask progression together with force-aware control for robust long-horizon manipulation. For additional material, please check: https://mertcookimg.github.io/bi-hil

5. Non-Markovian Long-Horizon Robot Manipulation via Keyframe Chaining

Yipeng Chen, Wentao Tan, Lei Zhu, Fengling Li, Jingjing Li, Guoli Yang, Heng Tao Shen — arXiv 2026, arXiv 2603.01465

学习判别式嵌入空间自动选关键帧，用进度感知查询按当前阶段检索历史帧并作为交错视觉 token 输入；ManiSkill 上四个非马尔可夫任务。

Abstract. Existing Vision-Language-Action (VLA) models often struggle to generalize to long-horizon tasks due to their heavy reliance on immediate observations. While recent studies incorporate retrieval mechanisms or extend context windows to handle procedural tasks, they often struggle to capture Non-Markovian dependencies, where optimal actions rely solely on specific past states rather than the current observation. To address this, we introduce Keyframe-Chaining VLA, a framework that extracts and links key historical frames to model long-horizon dependencies. Specifically, we propose an automatic keyframe selector that learns a discriminative embedding space, effectively identifying distinct state transitions. To capture task-critical information, we design a progress-aware query mechanism that dynamically retrieves historical frames based on their temporal relevance to the current execution phase. These selected keyframes are integrated into the VLA as interleaved visual tokens, explicitly grounding the policy in the long-horizon temporal context. Finally, we introduce a suite of four Non-Markovian manipulation tasks built upon the ManiSkill simulator to measure task success rates. Experimental results demonstrate that our method achieves superior performance, effectively tackling robot manipulation tasks characterized by long-horizon temporal dependencies. Code is available at https://github.com/cytoplastm/KC-VLA.

稠密与压缩的视觉历史 / Dense and Compressed Visual History

每帧都进，靠压缩、token 化、KV 复用或采样控制成本。这一族的近期趋势是「记忆几乎免费」：NativeMEM 每帧 1 token、TempoFit 免训练复用 K/V、StreamPI 零新增参数、DySta 复用静态 token 的 KV 缓存。EventVLA 的三倍延迟是这一族要解决的问题。

1. StreamPI: Streaming Multimodal Temporal Modeling for Vision-Language-Action Models

Zhe Liu, Jinghua Hou, Yuxiang Lu, Zhenya Yang, Xianzhe Fan, Junwei Luo, Junyi Li, Ruihua Han et al. — arXiv 2026, arXiv 2608.26067

零新增参数的流式多帧建模：以（观测，指令）对为原子时间单元，对内双向、对间因果注意力，随机间隔流式训练；在 $\pi_{0.5}$ 上超越单帧。

Abstract. Vision-Language-Action (VLA) models have demonstrated effectiveness in robot manipulation, yet state-of-the-art models such as pi0.5 operate under a single-frame paradigm, limiting their ability to retain past observations and develop precise spatial perception. In this paper, we propose StreamPI, a streaming multimodal temporal modeling framework that equips single-frame VLA with temporal reasoning capability without introducing any additional parameters. One core design is instruction-anchored temporal modeling. It treats each (visual observation, language instruction) pair as an atomic temporal unit: bidirectional attention within each pair enables cross-modal fusion, while causal attention across pairs preserves autoregressive streaming inference. This ensures the language instruction serves as a persistent semantic anchor throughout task execution. To bridge the gap between synchronous training and asynchronous real-robot deployment, we introduce a andom-interval streaming training strategy: a proper inter-frame interval (e.g., every 3 frames) enables faster and smoother action execution. Beyond this, randomizing the interval further improves robustness to frame-timing perturbations, supporting asynchronous deployment in practice. Furthermore, by leveraging the length extrapolation capability of the LLM backbone, StreamPI seamlessly inherits pretrained single-frame weights and supports flexible single-frame and multi-frame inference. Experiments on real-robot tasks spanning memory-dependent and precise perception scenarios, as well as the simulation benchmark LIBERO, demonstrate that StreamPI outperforms pi0.5 across diverse tasks.

2. Remember Smarter: Visual History Compressor and Hyperbolic Experience Space for Robotic Memory

Dai Zhou, Jiexi Yan, Tong Li, Yuxuan Wang, Cheng Deng — arXiv 2026, arXiv 2608.15269

视觉历史压缩（空间双向 Mamba + 时间因果 Mamba）+ 双曲空间经验记忆（Poincaré VAE），异步转成测地线提示 token；LIBERO-Plus 53.6→70.6。

Abstract. Long-horizon robot policies require compact access to recent observations and reusable experience without expanding the vision-language-action (VLA) context. We introduce Remember Smarter (RS), a plug-and-play module with complementary visual-history and hyperbolic experience-memory branches. Its visual branch compresses multi-view patch histories using bidirectional spatial Mamba and causal temporal Mamba, then exposes the resulting memory to action-facing hidden states through residual cross-attention while leaving the VLM visual-token stream unchanged. Its experience branch stores successful final-layer VLM states in a Poincare VAE space, organizes them hierarchically, and asynchronously converts retrieved experience into geodesic prompt tokens without blocking action inference. When adapted to pi0, RS increases total success on LIBERO-Plus from 53.6% to 70.6% and achieves substantial performance gains in real-robot experiments designed to evaluate memory retention and experience utilization.

3. AtlasVLA: Persistent World-Ego State Modeling for Vision-Language-Action Models

Guiyu Zhao, Longteng Guo, Yanghong Mei, Zilin Zhu, Yu Zhang, Bin Cao, Mingming Yu, Xingjian He et al. — arXiv 2026, arXiv 2608.06729

持久世界-自我状态：4D 体素哈希空间记忆解决单腕相机的视野盲区 + 自我工作状态追踪任务进度；仅腕相机即超多视角基线（LIBERO-Long +9.4，真机 +17.5）。

Abstract. While Vision-Language-Action (VLA) models have advanced embodied AI, their fundamentally reactive paradigm severely limits performance in partially observable and long-horizon tasks. When restricted to a single wrist-mounted camera, they inevitably suffer from perception forgetting as objects exit the field of view, and temporal task-progress forgetting} during multi-step execution. To overcome these bottlenecks, we propose AtlasVLA, a novel framework that transitions from direct reactive manipulation to proactive reasoning through a persistent world-ego state. AtlasVLA features a dual-memory architecture: a 4D Persistent World State Memory that lifts transient 2D observations into a globally updated, voxel-hashed spatial state to resolve visual blind spots, and an Ego-Working State Memory that tracks historical ego state and task progress. By conditioning a diffusion transformer (DiT) on this joint World-Ego state, AtlasVLA enables robust spatial reasoning. Extensive evaluations across LIBERO, RLBench, and real-world benchmarks demonstrate that AtlasVLA achieves state-of-the-art performance using solely a wrist camera. Remarkably, it decisively outperforms multi-view baselines, yielding absolute success rate improvements of 9.4% on LIBERO-Long and 17.5% in real-world long-horizon tasks.

4. BridgeVLA++: A Data-Efficient, Generalizable, and Memory-Augmented Vision-Language-Action Framework for 3D Manipulation

Peiyan Li, Yuze Zhu, Yixiang Chen, Qisen Ma, Yuan Xu, Jiabing Yang, He Guan, Yan Huang et al. — arXiv 2026, arXiv 2608.05042

3D VLA（点云→多视角热图）加统一时空记忆，在两个记忆依赖基准上取得 SOTA 且不损数据效率。

Abstract. Leveraging pre-trained vision-language models (VLMs) to construct vision-language-action (VLA) models has emerged as a promising paradigm for 3D robot manipulation. However, existing 3D VLA methods remain data-hungry, exhibit limited generalization under distribution shifts, and lack explicit memory of past observations. These limitations hinder their application to data-scarce, open-world, and memory-dependent manipulation scenarios. Our previous work, BridgeVLA, improves data efficiency and generalization by preserving the input–output alignment of a pre-trained VLM during 3D action learning: raw point clouds are projected into multi-view images, and intermediate heatmaps are predicted before generating robot actions. In this work, we develop BridgeVLA++ by equipping BridgeVLA with a unified spatio-temporal memory architecture that models persistent spatial context and temporal interaction history. The resulting memory-augmented framework can reason over observation histories while preserving BridgeVLA's data efficiency and generalization capabilities. Extensive experiments show that our framework achieves strong performance on spatial manipulation tasks while exhibiting robust generalization. BridgeVLA++ further achieves state-of-the-art performance on two challenging memory-dependent manipulation benchmarks without sacrificing the data efficiency and generalization of the original BridgeVLA. In addition, BridgeVLA++ performs effectively in bimanual manipulation settings and is validated on an additional real-world robotic platform, demonstrating its scalability across tasks, environments, and robotic platforms. These results establish BridgeVLA++ as a unified 3D vision-language-action framework that simultaneously supports data-efficient learning, robust generalization, and effective memory-aware robot manipulation. Project website: https://bridgevla-plus.github.io/.

5. FibVLA: An Efficient Temporal Vision-Language-Action Model with Fibonacci Sampling

Li Lin, Wujun Xu, Weiwei Meng, Kaiwen Xia, Kang Hao Cheong, Shuai Wang — arXiv 2026, arXiv 2607.29596

对本体状态与视觉帧做对数回溯采样 + 斐波那契递归推理，以低冗余捕获长时依赖。

Abstract. Vision-language-action models (VLAs), which leverage the cognition of multimodal information to infer physical-world actions, provide a generalized solution for embodied AI applications. Conventional VLAs usually concentrate on current digital cognition. While some efforts are made to enhance VLAs' reasoning capabilities by capturing temporal information, encoding the long-context history causes an efficiency-decreasing issue. To reconcile the conflict between capturing temporal information and maintaining inference efficiency in VLAs, this paper introduces FibVLA, an efficient framework featuring temporal perception of long-context history. Specifically, we leverage logarithmic hindsight sampling to both proprioceptive states and visual frames to capture long-term temporal dependencies with minimal redundancy. For the action expert, we introduce the flow matching to produce action distributions, and the Fibonacci recurrent inference strategy to generate long-range planning steps based on real-time closed-loop feedback. Experiments demonstrate that FibVLA significantly improves action smoothness and success rates without retraining large-scale visual encoders. Efficiency analysis demonstrates superior real-time responsiveness compared to video-based baselines in real-world evaluations.

6. Dual Latent Memory in Vision-Language-Action Models for Robotic Manipulation

Hongyu Qu, Jianzhe Gao, Xiaobin Hu, Shaohuan Yang, Xinlei Yu, Rui Yan, Wenguan Wang, Xiangbo Shu et al. — arXiv 2026, arXiv 2607.07608

LaMem-VLA：把历史重建成短期/长期 latent 记忆 token 并直接编织进 VLA 的连续嵌入序列（curator/seeker/condenser/weaver）。

Abstract. Mainstream Vision-Language-Action (VLA) models predict actions primarily from the current observation under a Markovian assumption, thus struggling with long-horizon, temporally dependent tasks. Existing memory-augmented VLAs either expand the observation window or retrieve history from the memory bank as auxiliary policy-side context. However, they leave memory outside the native latent embedding space of VLA reasoning, preventing historical experience from being fluidly interleaved with multimodal reasoning and action formation. To this end, we introduce LaMem-VLA, a latent-memory-native framework that reconstructs historical experience into latent memory tokens and directly interweaves them with VLA reasoning. At its core, LaMem-VLA introduces four coordinated components: (i) a curator that organizes historical experience into two complementary short-term and long-term memory vaults; (ii) a seeker that queries both vaults using the multimodal cognition to retrieve context-relevant evidence; (iii) a condenser that reconstructs the retrieved evidence into compact short-term and long-term latent memory tokens; and (iv) a weaver that injects these memory tokens with the current observation and instruction into one continuous embedding sequence. By representing, retrieving, and consuming historical experience entirely in the same continuous latent space, LaMem-VLA enables memory to directly participate in VLA reasoning and guide action generation under a bounded context. Extensive experiments on SimplerEnv and LIBERO demonstrate the superiority of our LaMem-VLA.

7. NativeMEM: Native Memory Compression for Long-Horizon Robotic Manipulation

Ziye Wang, Modi Shi, Chaojun Ni, Jiazhi Yang, Mengdi Li, Zhizhong Su, Tianwei Lin, Hongyang Li — arXiv 2026, arXiv 2607.06678

用 VLA 自己的视觉编码器把每帧每视角压成 1 个记忆 token，追加到输入序列即可长时记忆；仿真 32.4→84.0，真机最高 98.7，20% 数据即可竞争。

Abstract. How can pretrained Vision-Language-Action (VLA) models retain long-horizon visual histories with high-frequency updates without sacrificing efficiency? Existing approaches rely on external memory management, which restrains either the memory horizon or the reactiveness of pretrained policies. To this end, we present NativeMEM, a VLA policy that features long-term and real-time updated memory. At its core is an efficient memory encoding scheme, Native Memory Compression, which repurposes the VLA's own vision encoder to compress each historical frame from each camera view into a single token. Appended to the input sequence, these memory tokens enable the pretrained VLA to attend over long-term history with negligible latency overhead, requiring neither an external planner nor a freshly initialized memory module. To align the memory tokens with the pretrained policy, we first develop a generic memory tokenizer under the supervision of a frozen VLA on memory-demanding data, and then unfreeze the VLA for task-specific fine-tuning. NativeMEM consistently outperforms prior methods, boosting success rates from 32.4% to 84.0% in simulation and up to 98.7% on real robots, while maintaining low inference latency and GPU memory usage. Notably, NativeMEM exhibits high data efficiency by achieving competitive results with prior arts using only 20% of the training data.

8. HiMe: Hierarchical Embodied Memory for Long-Horizon Vision-Language-Action Control

Li Ji, Siyin Wang, Pengfang Qian, Xiaopeng Yu, Yihai Tian, Zhaoye Fei, Jingjing Gong, Xipeng Qiu — arXiv 2026, arXiv 2607.03449

分层具身记忆：高频 Executor + 工作记忆 Sentry + 长期 Planner，跨模态语义 schema 支持增/改/删，能按人类偏好自我修正知识。

Abstract. Current Vision-Language-Action (VLA) models excel at robotic manipulation but often struggle with non-Markovian tasks requiring long-term memory and reasoning due to their reliance on immediate observations. Existing solutions face a ''frequency-competence paradox,'' where stronger reasoning models are too slow for real-time control, while faster models lack sufficient reasoning capabilities. To resolve this architectural misalignment, we propose HiMe, a Hierarchical Embodied Memory framework that decouples embodied intelligence into a high-frequency Executor for execution, a Sentry for working memory, and a Planner for long-term strategy. We also introduce a dynamic knowledge system based on cross-modal semantic schemas and active management mechanisms, allowing robots to maintain memory plasticity through ''Add, Update, and Delete'' operations. This hierarchical design effectively balances the conflict between real-time execution and slow thinking planning, significantly improving success rates in long-horizon tasks. Experiments demonstrate that this approach not only outperforms flat memory baselines but also exhibits the novel ability to self-correct its internal knowledge based on human preferences.

9. Memory Retrieval in Visuomotor Policies for Long-Horizon Robot Control

Rutav Shah, Yisu Li, Femi Bello, Yuke Zhu, Roberto Martín-Martín — arXiv 2026, arXiv 2606.25136

HALO：注意力式记忆检索长达八分钟，用 VLM 生成的记忆依赖问答蒸馏先验抑制虚假相关，稀疏注意力限制检索范围以减少误差累积。

Abstract. General-purpose robots operating in partially observable environments, such as homes, require memory to support autonomy. They must recall diverse information from the past, such as where objects were placed, which tasks a human partner has completed, and when an appliance was turned on. Achieving this versatility requires a general memory retrieval mechanism. Transformer architectures that use attention over long contexts for

memory retrieval provide a promising approach, as they learn retrieval from data rather than relying on task-specific or hand-designed rules. However, directly incorporating them into imitation learning from offline data introduces two key challenges: (1) the policy may learn spurious correlations between past information and predicted actions, and (2) errors accumulate in memory due to prediction inaccuracies and their compounding interactions with the environment, causing model drift and cascading failures. To address both challenges, we introduce HALO, a visuomotor policy with an attention-based memory retrieval mechanism for long-horizon control. First, to suppress spurious correlations, HALO distills vision-language model (VLM) priors into the policy. It generates memory-dependent question–answer pairs from demonstration trajectories and trains jointly with a video question–answering objective, steering retrieval toward task-relevant information. Second, to reduce the impact of accumulated errors in memory during closed-loop control, HALO uses sparse attention that restricts retrieval to only the most relevant parts of the history. Together, these components enable more reliable long-horizon control by guiding the policy to retrieve task-relevant information from up to eight minutes of past experience. Project website: https://robin-lab.cs.utexas.edu/HALO

10. Scaling Short-Term Memory of Visuomotor Policies for Long-Horizon Tasks

Rutav Shah, Rajat Kumar Jenamani, Xiaohan Zhang, Lingfeng Sun, Roberto Martín-Martín, Yuke Zhu, Deva Ramanan, Karl Schmeckpeper — arXiv 2026, arXiv 2606.16178

PRISM：门控注意力过滤检索信息 + 分层压缩，把视觉运动策略的短期记忆扩展到两分钟；附 ReMemBench（8 任务、4 类短期记忆）。

Abstract. Many robotic tasks require short-term memory, whether it's retrieving an object that's no longer visible or turning off an appliance after a set period. Yet, most visuomotor policies trained via imitation learning rely only on immediate sensory input without using past experiences to guide decisions. We present PRISM, a transformer-based architecture for visuomotor policies to effectively use short-term memory via two key components: (i) gated attention, which filters retrieved information to suppress irrelevant details, improving performance by reducing the spurious correlations between the history and current action prediction, (ii) a hierarchical architecture that first compresses local information into compact tokens and then integrates them to capture temporally extended dependencies, improving its compute and memory footprint. Together, these mechanisms enable us to scale short-term memory in visuomotor policies for up to two minutes. To systematically evaluate memory in visuomotor control, we introduce ReMemBench – a benchmark of eight diverse household manipulation tasks spanning four categories of short-term memory – designed to foster general memory mechanisms rather than siloed, task-specific solutions. PRISM consistently outperforms prior works, including recurrent architectures, transformers, and their variants – achieving an absolute improvement of 5%–12% over the strongest baseline. On the RoboCasa and LIBERO benchmarks, it achieves absolute improvements of 11%–15% over its no-memory variant and fine-tuned Vision-Language-Action baselines such as GR00T-N1-3B and OpenVLA, despite not leveraging any large-scale pretraining. Together, PRISM and ReMemBench establish a foundation for developing and evaluating short-term memory-augmented visuomotor policies that scale to long-horizon tasks. Additional materials are available at https://shahrutav.github.io/short-term-memory

11. MemoryVLA++: Temporal Modeling via Memory and Imagination in Vision-Language-Action Models

Hao Shi, Weiye Li, Bin Xie, Yulin Wang, Renping Zhou, Tiancai Wang, Xiangyu Zhang, Ping Luo et al. — arXiv 2026, arXiv 2606.09827

MemoryVLA 加上潜空间世界模型「想象」未来，记忆与想象共同形成时间感知 token；真机记忆依赖任务 +26%、想象依赖任务 +28%。

Abstract. Temporal modeling is essential for robotic manipulation, as effective control requires both memory of past interactions and imagination of future states. However, most VLA models rely primarily on the current observation and therefore struggle with long-horizon, temporally dependent tasks. Cognitive science suggests that humans rely on working memory to buffer short-lived context, the hippocampal system to preserve episodic memory of past experience, and internal models to imagine possible future state evolution. Inspired by these mechanisms, we propose MemoryVLA++, a full temporal modeling framework that equips VLA models with memory and imagination for robotic manipulation. A pretrained VLM encodes the current observation into perceptual and cognitive tokens, forming working memory. These tokens query a Perceptual-Cognitive Memory Bank to retrieve relevant historical context. This bank stores low-level details and high-level semantics from past interactions, and is updated through redundancy-aware consolidation. A world model imagines future states in a denoising latent space, and the imagined latents are integrated under memory guidance to form full temporal-aware tokens. The resulting tokens condition a diffusion action expert to predict temporally consistent action sequences. We conduct extensive experiments on 5 simulation benchmarks and 3 categories of real-robot tasks across 3 robots, covering general manipulation, long-horizon temporal tasks, robustness, and generalization. Our method achieves strong performance across Libero, SimplerEnv, Mikasa-Robo, Calvin, Libero-Plus, and diverse real-robot tasks, validating the effectiveness of full temporal modeling with memory and imagination. For example, on real robots, it achieves +9%, +26%, +28% gains on general, memory-dependent, and imagination-dependent tasks. Project Page: https://shihao1895.github.io/MemoryVLA-PP-Web

12. ST-VLA: Enabling 4D-Aware Spatiotemporal Understanding for General Robot Manipulation

You Wu, Zixuan Chen, Cunxu Ou, Wenxuan Wang, Wenbo Huang, Lin Cao, Yangtao Chen, Weichao Qiu et al. — arXiv 2026, arXiv 2603.13788

分层 VLA 用统一 3D-4D 表征（3D 轨迹 + 4D 时空掩码）桥接推理与控制；偏时空感知而非显式记忆。

Abstract. Robotic manipulation in open-world environments requires reasoning across semantics, geometry, and long-horizon action dynamics. Existing hierarchical Vision-Language-Action (VLA) frameworks typically use 2D representations to connect high-level reasoning with low-level control, but lack depth awareness and temporal consistency, limiting robustness in complex 3D scenes. We propose ST-VLA, a hierarchical VLA framework using a unified 3D-4D representation to bridge perception and action. ST-VLA converts 2D guidance into 3D trajectories and generates smooth spatial masks that capture 4D spatio-temporal context, providing a stable interface between semantic reasoning and continuous control. To enable effective learning of such representations, we introduce ST-Human, a large-scale human manipulation dataset with 14 tasks and 300k episodes, annotated with 2D, 3D, and 4D supervision via a semi-automated pipeline. Using ST-Human, we train ST-VLM, a spatio-temporal vision-language model that generates spatially grounded and temporally coherent 3D representations to guide policy execution. The smooth spatial masks focus on task-relevant geometry and stabilize latent representations, enabling online replanning and long-horizon reasoning. Experiments on RLBench and real-world manipulation tasks show that \method significantly outperforms state-of-the-art baselines, improving zero-shot success rates by 44.6% and 30.3%. These results demonstrate that offloading spatio-temporal reasoning to VLMs with unified 3D-4D representations substantially improves robustness and generalization for open-world robotic manipulation. Project website: https://oucx117.github.io/ST-VLA/.

13. AnchorVLA4D: an Anchor-Based Spatial-Temporal Vision-Language-Action Model for Robotic Manipulation

Juan Zhu, Zhanying Shao, Xiaoqi Li, Ethan Morgan, Jiadong Xu, Hongwei Fan, Hao Dong — arXiv 2026, arXiv 2603.12730

最简的空间-时间锚：把初始帧作为锚图像与当前帧一起输入，加轻量空间编码器；对应 EventVLA 的「初始帧锚」组件。

Abstract. Since current Vision-Language-Action (VLA) systems suffer from limited spatial perception and the absence of memory throughout manipulation, we investigate visual anchors as a means to enhance spatial and temporal reasoning within VLA policies for robotic manipulation. Conventional VLAs generate actions by conditioning on a single current frame together with a language instruction. However, since the frame is encoded as a 2D image, it does not contain detailed spatial information, and the VLA similarly lacks any means to incorporate past context. As a result, it frequently forgets objects under occlusion and becomes spatially disoriented during the manipulation process. Thus, we propose AnchorVLA4D, a simple spatial-temporal VLA that augments the visual input with an anchor image to preserve the initial scene context throughout execution, and adds a lightweight spatial encoder that jointly processes the anchor and current frames to expose geometric relationships within an episode. Built on a Qwen2.5-VL backbone with a diffusion-based action head, AnchorVLA4D requires no additional sensing modalities (e.g., depth or point clouds) and introduces negligible inference overhead. Combining anchoring with a frozen pretrained spatial encoder yields further gains, realizing a 13.6% improvement on the Simpler WidowX benchmark and confirming the approach on real-world tasks, where it achieved an average success rate of 80%.

14. TempoFit: Plug-and-Play Layer-Wise Temporal KV Memory for Long-Horizon Vision-Language-Action Manipulation

Jun Sun, Boyu Yang, Jiahao Zhang, Ning Ma, Chencheng Wu, Siqing Zhang, Yiou Huang, Qiufeng Wang et al. — arXiv 2026, arXiv 2603.07647

免训练的时间改装：复用中间层前缀 K/V 做逐层 FIFO 记忆，K-to-K 检索 + 帧间隙时间偏置，不加 token、不加参数；LIBERO-LONG +4.0。

Abstract. Pretrained Vision-Language-Action (VLA) policies have achieved strong single-step manipulation, but their inference remains largely memoryless, which is brittle in non-Markovian long-horizon settings with occlusion, state aliasing, and subtle post-action changes. Prior approaches inject history either by stacking frames, which scales visual tokens and latency while adding near-duplicate pixels, or by learning additional temporal interfaces that require (re-)training and may break the original single-frame inference graph. We present TempoFit, a training-free temporal retrofit that upgrades frozen VLAs through state-level memory. Our key insight is that prefix attention K/V already form a model-native, content-addressable runtime state; reusing them across timesteps introduces history without new tokens or trainable modules. TempoFit stores layer-wise FIFO prefix K/V at selected intermediate layers, performs parameter-free K-to-K retrieval with Frame-Gap Temporal Bias (FGTB), a fixed recency bias inspired by positional biases in NLP, to keep decisions present-dominant, and injects the retrieved context via pre-attention residual loading with norm-preserving rescaling to avoid distribution shift under frozen weights. On LIBERO-LONG, TempoFit improves strong pretrained backbones by up to +4.0% average success rate while maintaining near-real-time latency, and it transfers consistently to CALVIN and real-robot long-horizon tasks.

15. Global Prior Meets Local Consistency: Dual-Memory Augmented Vision-Language-Action Model for Efficient Robotic Manipulation

Zaijing Li, Bing Hu, Rui Shao, Gongwei Chen, Dongmei Jiang, Pengwei Xie, Jianye Hao, Liqiang Nie — arXiv 2026, arXiv 2602.20200

OptimusVLA：全局先验记忆（用相似轨迹的任务先验替代高斯噪声起点）+ 局部一致性记忆（建模已执行动作推断进度）；LIBERO 98.6%，推理 2.9 倍加速。

Abstract. Hierarchical Vision-Language-Action (VLA) models have rapidly become a dominant paradigm for robotic manipulation. It typically comprising a Vision-Language backbone for perception and understanding, together with a generative policy for action generation. However, its performance is increasingly bottlenecked by the action generation proceess. (i) Low inference efficiency. A pronounced distributional gap between isotropic noise priors and target action distributions, which increases denoising steps and the incidence of infeasible samples. (ii) Poor robustness. Existing policies condition solely on the current observation, neglecting the constraint of history sequence and thus lacking awareness of task progress and temporal consistency. To address these issues, we introduce OptimusVLA, a dual-memory VLA framework with Global Prior Memory (GPM) and Local Consistency Memory (LCM). GPM replaces Gaussian noise with task-level priors retrieved from semantically similar trajectories, thereby shortening the generative path and reducing the umber of function evaluations (NFE). LCM dynamically models executed action sequence to infer task progress and injects a learned consistency constraint that enforces temporal coherence and smoothness of trajectory. Across three simulation benchmarks, OptimusVLA consistently outperforms strong baselines: it achieves 98.6% average success rate on LIBERO, improves over pi_0 by 13.5% on CALVIN, and attains 38% average success rate on RoboTwin 2.0 Hard. In Real-World evaluation, OptimusVLA ranks best on Generalization and Long-horizon suites, surpassing pi_0 by 42.9% and 52.4%, respectively, while delivering 2.9x inference speedup.

16. Efficient Long-Horizon Vision-Language-Action Models via Static-Dynamic Disentanglement

Weikang Qiu, Huashuo Lei, Tinglin Huang, Rex Ying — arXiv 2026, arXiv 2602.03983

DySta：把视觉 token 拆成静态/动态，跨帧只保留一份静态 token 并复用其 KV 缓存；多帧整合 +24.5%，推理 2 倍加速。

Abstract. Vision-Language-Action (VLA) models have recently emerged as a promising paradigm for generalist robotic control. Built upon vision-language model (VLM) architectures, VLAs predict actions conditioned on visual observations and language instructions, achieving strong performance and generalization across tasks. However, VLAs face two major challenges: a limited context window for input frames and inefficient inference due to the quadratic attention complexity and large parameter counts. To this end, we propose DySta, a framework that disentangles visual inputs into multi-level static and dynamic tokens, which enables (1) retaining a single copy of static tokens across frames to significantly reduce context length, and (2) reusing the key-value (KV) cache of static tokens through a lightweight recache gate that updates only when necessary. This design enables efficient multi-frame integration and efficient inference. In addition, we introduce a new benchmark that more effectively evaluates the multi-frame integration ability of VLAs. Experiments show that Dysta improves multi-frame integration by 24.5% across metrics on our benchmark and 23.3% in absolute success rate on real-world memory-dependent tasks, while accelerating inference by 2.0x (with +2.3% success rate) on simulation benchmarks and 2.2x (with +10.6% success rate) on real-world general tasks.

17. LoLA: Long Horizon Latent Action Learning for General Robot Manipulation

Xiaofan Wang, Xingyu Gao, Jianlong Fu, Zuolei Li, Dean Fortier, Galen Mullins, Andrey Kolobov, Baining Guo — arXiv 2025, arXiv 2512.20166

VLM 编码长时多视角历史 + 状态感知 latent 重表示，把 VL 表征锚定到物理尺度。

Abstract. The capability of performing long-horizon, language-guided robotic manipulation tasks critically relies on leveraging historical information and generating coherent action sequences. However, such capabilities are often overlooked by existing Vision-Language-Action (VLA) models. To solve this challenge, we propose LoLA (Long Horizon Latent Action Learning), a framework designed for robot manipulation that integrates long-term multi-view observations and robot proprioception to enable multi-step reasoning and action generation. We first employ Vision-Language Models to encode rich contextual features from historical sequences and multi-view observations. We further introduces a key module, State-Aware Latent Re-representation, which transforms visual inputs and language commands into actionable robot motion space. Unlike existing VLA approaches that merely concatenate robot proprioception (e.g., joint angles) with VL embeddings, this module leverages such robot states to explicitly ground VL representations in physical scale through a learnable "embodiment-anchored" latent space. We trained LoLA on diverse robotic pre-training datasets and conducted extensive evaluations on simulation benchmarks (SIMPLER and LIBERO), as well as two real-world tasks on Franka and Bi-Manual Aloha robots. Results show that LoLA significantly outperforms prior state-of-the-art methods (e.g., pi0), particularly in long-horizon manipulation tasks.

18. HiF-VLA: Hindsight, Insight and Foresight through Motion Representation for Vision-Language-Action Models

Minghui Lin, Pengxiang Ding, Shu Wang, Zifeng Zhuang, Yang Liu, Xinyang Tong, Wenxuan Song, Shangke Lyu et al. — arXiv 2025, arXiv 2512.09928

以运动为时间上下文的紧凑表征：后见先验编码过去动态、前瞻推理预测未来运动，边想边做。

Abstract. Vision-Language-Action (VLA) models have recently enabled robotic manipulation by grounding visual and linguistic cues into actions. However, most VLAs assume the Markov property, relying only on the current observation and thus suffering from temporal myopia that degrades long-horizon coherence. In this work, we view motion as a more compact and informative representation of temporal context and world dynamics, capturing inter-state changes while filtering static pixel-level noise. From this perspective, HiF-VLA equips a motion-centric world model for the VLA, enabling agents to reason about temporal dynamics for future evolution during action generation. Building on this idea, we propose HiF-VLA (Hindsight, Insight, and Foresight for VLAs), a unified framework that leverages motion for bidirectional temporal reasoning. HiF-VLA encodes past dynamics through hindsight priors, anticipates future motion via foresight reasoning, and integrates both through a hindsight-modulated joint expert to enable a ''think-while-acting'' paradigm for long-horizon manipulation. As a result, HiF-VLA surpasses strong baselines on LIBERO-Long and CALVIN ABC-D benchmarks, while incurring negligible additional inference latency. Furthermore, HiF-VLA achieves substantial improvements in real-world long-horizon manipulation tasks, demonstrating its broad effectiveness in practical robotic settings.

19. CycleManip: Enabling Cyclic Task Manipulation via Effective Historical Perception and Understanding

Yi-Lin Wei, Haoran Liao, Yuhao Lin, Pengyue Wang, Zhizhao Liang, Guiliang Liu, Wei-Shi Zheng — arXiv 2025, arXiv 2512.01022

周期性任务（摇瓶、敲钉）需要知道做了几次：代价感知的历史采样 + 多任务学习，附周期任务基准。

Abstract. In this paper, we explore an important yet underexplored task in robot manipulation: cycle-based manipulation, where robots need to perform cyclic or repetitive actions with an expected terminal time. These tasks are crucial in daily life, such as shaking a bottle or knocking a nail. However, few prior works have explored this task, leading to two main challenges: 1) the imitation methods often fail to complete these tasks within the expected terminal time due to the ineffective utilization of history; 2) the absence of a benchmark with sufficient data and automatic evaluation tools hinders development of effective solutions in this area. To address these challenges, we first propose the CycleManip framework to achieve cycle-based task manipulation in an end-to-end imitation manner without requiring any extra models, hierarchical structure or significant computational overhead. The core insight is to enhance effective history perception by a cost-aware sampling strategy and to improve historical understanding by multi-task learning. Second, we introduce a cycle-based task manipulation benchmark, which provides diverse cycle-based tasks, and an automatic evaluation method. Extensive experiments conducted in both simulation and real-world settings demonstrate that our method achieves high success rates in cycle-based task manipulation. The results further show strong adaptability performance in general manipulation, and the plug-and-play ability on imitation policies such as Vision-Language-Action (VLA) models. Moreover, the results show that our approach can be applied across diverse robotic platforms, including bi-arm grippers, dexterous hands, and humanoid robots.

20. AVA-VLA: Improving Vision-Language-Action models with Active Visual Attention

Lei Xiao, Jifeng Li, Juntao Gao, Feiyang Ye, Yan Jin, Jingjing Qian, Jing Zhang, Yong Wu et al. — arXiv 2025, arXiv 2511.18960

从 POMDP 视角引入循环信念状态，并据此对当前观测做主动视觉注意力重加权；EventVLA 把它归为循环范式代表。

Abstract. Vision-Language-Action (VLA) models have shown remarkable progress in embodied tasks recently, but most methods process visual observations independently at each timestep. This history-agnostic design treats robot manipulation as a Markov Decision Process, even though real-world robotic control is inherently partially observable and requires reasoning over past interactions. To address this mismatch, we reformulate VLA policy learning from a Partially Observable Markov Decision Process perspective and propose AVA-VLA, a framework that conditions action generation on a recurrent state that serves as a neural approximation to the agent's belief over task history. Built on this recurrent state, we introduce Active Visual Attention (AVA), which dynamically reweights visual tokens in the current observation to focus on regions most relevant given both the instruction and execution history. Extensive experiments show that AVA-VLA achieves state-of-the-art performance on standard robotic benchmarks, including LIBERO and CALVIN, and transfers effectively to real-world dual-arm manipulation tasks. These results demonstrate the effectiveness of temporally grounded active visual processing for improving VLA performance in robotic sequential decision-making. The project page is available at https://liauto-dsr.github.io/AVA-VLA-Page.

21. ContextVLA: Vision-Language-Action Model with Amortized Multi-Frame Context

Huiwon Jang, Sihyun Yu, Heeseung Kwon, Hojin Jeon, Younggyo Seo, Jinwoo Shin — arXiv 2025, arXiv 2510.04246

把过去多帧压成单个上下文 token，以低成本获得多帧训练的收益；观察到 VLM 骨干比普通 BC 更能利用多帧。

Abstract. Leveraging temporal context is crucial for success in partially observable robotic tasks. However, prior work in behavior cloning has demonstrated inconsistent performance gains when using multi-frame observations. In this paper, we introduce ContextVLA, a policy model that robustly improves robotic task performance by effectively leveraging multi-frame observations. Our approach is motivated by the key observation that Vision-Language-Action models (VLA), i.e., policy models built upon a Vision-Language Model (VLM), more effectively utilize multi-frame observations for action generation. This suggests that VLMs' inherent temporal understanding capability enables them to extract more meaningful context from multi-frame observations. However, the high dimensionality of video inputs introduces significant computational overhead, making VLA training and inference inefficient. To address this, ContextVLA compresses past observations into a single context token, allowing the policy to efficiently leverage temporal context for action generation. Our experiments show that ContextVLA consistently improves over single-frame VLAs and achieves the benefits of full multi-frame training but with reduced training and inference times.

22. HAMLET: Switch your Vision-Language-Action Model into a History-Aware Policy

Myungkyu Koo, Daewon Choi, Taeyoung Kim, Kyungmin Lee, Changyeon Kim, Younggyo Seo, Jinwoo Shin — arXiv 2025, arXiv 2510.00695

moment token（时间对比初始化）+ 轻量记忆模块，把 GR00T N1.5 变成历史感知策略；真机历史依赖任务 76.4%（+47.2）。

Abstract. Inherently, robotic manipulation tasks are history-dependent: leveraging past context could be beneficial. However, most existing Vision-Language-Action models (VLAs) have been designed without considering this aspect, i.e., they rely solely on the current observation, ignoring preceding context. In this paper, we propose HAMLET, a scalable framework to adapt VLAs to attend to the historical context during action prediction. Specifically, we introduce moment tokens that compactly encode perceptual information at each timestep. Their representations are initialized with time-contrastive learning, allowing them to better capture temporally distinctive aspects. Next, we employ a lightweight memory module that integrates the moment tokens across past timesteps into memory features, which are then leveraged for action prediction. Through empirical evaluation, we show that HAMLET successfully transforms a state-of-the-art VLA into a history-aware policy, especially demonstrating significant improvements on long-horizon tasks that require historical context. In particular, on top of GR00T N1.5, HAMLET achieves an average success rate of 76.4% on history-dependent real-world tasks, surpassing the baseline performance by 47.2%. Furthermore, HAMLET pushes prior art performance from 64.1% to 66.4% on RoboCasa Kitchen (100-demo setup) and from 95.6% to 97.7% on LIBERO, highlighting its effectiveness even under generic robot-manipulation benchmarks.

23. MemoryVLA: Perceptual-Cognitive Memory in Vision-Language-Action Models for Robotic Manipulation

Hao Shi, Bin Xie, Yingfei Liu, Lin Sun, Fengrong Liu, Tiancai Wang, Erjin Zhou, Haoqiang Fan et al. — arXiv 2025, arXiv 2508.19236

> **arXiv 2508.19236** · arXiv 2025 · 提交 2025-08-26 · 分类 `dense` / 稠密与压缩的视觉历史
> *Hao Shi, Bin Xie, Yingfei Liu, Lin Sun, Fengrong Liu, Tiancai Wang, Erjin Zhou, Haoqiang Fan, Xiangyu Zhang, Gao Huang*

arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

MemoryVLA（ICLR 2026）在 CogACT 式的「7B Prismatic VLM + DiT 扩散动作专家」外挂一个每步写入的双流记忆库 PCMB：感知流每条 256 个压缩视觉 token，认知流每条 1 个 LLM EOS token，交叉注意力检索、门控融合，满了就合并最相似的相邻两条。只用单张第三人称 RGB：SimplerEnv-Bridge 71.9%（比 CogACT-Large +14.6）、Fractal 72.7%、LIBERO 五套 96.5%、MIKASA-Robo 41.2%（比 π_0 +11.8）、真机 12 任务 84.0%，长程时序任务比 CogACT +26；推理延迟只比无记忆基座多 3.6%（0.187→0.194 s）。

解决什么问题

主流 VLA（OpenVLA、π_0）只看当前帧，而操作任务非马尔可夫：Push Buttons 按前按后画面几乎一样，单帧无法判断是否已按过。直接把多帧拼给 VLM 有两个问题：自注意力二次复杂度限制上下文；多帧输入偏离 VLM 单帧机器人预训练分布。已有时序建模各有代价：Octo、RoboVLMs 的交错视频格式实现复杂、算力大；RoboFlamingo 用 LSTM 传一个粗 latent，丢掉细粒度感知；TraceVLA 把轨迹画到图上，丢掉语义；UniVLA 把历史动作放进 prompt，只是 CoT。MemoryVLA 主张同时保留细粒度感知与高层语义，放在 VLM 之外的库里按需检索。

方法

骨干：Prismatic 7B VLM（DINOv2 + SigLIP 并联 → LLaMA-7B），OXE 预训练；输入单张 224×224 第三人称 RGB + 指令。视觉 token 经 SE-bottleneck 压成 N_p = 256 个感知 token p；LLM 的 EOS 位置输出作为 1 个 4096 维认知 token c。二者合称工作记忆。

存什么、何时写：PCMB 有感知、认知两条流，每流最多 L 条。每一步都写，写入的是门控融合之后的 p̃、c̃——库里存的是已含检索结果的特征，不是原图。

如何读：以当前 p、c 分别为 query，对本流 L 条记忆做 scaled dot-product 注意力，key 加按 episode 时间步的正弦编码 TE(t_i)，value 不加（Eq. 5–6）；接 FFN 成一层 Transformer，堆两层得 H^p、H^c。融合 g^x = σ(MLP([x; H^x]))，x̃ = g^x ⊙ H^x + (1 − g^x) ⊙ x（Eq. 7–8）。

容量：超过 L 时在每流内算相邻条目余弦相似度，把最相似的一对取平均合并（Eq. 9），不是 FIFO。仿真与真机通用任务 L = 16，真机长程时序任务 L = 256。

动作头：约 300M 参数 DiT，DDIM 10 步、CFG 1.5，预测 T = 16 步 7-DoF 动作（Δ平移、Δ欧拉角、gripper）。每个去噪步先与 c̃ 拼接过 cognition-attention，再对 p̃ 做 perception-attention；只有动作 MSE 损失，没有记忆专属损失。

训练：8×A100 FSDP，全局 batch 256，lr 2×10^{-5}。dataloader 必须保持 episode 内时序：Bridge/Fractal 流式加载连续帧；LIBERO/MIKASA 每次在单个 episode 内采 16 帧并保持顺序，与 L 对齐。真机数据按末端位移 > 0.01 m 或转角 > 0.4 rad 稀疏取帧，最大间隔 120 帧。

关键数字

基准（协议）	MemoryVLA	对照	备注
SimplerEnv-Bridge，WidowX，每任务 24 trials	71.9	CogACT-Large 57.3；π_0-Beta* 68.4	* open-pi-zero 复现，用本体状态

基准（协议）	MemoryVLA	对照	备注
SimplerEnv-Fractal 总体（VM 77.7 / VA 67.7），336 变体	72.7	CogACT 68.1（74.8 / 61.3）	Open/Close Drawer VA +24.9
LIBERO 五套均值，每任务 50 trials	96.5	CogACT 93.2（复现）；π_0* 94.2	Long-10 93.4，Long-90 95.6
MIKASA-Robo 五任务，每任务 100 episodes	41.2	π_0 29.4；CronusVLA 18.0（复现）	ShellGameTouch 88 vs 33；Intercept 24 低于 π_0 42
真机 General 六任务，15 trials	85	CogACT 76	
真机 Long-horizon 六任务，10–15 trials，分步计分	83	CogACT 57；π_0 52	Seq. Push Buttons 58 vs 15
推理，RTX 4090，bf16，300 次	0.194 s / 16.6 GB	基座 0.187 s / 15.8 GB	HGX H20 上 0.246 vs 0.236 s

Bridge 消融（4 任务 × 24 trials 均值）：

因素	变体 → 成功率
记忆类型	仅认知 63.5 / 仅感知 64.6 / 双流 71.9
记忆长度 L	4: 67.7 / 16: 71.9 / 64: 67.7
检索时间编码	无 69.8 / 有 71.9
融合	相加 67.7 / 门控 71.9
整合	FIFO 66.7 / 相邻合并 71.9
真机 Clean Table & Count 的 L	64: 78 / 256: 84 / 512: 81

局限与注意

论文没有单列局限，结论只给未来方向（把长期记忆对齐到 LLM 输入空间做 embedding 级 CoT；终身记忆整合）。附录 B 承认相机视角变化下掉得厉害：Pick Coke Can 92.0 → 42.0。

我的读法： - 这是稠密记忆：每步都写、没有事件选择，压缩靠合并相邻相似项，而且写入的是融合后的特征，库内容递归地包含自身检索结果。EventVLA 对「冗余淹没稀疏证据」的批评针对的正是这类设计。 - 消融差距很小：Bridge 96 次 trial 上 69.8 vs 71.9 差 2 个点约等于 2 次 trial；Bridge、LIBERO、MIKASA 都是每隔若干步验证并报告最佳验证步，等于在评测集上选 checkpoint。 - 基线处理总体公平且对自己不利：π_0 在 Bridge 用本体状态、在 LIBERO 用腕相机 + 本体，MemoryVLA 只用第三人称 RGB；CogACT 重新复现。 - MIKASA-Robo 均值领先，但 Intercept Medium 24 低于 π_0 的 42，RememberColor9 只有 20；增益主要来自 ShellGameTouch。 - 真机长程任务分步给分（30/30/30 + 10 奖励之类）、每任务 10 次，83 是分数不是二值成功率；L 在长程任务上改成 256，逐套调。 - LIBERO/MIKASA 训练时记忆一次只见同一 episode 内采出的 16 帧，测试时 episode 长达 505 步、记忆持续累积合并，训练/测试记忆跨度不一致，论文未讨论。 - 表 15 的 82.5 Hz 是按 16 步 chunk 折算的动作吞吐，不是控制频率。

与核心论文的关系

在写入策略上与 EventVLA 处于两端：EventVLA 只把预测为关键帧的原图写进 5 槽 FIFO，MemoryVLA 每步写融合特征再靠合并压缩。EventVLA 把它当帧缓冲范式的代表基线：RMBench 41.7%（QwenOFT 骨干）/ 19.4%（OpenVLA 骨干），对比 EventVLA 仅锚帧 67.8%；RoboTwin-MeM 10.8% 对比 75.2%。KEMO 也拿它当记忆基

线，六个真机双臂任务聚合 TSR 1.4% 对比 KEMO 51.4%，但那是单相机 Prismatic 骨干对三相机 $\pi_{0.5}$。两处复现都换了骨干和输入，测的不只是记忆机制。

机制上最接近 TRACE：都是挂在骨干之外、注意力读出、带门的 latent 记忆。差别在：TRACE 固定 K = 4/6 个 512 维槽，用机器人状态轨迹的路径签名寻址，每步门控写入，配三个稳定损失；MemoryVLA 每流 L = 16（或 256）条，寻址靠内容相似度（当前 token 作 query）加正弦时间编码，写入无条件、门在读端，没有记忆专属损失，溢出靠合并而非覆盖。容量：MemoryVLA 16 × (256 + 1) ≈ 4112 个 token，TRACE 4–6 个向量，EventVLA 5 张原图。成本：MemoryVLA 报 +3.6% 延迟，EventVLA 报 2.91→0.94 Hz，硬件和模型都不同，只能说 MemoryVLA 落在便宜的一端。SAI 的 30 步 GRU 历史 token 与 MemoryVLA 的认知流（每步 1 个 token）同为单向量摘要，一个递归、一个检索。

关联阅读

- CogACT（2411.19650）：MemoryVLA 的基座架构与主要对照。
- CronusVLA（2506.19816）：滑窗聚合多帧 VLM 特征，MemoryVLA 在 MIKASA-Robo 上复现的时序基线（18.0）。
- MIKASA-Robo（2502.10550）：唯一用到的公开记忆基准。
- EventVLA（2606.20092）：把 MemoryVLA 当帧缓冲基线，RMBench 41.7%。
- KEMO（2606.23589）：真机双臂任务上把 MemoryVLA 当记忆基线，聚合 TSR 1.4%。

24. CronusVLA: Towards Efficient and Robust Manipulation via Multi-Frame Vision-Language-Action Modeling

Hao Li, Shuai Yang, Yilun Chen, Xinyi Chen, Xiaoda Yang, Yang Tian, Hanqing Wang, Tai Wang et al. — arXiv 2025, arXiv 2506.19816

单帧预训练 + 多帧后训练，用特征分块聚合历史；提出 SimplerEnv-OR 观测扰动基准。

Abstract. Recent vision-language-action (VLA) models built on pretrained vision-language models (VLMs) have demonstrated strong performance in robotic manipulation. However, these models remain constrained by the single-frame image paradigm and fail to fully leverage the temporal information offered by multi-frame histories, as directly feeding multiple frames into VLM backbones incurs substantial computational overhead and inference latency. We propose CronusVLA, a unified framework that extends single-frame VLA models to the multi-frame paradigm. CronusVLA follows a two-stage process: (1) Single-frame pretraining on large-scale embodied datasets with autoregressive prediction of action tokens, establishing an effective embodied vision-language foundation; (2) Multi-frame post-training, which adapts the prediction of the vision-language backbone from discrete tokens to learnable features, and aggregates historical information via feature chunking. CronusVLA effectively addresses the existing challenges of multi-frame modeling while enhancing performance and observational robustness. To evaluate the robustness under temporal and spatial disturbances, we introduce SimplerEnv-OR, a novel benchmark featuring 24 types of observational disturbances and 120 severity levels. Experiments across three embodiments in simulated and real-world environments demonstrate that CronusVLA achieves leading performance and superior robustness, with a 70.9% success rate on SimplerEnv, a 26.8% improvement over OpenVLA on LIBERO, and the highest robustness score on SimplerEnv-OR. These results highlight the potential of efficient multi-frame adaptation in VLA models for more powerful and robust real-world deployment.

25. TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies

Ruijie Zheng, Yongyuan Liang, Shuaiyi Huang, Jianfeng Gao, Hal Daumé, Andrey Kolobov, Furong Huang, Jianwei Yang — ICLR 2025, arXiv 2412.10345

视觉轨迹提示：把状态-动作轨迹画到图像上作为提示，增强时空感知；早期（2024）的历史注入方式。

Abstract. Although large vision-language-action (VLA) models pretrained on extensive robot datasets offer promising generalist policies for robotic learning, they still struggle with spatial-temporal dynamics in interactive robotics, making them less effective in handling complex tasks, such as manipulation. In this work, we introduce visual trace prompting, a simple yet effective approach to facilitate VLA models' spatial-temporal awareness for action prediction by encoding state-action trajectories visually. We develop a new TraceVLA model by finetuning OpenVLA on our own collected dataset of 150K robot manipulation trajectories using visual trace prompting. Evaluations of TraceVLA across 137 configurations in SimplerEnv and 4 tasks on a physical WidowX robot demonstrate state-of-the-art performance, outperforming OpenVLA by 10% on SimplerEnv and 3.5x on real-robot tasks and exhibiting robust generalization across diverse embodiments and scenarios. To further validate the effectiveness and generality of our method, we present a compact VLA model based on 4B Phi-3-Vision, pretrained on the Open-X-Embodiment and finetuned on our dataset, rivals the 7B OpenVLA baseline while significantly improving inference efficiency.

latent 状态、循环与槽记忆（TRACE 一族） / Latent, Recurrent and Slot Memory

历史被压成固定大小的向量或槽并递归更新。TRACE 用轨迹签名寻址槽；Chronos、TFP、RB-VLA 主张「历史应是策略的 latent 状态」；μVLA 隔离出最小循环记忆的能力边界；GMP 与 PTP 处理长历史带来的虚假相关。

1. FM-VLA: Force-based Memory for Vision-Language-Action Models in Contact-Rich Manipulation

Ruicheng Li, Qixiu Li, Ruichun Ma, Yu Deng, Lin Luo, Zhiying Du, Jianfeng Xiang, Huizhi Liang et al. — arXiv 2026, arXiv 2607.18231

力历史经 VAE 压成力记忆 token 条件动作专家：视觉分不清的重复接触事件（按几次按钮、擦几次盘子）靠力记住，成功率 80%+。

Abstract. Vision-language-action (VLA) models have achieved impressive generalization in robotic manipulation, and recent memory-augmented VLAs have relaxed the Markovian assumption by conditioning on past images or language summaries. Vision-based memory approaches address this by conditioning on sampled past image frames, but they are computationally expensive and fundamentally limited when temporal events are visually ambiguous, e.g., pushing a button multiple times with small movements. We propose FM-VLA, a VLA model with force-based memory, enabling temporal context reasoning for non-Markovian, contact-rich manipulation. We encode force histories into compact force memory tokens with a variational autoencoder (VAE) pretrained with force time series reconstruction. By projecting force latent representations and short state history as additional conditioning tokens to the action expert module, we enable VLAs to leverage accumulated contact event history to guide manipulation. We evaluate FM-VLA on three memory-dependent tasks, including finding a hidden block, pressing a button, and wiping a dish for a specific number of times. Our lightweight force memory achieves over 80% success rate with minimal inference overhead, significantly outperforming baseline approaches. Project page: https://qft-333.github.io/FM-VLA-Page/

2. TFP: Temporally Conditioned Memory-Fusion Policies for Visuomotor Learning

Yushen Liang, Yue Peng, Baosheng Jin, Tianluo Zhang, Xinyu Zhang, Shuyi Zhou, Zhuoran Chen, Xinqi Liu et al. — arXiv 2026, arXiv 2607.08283

液态时间常数动力学维护回合内任务进度信念，自适应调制注入流匹配解码器；事件附近写入增益约为非事件阶段的 6 倍，隐状态干预证明信念因果影响动作。

Abstract. Vision–Language–Action (VLA) policies such as MATH7 and OpenVLA perform well on many manipulation tasks, but they are often reactive: the next action is predicted from the current observation, instruction, and proprioceptive state. This assumption breaks down in stage-dependent manipulation, where visually similar states may require different actions depending on latent task progress and previous interaction outcomes. We argue that such tasks require not only memory, but dynamics-aware belief updates: the policy should preserve task progress during stable or occluded phases and revise its belief near contact, release, or subgoal transitions. We introduce Temporally Conditioned Memory-Fusion Policies (TFP), a lightweight memory-action framework for VLA backbones. TFP maintains an episode-local task-progress belief with Liquid Time-Constant dynamics and injects the updated belief directly into the flow-matching action decoder through adaptive modulation. This lets temporally accumulated context shape the generated action chunk, rather than serving only as passive history context. With a 3.3B-parameter model, TFP improves the average success rate from 96.9% to 98.75% on LIBERO and from 91.4% to 93.77% on LIBERO-plus. On the memory-focused MIKASA ShellGameTouch diagnostic, TFP achieves success up to 75.0%. Mechanistic analyses show that write-gain changes near manipulation events are about 6 times larger than far non-event phases, and hidden-state interventions show that the belief causally modulates generated action chunks. These results suggest that compact, event-sensitive memory dynamics can improve VLA policies under occlusion, visual perturbation, and stage-dependent task structure.

3. ChronoFlow-Policy: Unifying Past-Current-Future Interaction Flow in Visuomotor Policy Learning

Bokai Lin, Yifu Xu, Xinyu Zhan, Hongjie Fang, Jialin Tian, Fu-Cheng Zhang, Yong-Lu Li, Cewu Lu et al. — arXiv 2026, arXiv 2606.31493

用物体与夹爪的稀疏 3D 关键点统一表示过去-当前-未来交互流，与动作共同训练的扩散策略。

Abstract. Visual signals play a crucial role in policy learning by enabling models to capture object motion and interaction dynamics. Just as humans reason about actions using both past experience and anticipated outcomes, effective policies should integrate past interactions with future predictions. However, existing visuomotor policies typically model either historical context or future dynamics in isolation, lacking a unified temporal representation of interaction dynamics. In this work, we introduce ChronoFlow, a temporally unified representation that captures past, current, and future interaction dynamics through sparse 3D keypoints of both objects and the gripper. Based on this representation, we propose ChronoFlow-Policy, a diffusion-based visuomotor policy that jointly learns ChronoFlow and action sequences through a co-training objective. Experiments on 14 simulated tasks and 5 real-world manipulation tasks demonstrate that ChronoFlow-Policy consistently outperforms strong diffusion-policy baselines and improves robustness in long-horizon and non-Markovian manipulation scenarios. Our project page is available at https://the-kamisato-sii.github.io/ChronoFlow-Policy-project-page/.

4. Chronos: A Physics-Informed Full-History Framework for Non-Markovian Long-Horizon Manipulation

Yulin Zhou, Yimeng Wang, Nengyu Wang, Shaojia Xing, Shiyun Tu, Xiang Li, Jingkai Zhang, Ningbo Jiang et al. — arXiv 2026, arXiv 2606.30318

> **arXiv 2606.30318** · arXiv 2026 · 提交 2026-06-29 · 分类 `latent` / latent 状态、循环与槽记忆（TRACE 一族）
> *Yulin Zhou, Yimeng Wang, Nengyu Wang, Shaojia Xing, Shiyun Tu, Xiang Li, Jingkai Zhang, Ningbo Jiang, Yuankai Lin, Hua Yang, Xiangrui Zeng, Zhouping Yin*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

Chronos 把整条观测历史当作策略的 latent 状态：每个物理控制步压成一个 state token，用 Mamba 型选择性 SSM 沿全轨迹因果传播，再以 IMLE 粗先验 + 预测加速度场的二阶 Schrödinger 桥生成动作块。0.3B 参数在 RMBench 七任务平均 73.6%，比 π0.5（11.2%）高 62.4 个百分点，比记忆 VLA Mem-0（50.8%）高 22.8 个百分点。

解决什么问题

观测混叠：杯子拿走再放回后，图像和点云与初始几乎一致，但任务阶段不同，正确动作也不同。Markov 策略 π(o_t, p_t) 与固定窗口 π(o_{t−K:t}) 只在条件变量是充分统计量时成立，RMBench 类任务正是构造成让这一前提失效。论文点名两类先前做法的短板：外挂记忆模块的 VLA（Mem-0 的 anchor + sliding、MemoryVLA）在历史需要连续调制低层动作、而非作为离散线索被检索时仍会失败；MTIL 虽是全历史 SSM 模仿，但 hidden state 被 detach，晚期损失无法修正早期阶段表征。

方法

- **存什么**：SSM 内部循环状态 h_t 与输出上下文 y_t。输入 token x_t = LN(W_x[g_t^obs; g_t^prop] + b_x)，观测特征来自可训练 PointNet 式点云编码器（RoboTwin 2.0、RMBench）或冻结 DINOv2 / ResNet18 + 可训练 adapter（ALOHA、真机）。token 序列长度等于示教轨迹长度 L。
- **何时写、如何读**：每步无门控写入，h_t = Ā_t h_{t−1} + B̄_t x_t，y_t = C_t h_t + D x_t，Δ_t、B_t、C_t 由当前 token 选择性生成；读出即 y_t 进动作条件 c_t = LN(W_c[y_t; x_t] + b_c)。没有槽、没有寻址，容量就是 SSM 状态维度。
- **动作头**：IMLE 生成器 G_φ(z, c_t) 采 K 个候选，只让离专家块最近者受拉（L_IMLE = E[min_k d(G(z^k, c), q*)]），得到粗先验 q_0。二阶桥再精化：参考路径 q_ref(s) = q_0 + α(s)(q_1 − q_0)，α(s) = 3s² − 2s³ 使端点速度为零；目标加速度 a_tar = a_spline(s) + K_p(q_ref − q) + K_d(v_ref − v)，由 Schrödinger → Madelung → Kostin 耗散势推出，落地是围绕三次样条的 PD 稳定器；噪声调度 σ(s) = 16σ_max s²(1−s)²，位置与速度扰动在两端同时为零。总损失 L = L_IMLE + λ_acc L_acc + λ_bc L_BC。
- **训练与推理**：感知按块编码后拼接、整体过一次 SSM，SSM 与动作头不 detach、全序列反传。推理为因果单步 SSM.step，IMLE 隐变量用时间相关采样保持连续，加速度场 symplectic Euler 积分 N 步，3–5 步最佳。只有模仿损失，无记忆专用辅助损失。

关键数字

基准 / 协议	对比项	数字
RMBench 7 任务，50 demo，每任务 100 次；基线为论文报告值	DP / ACT / π0.5 / X-VLA / Mem-0 / Chronos 平均	5.8 / 7.4 / 11.2 / 11.1 / 50.8 / **73.6%**

基准 / 协议	对比项	数字
RMBench 分任务（Rearrange / Put Back / Swap Blocks / Swap T / Battery Try / Cover / Press Button）	Chronos	98 / 98 / 99 / 93 / 29 / 96 / 2%
同上	Mem-0	89 / 90 / 67 / 14 / 28 / 68 / 0%
RoboTwin 2.0 Easy 8 任务，50 demo，100 次	DP / ACT / RDT-1B / π0 / DP3 / Chronos 平均	35.1 / 30.6 / 35.8 / 48.8 / 59.5 / **70.0%**；Put Bottles Dustbin 输 DP3（54 vs 60）
ALOHA 双臂插入，50 demo，50 次；ACT、MTIL 为复现	ACT / MTIL / 回归头（无 IMLE、无 SB）/ 换 diffusion / 换 flow / 无 SB / 无 IMLE / 一阶噪声调度 / 完整	50 / 76 / 76 / 66 / 72 / 86 / 84 / 72 / **90%**
ALOHA 推理积分步数 N = 1 / 3 / 5 / 15	同一模型	84 / 90 / 90 / 88%
真机双 UR3 + 单 D435 RGB，每任务 50 次	π0.5 vs Chronos：Put Back Blocks / Swap T（无记忆对照）/ Swap T-Mem / Cover Blocks	0 vs 98、28 vs 96、0 vs 98、0 vs 20%；三个记忆任务合计 0/150 vs 108/150（72%）
模型规模（论文表 I）	Mem-0 > 10B，π0.5 > 3.3B，DP3 264.4M，Chronos 0.3B	仅作规模语境

局限与注意

- **基线不对等**：RMBench 与 RoboTwin 2.0 上除 Chronos 外全是 Reported 数字，作者明说未在同一 seed、硬件、编码器、代码库下重训。更关键的是 Chronos 在这两个基准用点云输入，π0.5、X-VLA、Mem-0 是 RGB VLA，”少 10 倍参数却高 62 点”混合了记忆机制与输入模态两个因素。真正受控的只有 ALOHA 上共享同一历史编码器的动作头消融，以及双方同一单目相机的真机对比。
- **缺记忆消融**：没有”去掉 SSM / 换短窗口”对照，也没有截断反传 vs 全序列反传的直接对比。MTIL 只在 ALOHA 出现且与 Chronos 回归头打平（76%），说明该任务瓶颈在动作头不在记忆。”历史应当是策略状态”来自 RMBench 大幅领先和 PCA 可视化（y_t 能分开 x_t 混叠的阶段），是间接证据。
- **物理推导的份量**：Schrödinger–Madelung–Kostin 链条最后落成 PD 稳定器 + 三次样条 + 四次噪声调度，作者自认”不是解一般桥问题，而是控制导向的投影”。真正有效的消融是一阶 vs 四次调度（72 → 90%）。
- **未评估**：延迟与吞吐未报告；训练显存只有大 O 分析；无跨 seed 方差。Press Button 2%、Battery Try 29% 说明 OCR 与接触精度仍是瓶颈；真机 Cover Blocks 20% 被归因为 ResNet18 弱化颜色–位置绑定。训练需要整条示教轨迹按物理时间对齐。

与核心论文的关系

- **最接近 TRACE**，同属 latent 状态记忆、都可挂在 compact 策略上，但设计相反：TRACE 是固定 K 槽（4 或 6 × 512-D）、按机器人状态轨迹的 path signature（深度 3，5219-D）寻址、门控写入、三项稳定器损失、适配器接入不改骨干与损失；Chronos 无槽无寻址，整个 SSM 状态就是记忆，每步无门控写入，只靠模仿损失，但把动作头整体换成 IMLE + 二阶桥。TRACE 用顺序反转负对照（37.8 路由相似度）检验记忆是否真依赖顺序，Chronos 没有等价负对照，只有真机 Swap T 与 Swap T-Mem 的任务级对照（π0.5 28% vs 0%，Chronos 96% vs 98%）。
- **与 EventVLA**：两篇都报 RMBench 但不能横向比。EventVLA 报 anchors-only 67.8%、Mem-0 42.0%、π0.5 10.4%；Chronos 报自身 73.6%、Mem-0 50.8%、π0.5 11.2%，同一基线在两篇里数字不同，任务子集或口径必然不同。机制上是对立设计：EventVLA 存稀疏原始关键帧（5 槽 FIFO，KEM 头预测 50 步 chunk 内的关键帧，

阈值 0.55），事件触发写入、可视可查；Chronos 存稠密连续状态，每步写入、不可读。EventVLA 的隐式 latent bank 消融只有 24.9%，看似与 Chronos 矛盾，但那是挂在 VLA 上的外置库，Chronos 的 SSM 是全轨迹反传训练的策略主干。
- **与 SAI**：SAI 的 30 步 GRU 历史 token 是短窗口循环记忆（过早释放 64.5% → 16.1%），Chronos 是全历史循环；两者都表明循环状态不加记忆损失也能起作用。

关联阅读

- MTIL（arXiv 2505.12410）：Chronos 的直接前身，Mamba 全历史模仿但 detach hidden state，本文用它做截断信用分配的对照。
- RMBench（arXiv 2603.01229）：主实验基准及 Mem-0 基线的出处，anchor + sliding 记忆设计。
- IMLE Policy（arXiv 2502.12371）：粗先验所用的 IMLE 视觉运动策略，解释为何用最近样本匹配替代回归。
- MemoryVLA（arXiv 2508.19236）：感知–认知记忆库，代表 Chronos 所批评的外挂记忆模块路线。
- Mamba（arXiv 2312.00752）：选择性 SSM 原始论文，Chronos 的历史编码器。

5. Remember what you did?: Learning Behavioral Memories for Partially Observable Object Manipulation

Kuancheng Wang, Seungho Yeom, Jinglin Cao, Yuheng Zhi, Nikhil Shinde, Michael Yip — arXiv 2026, arXiv 2606.21188

CAMP：把机器人自己的动作历史压缩成行为记忆，自监督地追踪任务进度并从失败尝试中学习；附 Memory-T-Bench / Memory-Manip-Bench。

Abstract. Long horizon, contact-rich manipulation is inherently partially observable. This is as a single visual observation rarely captures a robot's full action context, including prior attempts, interactions, or progress. Consequently, standard visuomotor policies or vision-language-action models are prone to struggle in such tasks due to a lack of memory. To address this, we introduce Compressed Action Memory Policy (CAMP) based on the insight that a robot's own action history serves as a highly informative, self-supervised signal, enabling the policy to learn a robust, compact history representation. In our approach, we train a memory module to maintain a compressed representation of past actions, forcing it to encode a latent behavioral memory of all the robot's past interactions that can then be used to better contextualize future actions. This allows our approach to implicitly track generalized task progress and learn from failed attempts without any additional supervision, or external oversight. We evaluate CAMP across four real-robot setups and two novel simulation benchmarks: Memory-T-Bench and Memory-Manip-Bench. By demonstrating substantial gains over state-of-the-art baselines, CAMP is, to our knowledge, the first policy to demonstrate substantial success on contact-rich partially observable manipulation tasks purely through learned memory.

6. MATH8VLA: On Recurrent Memory for Partially Observable Manipulation in VLA Models

Egor Cherepanov, Nikita Kachaev, Daniil Zelezetsky, Aydar Bulatov, Artem Pshenitsyn, Yuri Kuratov, Alexey Skrynnik, Aleksandr I. Panov et al. — arXiv 2026, arXiv 2606.12497

> **arXiv 2606.12497** · arXiv 2026 · 提交 2026-06-10 · 分类 `latent` / latent 状态、循环与槽记忆（TRACE 一族）
> *Egor Cherepanov, Nikita Kachaev, Daniil Zelezetsky, Aydar Bulatov, Artem Pshenitsyn, Yuri Kuratov, Alexey Skrynnik,*

Aleksandr I. Panov, Alexey K. Kovalev
arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

μVLA 是一项受控隔离研究：在 OpenVLA-OFT 里只加 m 个跨步携带的可学习记忆 token，用 TBPTT 端到端训练，无辅助损失、无结构改动，让循环成为唯一变量。MIKASA-Robo 五个训练任务平均成功率 0.42 → 0.84（m=64，K=2）。最值得记住的是 TBPTT 长度的 U 形曲线：K=2 在 RememberColor5 达 0.93，K=1 与 K=8 只有 0.40 / 0.35。

解决什么问题

VLA 从当前观测预测动作块，部分可观测下决策所需的信息已不可见。已有记忆 VLA 同时引入循环、检索、压缩、辅助目标、层级记忆或专用结构改动（ReMem-VLA 的 EMA 更新加过去观测预测损失、AVA-VLA 的信念状态加额外损失、MemoryVLA 的记忆库），循环本身的贡献无法从这些机械中剥离。论文回答四个问题：能否在微调阶段给无记忆预训练 VLA 加循环；跨步梯度是否必要、截断多长合适；记忆语义迁移时表现如何；全可观测任务是否退化。

方法

- **存什么**：m 个 d=4096 维记忆 token M_t，插在 OpenVLA-OFT 输入序列的 PROPRIO 与 TEXT 之间。M_0 = M^init 为共享可学习参数；t ≥ 1 时 M_t 取上一步前向中记忆位置的隐藏状态，一次前向同时完成读与写。m=64 为主设定，m=1 为带宽探针。
- **注意力掩码守卫**：OpenVLA-OFT 是双向注意力，记忆若能看动作 token，会退化为 M_t = ϕ(ACTION_t) 的自指解，什么环境信息都不存。把 context→action 块置零，只让动作 token 读全上下文。去掉守卫后 RememberColor5 0.44 → 0.25，而运动主导的 ShellGamePush 0.77 → 0.94，记忆转向编码动作轨迹。
- **数据管线**：round-robin 分集加载器，B 条独立流各自逐步走完一条 episode，is_first 标志按流重置到 M^init，取代原 RLDS 打乱单步采样。
- **写入规则**：TBPTT 累积 K 步 L1 chunk 损失后一次反传，仅在截断边界 detach；EMA 变体 M_{t+1} = α M'_t.detach() + (1−α) M_t.detach()，α=0.1，反传限于单步。
- **训练**：从 OpenVLA 权重起，LoRA r=32 加记忆位置嵌入与 M^init，AdamW lr 5e-4，8×H100，最多 150k 步，H=8，两视角 224×224。
- **推理**：receding-horizon，每个环境步重查询、只执行 chunk 首动作，使记忆更新节奏与训练一致。同一 K=8 检查点改成 H=8 开环执行，LIBERO Long-10 从 95.8 跌到 5.4，Goal 从 96.6 跌到 35.8。
- **监督**：只有动作 L1 损失。

关键数字

基准 / 协议	对比项	数字
MIKASA-Robo-VLA 5 个训练任务，每环境 100 个固定 seed 回合	π0.5 / OpenVLA-OFT / OFT 分集加载器 / m=1,K=8 / m=64,K=8 / K=1 / EMA / **K=2** / 首帧 oracle	0.46 / 0.42 / 0.48 / 0.54 / 0.57 / 0.57 / 0.57 / **0.84** / 0.85
RememberColor5	OFT / OFT 分集 / K=1 / K=8 / EMA / K=2 / 首帧 oracle	0.04 / 0.09 / 0.40 / 0.35 / 0.44 / **0.93** / 0.96
TakeItBack（线索不在首帧）	OFT 分集 / 首帧 oracle / K=2	0.87 / 0.94 / **0.99**
11 个同记忆语义留出任务平均	OFT / OFT 分集 / K=2 / 首帧 oracle	0.01 / 0.07 / **0.23** / 0.24

基准 / 协议	对比项	数字
7 个新记忆语义留出任务平均	OFT 分集 / 其余循环变体 / K=2 / 首帧 oracle	0.07 / 0.09–0.11 / 0.16 / 0.19
LIBERO 四套件平均（每子任务 50 回合）	OpenVLA-OFT / MemoryVLA / CronusVLA / μVLA K=8 / μVLA EMA	97.1 / 96.5 / 86.2 / 96.2 / 44.8%
因果干预（K=2，100 回合）	记忆换高斯噪声：RC5 / TakeItBack；冻结首步记忆：RC5 / InterceptMedium	0.94 → 0.09 / 0.99 → 0.21；0.36 / 0.07
阶段长度外推 RC5（训练 N ≤ 5）	K=2 在 N=3 / N=20；K=8 全程；EMA N ≥ 20	0.93 / 丢约 75%；0.31–0.38；< 0.15
代价（RC5 评测）	OFT 开环 vs μVLA：前向延迟 / 闭环吞吐	61.3 vs 66.7–68.3 ms / 20.9 vs 9.1–9.3 Hz
训练时长（8×H100）	OFT 分集 / K=1 / EMA / K=2 / K=8	22h41m / 1d3h / 1d2h / 2d18h / 12d0h

局限与注意

- **正确基线是分集加载器版 OFT（0.48）而非原始 OFT（0.42）**，作者自己指出数据顺序就贡献 6 点；记忆带宽 m 从 0 到 1 加 6 点、1 到 64 只加 3 点，主要杠杆是 K。
- **K=2 的优势是分布内的**：阶段长度超出训练范围后 K=2 掉得最快，K=8 低而平，论文明说 TBPTT 不面向任意长程记忆。首帧 oracle 在线索位于首帧的任务上与 K=2 持平或更好（RC5 0.96 vs 0.93），只有 TakeItBack 这类线索不在首帧的任务证明循环是必要的。
- **新记忆语义几乎不迁移**：Rotate 家族全部接近零，RememberShape 只有部分迁移。”能力包络”的措辞诚实，但也意味着 m=64 单一循环状态不是通用记忆。
- **成本**：receding-horizon 让闭环吞吐减半以上；K=8 训练 12 天，K=2 近 3 天，是无记忆基线的近 3 倍。7B 骨干加 LoRA r=32，可训练参数量未单独给出。
- **未评估**：仅仿真、脚本 oracle 示教，无真机；LIBERO 只报 K=8 而未报最强的 K=2；与 MemoryVLA、CronusVLA 的对比取自他人论文数字，训练配方不同，不能当作严格胜出。

与核心论文的关系

- **与 TRACE 同属 latent 循环记忆但更极简**：TRACE 用固定 K 槽、path signature 寻址、门控写入、三项稳定器损失、适配器外挂在 ACT / DP；μVLA 把记忆放进 7B 骨干的自注意力，无寻址（记忆 token 经注意力自行读写）、每步无门控写入、容量 m×4096、零辅助损失。两者都表明训练信号决定成败：TRACE 靠稳定器损失，μVLA 靠跨步梯度（K=1 detach 后只有 0.57）。μVLA 的 EMA 变体（0.57）与 ReMem-VLA 的 detached EMA 更新同类，被判定不足以精确保持线索。
- **与 EventVLA 是两个极端**：EventVLA 存 5 槽原始关键帧、KEM 头显式预测何时写、吞吐 2.91 → 0.94 Hz；μVLA 存不可读的连续 token、每步写，吞吐 20.9 → 9.1 Hz 主要来自 receding-horizon 而非记忆本身（前向延迟只增 5–7 ms）。μVLA 观察到记忆 token 的余弦变化在阶段转换处出现尖峰，并提出可作为学到的关键帧信号，与 KEM 思路呼应。EventVLA 的隐式 latent bank 消融（24.9%）与 μVLA 的正面结果不矛盾：μVLA 的循环状态经 TBPTT 端到端训练且有动作拷贝守卫。
- **与 SAI**：SAI 的 30 步 GRU 历史 token 也是不加辅助损失的最小循环；μVLA 的隔离结论（跨步梯度加训练/推理节奏对齐）为这类设计给出可迁移的经验规则。

关联阅读

- Recurrent Memory Transformer（arXiv 2207.06881）：μVLA 的原型，记忆 token 跨段携带；本文把它降到环境步级并加动作守卫。
- MIKASA-Robo（arXiv 2502.10550）：主基准，按线索回忆 / 遮挡 / 序列预测三类记忆结构组织任务。
- ReMem-VLA（arXiv 2603.12942）：帧级与 chunk 级循环 query 加 EMA 更新与辅助损失，本文 EMA 变体的对照原型。
- CronusVLA（arXiv 2506.19816）：多帧后训练的压缩上下文路线，LIBERO 与 MIKASA 表中的语境对比。
- RoboMME（arXiv 2603.04639）：14 种 π0.5 记忆变体的受控研究，与本文同属隔离记忆机制的方法学。

7. Action-Effect Memory Pretraining for Robot Manipulation

Yijing Zhou, Qiwei Liang, Sitong Zhuang, Jiaxi Li, Xianpeng Wang, Boyang Cai, Yunyang Mo, Renjing Xu — arXiv 2026, arXiv 2606.12499

AEM：交错视觉-动作特征做掩码建模，学习动作条件的状态演化，Mamba 输出单向量作为紧凑历史表征。

Abstract. We present AEM, an Action-Effect Memory pretraining framework for robot manipulation that learns compact temporal representations from vision-action history. Unlike prior robot representation pretraining methods that mainly focus on single-frame visual encoding, AEM targets the temporal nature of manipulation, where the current observation alone is often insufficient under partial observability. AEM models manipulation as an action-driven interaction process by interleaving visual and action features and applying masked modeling to recover missing content from incomplete histories, thereby learning action-conditioned state evolution. The Mamba-encoded output of the final vision token is used as a compact history representation, serving as the global context for decoding and downstream control. This design preserves a single-vector temporal bottleneck while keeping inference efficient. We evaluate AEM with Diffusion Policy and Flow Policy. AEM consistently improves manipulation performance in both simulation and real-world settings, outperforming baselines across clean scenes, cluttered and random scenes, and non-Markovian tasks. Ablation studies further show that history-aware pretraining surpasses single-frame pretraining and direct frame stacking, while reducing inference latency and computational cost.

8. DSSP: Diffusion State Space Policy with Full-History Encoding

Zhiyuan Guan, Jianshu Hu, Han Fang, Yunpeng Jiang, Yize Huang, Shujia Li, Xiao Li, Yutong Ban — arXiv 2026, arXiv 2605.14598

SSM 历史编码器压缩全部观测流 + 动力学感知辅助目标，与近期观测形成分层条件；扩散骨干也用 SSM。

Abstract. Diffusion-based imitation learning has shown strong promise for robot manipulation. However, most existing policies condition only on the current observation or a short window of recent observations, limiting their ability to resolve history-dependent ambiguities in long-horizon tasks. To address this, we introduce DSSP, a history-conditioned Diffusion State Space Policy that enables efficient, full-history conditioning for robot manipulation. Leveraging the continuous sequence modeling properties of State Space Models (SSMs), our history encoder effectively compresses the entire observation stream into a compact context representation. To ensure this context preserves critical information regarding future state evolution, the encoder is optimized with a dynamics-aware auxiliary training objective. This high-level context representation is then seamlessly fused with recent state observations to form a hierarchical conditioning mechanism for action generation. Furthermore, to maintain architectural consistency and minimize GPU memory overhead, we also instantiate the diffusion backbone itself using an SSM. Extensive experiments across simulation benchmarks and real-world manipulation tasks show that

DSSP achieves state-of-the-art performance with a significantly smaller model size, demonstrating superior efficiency of the hierarchical conditioning in capturing crucial information as the history length increases.

9. Gated Memory Policy: In-Context Memorization and Adaptation

Yihuai Gao, Jeff Jinyun Liu, Shuang Li, Shuran Song — arXiv 2026, arXiv 2604.18933

学习「何时回忆」（记忆门）与「回忆什么」（轻量交叉注意力），对历史动作注入扩散噪声防过拟合；MemMimic 上比长历史基线 +30.1。

Abstract. Robotic manipulation tasks exhibit varying memory requirements, ranging from Markovian tasks that require no memory to non-Markovian tasks that demand in-context memorization of historical information within a single trial or in-context adaptation based on the outcomes of multiple past trials. Surprisingly, simply extending observation histories of a visuomotor policy often leads to a significant performance drop due to distribution shift and overfitting. To address these issues, we propose Gated Memory Policy (GMP), a visuomotor policy that learns both when to recall memory and what to recall. To learn when to recall memory, GMP employs a learned memory gate mechanism that selectively activates history context only when necessary, improving robustness and reactivity. To learn what to recall efficiently, GMP introduces a lightweight cross-attention module that constructs effective latent memory representations. To further enhance robustness, GMP injects diffusion noise into historical actions, mitigating sensitivity to noisy or inaccurate histories during both training and inference. On our proposed non-Markovian benchmark MemMimic, GMP achieves a 30.1% average success rate improvement over long-history baselines, while maintaining competitive performance on Markovian tasks in RoboMimic. All code, data and in-the-wild deployment instructions are available on our project website https://gated-memory-policy.github.io/.

10. MemoAct: Atkinson-Shiffrin-Inspired Hierarchical Memory-Augmented Policy for Robotic Manipulation

Liufan Tan, Jiale Li, Gangshan Jing — arXiv 2026, arXiv 2603.18494

Atkinson-Shiffrin 三层记忆：感觉记忆过滤、无损短期记忆精确追踪状态、压缩长期记忆长期保持；附 MemoryRTBench（RoboTwin 2.0，6 任务）。

Abstract. Memory-augmented robotic policies are essential in handling memory-dependent tasks. However, existing approaches typically rely on simply extending the observation window, struggling to simultaneously achieve precise task-state tracking and robust long-horizon retention. To overcome these challenges, inspired by the Atkinson–Shiffrin memory model, we propose MemoAct, a hierarchical memory-augmented policy that leverages distinct memory tiers to tackle specific bottlenecks. Specifically, sensory memory filters immediate perceptual inputs, lossless short-term memory supports precise task-state tracking, and compressed long-term memory facilitates robust long-horizon retention. To enrich the evaluation landscape, we construct MemoryRTBench based on RoboTwin 2.0, comprising 6 manipulation tasks that systematically evaluate policy memory capabilities across three dimensions: sequential, spatial, and episodic memory. Extensive experiments across simulated and real-world scenarios demonstrate that MemoAct achieves superior performance compared to both existing Markovian baselines and history-aware policies. The project page is available at https://tlf-tlf.github.io/MemoActPage/.

11. Beyond Short-Horizon: VQ-Memory for Robust Long-Horizon Manipulation in Non-Markovian Simulation Benchmarks

Honghui Wang, Zhi Jing, Jicong Ao, Shiji Song, Xuelong Li, Gao Huang, Chenjia Bai — arXiv 2026, arXiv 2603.09513

用 VQ-VAE 把过去本体状态离散成 token 作为任务阶段线索，过滤低层噪声；附 LLM 生成的保险箱开锁基准 RuleSafe。

Abstract. The high cost of collecting real-robot data has made robotic simulation a scalable platform for both evaluation and data generation. Yet most existing benchmarks concentrate on simple manipulation tasks such as pick-and-place, failing to capture the non-Markovian characteristics of real-world tasks and the complexity of articulated object interactions. To address this limitation, we present RuleSafe, a new articulated manipulation benchmark built upon a scalable LLM-aided simulation framework. RuleSafe features safes with diverse unlocking mechanisms, such as key locks, password locks, and logic locks, which require different multi-stage reasoning and manipulation strategies. These LLM-generated rules produce non-Markovian and long-horizon tasks that require temporal modeling and memory-based reasoning. We further propose VQ-Memory, a compact and structured temporal representation that uses vector-quantized variational autoencoders (VQ-VAEs) to encode past proprioceptive states into discrete latent tokens. This representation filters low-level noise while preserving high-level task-phase context, providing lightweight yet robust temporal cues that are compatible with existing Vision-Language-Action models (VLA). Extensive experiments on state-of-the-art VLA models and diffusion policies show that VQ-Memory consistently improves long-horizon planning, enhances generalization to unseen configurations, and enables more efficient manipulation with reduced computational cost. Project page: vqmemory.github.io

12. VPWEM: Non-Markovian Visuomotor Policy with Working and Episodic Memory

Yuheng Lei, Zhixuan Liang, Hongyuan Zhang, Ping Luo — arXiv 2026, arXiv 2603.04910

滑窗工作记忆 + Transformer 上下文压缩器把窗外观测递归压成固定数量情景嵌入，每步计算近似常数；MIKASA 上比 DP/VLA 高 20%+。

Abstract. Imitation learning from human demonstrations has achieved significant success in robotic control, yet most visuomotor policies still condition on single-step observations or short-context histories, making them struggle with non-Markovian tasks that require long-term memory. Simply enlarging the context window incurs substantial computational and memory costs and encourages overfitting to spurious correlations, leading to catastrophic failures under distribution shift and violating real-time constraints in robotic systems. By contrast, humans can compress important past experiences into long-term memories and exploit them to solve tasks throughout their lifetime. In this paper, we propose VPWEM, a non-Markovian visuomotor policy equipped with working and episodic memories. VPWEM retains a sliding window of recent observation embeddings as short-term working memory, and introduces a Transformer-based contextual memory compressor that recursively converts out-of-window observations into a fixed number of episodic memory embeddings. The compressor uses self-attention over a cache of past summary embeddings and cross-attention over a cache of historical observations, and is trained jointly with the policy. We instantiate VPWEM on diffusion policies to exploit both short-term and episode-wide information for action generation with nearly constant memory and computation per step. Experiments demonstrate that VPWEM outperforms state-of-the-art baselines including diffusion policies and vision-language-action (VLA) models by more than 20\% on the memory-intensive manipulation tasks in MIKASA and achieves an average 5\% improvement on the mobile manipulation benchmark MoMaRT. Code is available at https://github.com/HarryLui98/code_vpwem.

13. Recursive Belief Vision Language Action Models

Vaidehi Bagaria, Bijo Sebastian, Nirav Kumar Patel — arXiv 2026, arXiv 2602.20659

RB-VLA：自监督世界模型目标训练紧凑信念状态，VLM 每任务只查询一次给意图；无信念 32.5%→有信念 77.5%，延迟降至 1/5。

Abstract. Vision-language-action models must enable agents to execute long-horizon tasks under partial observability. However, most existing approaches remain observation-driven, relying on short context windows or repeated queries to vision-language models (VLMs). This leads to loss of task progress, action repetition under perceptual aliasing, and high inference latency. While semantic grounding is important, long-horizon manipulation fundamentally requires persistent, action-conditioned state representations. Current VLAs lack such representations and exhibit limited temporal and physical reasoning, making them ill-suited for multi-stage control. This paper introduces RB-VLA, a belief-centric architecture trained with self-supervised world-model objectives that maintains a compact latent state encoding task-relevant history, dynamics, and object interactions. Queried once per task, the VLM provides high-level intent, while the belief tracks task progress and enables phase-aware, causally grounded control under partial observability without storing raw observations or scaling memory with time. The belief and intent jointly condition a diffusion policy for robust closed-loop execution. RB-VLA outperforms prior VLAs on long-horizon benchmarks, achieving 52.5 percent and 37.5 percent higher success rates on multi-stage pick-and-place and stacking tasks, respectively, compared to pi_0. It also reduces inference latency by up to five times relative to baselines and eliminates memory growth across timesteps observed in existing VLAs. Ablations show the belief module is the primary driver of performance, increasing success rates from 32.5 percent without belief to 77.5 percent with belief.

14. Rethinking Progression of Memory State in Robotic Manipulation: An Object-Centric Perspective

Nhat Chung, Taisei Hanyu, Toan Nguyen, Huy Le, Frederick Bumgarner, Duy Minh Ho Nguyen, Khoa Vo, Kashu Yamazaki et al. — arXiv 2025, arXiv 2511.11478

物体中心的记忆状态：时空一致的 slot 身份 + slot 状态空间模型 + 关系编码器；附 LIBERO-Mem 非马尔可夫任务套件。

Abstract. As embodied agents operate in increasingly complex environments, the ability to perceive, track, and reason about individual object instances over time becomes essential, especially in tasks requiring sequenced interactions with visually similar objects. In these non-Markovian settings, key decision cues are often hidden in object-specific histories rather than the current scene. Without persistent memory of prior interactions (what has been interacted with, where it has been, or how it has changed) visuomotor policies may fail, repeat past actions, or overlook completed ones. To surface this challenge, we introduce LIBERO-Mem, a non-Markovian task suite for stress-testing robotic manipulation under object-level partial observability. It combines short- and long-horizon object tracking with temporally sequenced subgoals, requiring reasoning beyond the current frame. However, vision-language-action (VLA) models often struggle in such settings, with token scaling quickly becoming intractable even for tasks spanning just a few hundred frames. We propose Embodied-SlotSSM, a slot-centric VLA framework built for temporal scalability. It maintains spatio-temporally consistent slot identities and leverages them through two mechanisms: (1) slot-state-space modeling for reconstructing short-term history, and (2) a relational encoder to align the input tokens with action decoding. Together, these components enable temporally grounded,

context-aware action prediction. Experiments show Embodied-SlotSSM's baseline performance on LIBERO-Mem and general tasks, offering a scalable solution for non-Markovian reasoning in object-centric robotic policies.

15. ELMUR: External Layer Memory with Update/Rewrite for Long-Horizon RL Problems

Egor Cherepanov, Alexey K. Kovalev, Aleksandr I. Panov — arXiv 2025, arXiv 2510.07151

层内外部记忆 + LRU 更新/重写，有效时程扩展到注意力窗口的 10 万倍；MIKASA-Robo 23 任务中 21 个最好。

Abstract. Real-world robotic agents must act under partial observability and long horizons, where key cues may appear long before they affect decision making. However, most modern approaches rely solely on instantaneous information, without incorporating insights from the past. Standard recurrent or transformer models struggle with retaining and leveraging long-term dependencies: context windows truncate history, while naive memory extensions fail under scale and sparsity. We propose ELMUR (External Layer Memory with Update/Rewrite), a transformer architecture with structured external memory. Each layer maintains memory embeddings, interacts with them via bidirectional cross-attention, and updates them through an Least Recently Used (LRU) memory module using replacement or convex blending. ELMUR extends effective horizons up to 100,000 times beyond the attention window and achieves a 100% success rate on a synthetic T-Maze task with corridors up to one million steps. In POPGym, it outperforms baselines on more than half of the tasks. On MIKASA-Robo sparse-reward manipulation tasks with visual observations, it nearly doubles the performance of strong baselines, achieving the best success rate on 21 out of 23 tasks and improving the aggregate success rate across all tasks by about 70% over the previous best baseline. These results demonstrate that structured, layer-local external memory offers a simple and scalable approach to decision making under partial observability. Code and project page: https://elmur-paper.github.io/.

16. MEMBOT: Memory-Based Robot in Intermittent POMDP

Youzhi Liang, Eyan Noronha — arXiv 2025, arXiv 2509.11225

解耦信念推断与策略学习：SSM/LSTM 信念编码器在观测间歇缺失下维持 latent 状态，50% 观测可用时保持 80% 性能。

Abstract. Robotic systems deployed in real-world environments often operate under conditions of partial and often intermittent observability, where sensor inputs may be noisy, occluded, or entirely unavailable due to failures or environmental constraints. Traditional reinforcement learning (RL) approaches that assume full state observability are ill-equipped for such challenges. In this work, we introduce MEMBOT, a modular memory-based architecture designed to address intermittent partial observability in robotic control tasks. MEMBOT decouples belief inference from policy learning through a two-phase training process: an offline multi-task learning pretraining stage that learns a robust task-agnostic latent belief encoder using a reconstruction losses, followed by fine-tuning of task-specific policies using behavior cloning. The belief encoder, implemented as a state-space model (SSM) and a LSTM, integrates temporal sequences of observations and actions to infer latent state representations that persist even when observations are dropped. We train and evaluate MEMBOT on 10 robotic manipulation benchmark tasks from MetaWorld and Robomimic under varying rates of observation dropout. Results show that MEMBOT consistently outperforms both memoryless and naively recurrent baselines, maintaining up to 80% of peak performance under 50% observation availability. These findings highlight the effectiveness of explicit belief modeling in achieving robust, transferable, and data-efficient policies for real-world partially observable robotic systems.

17. MTIL: Encoding Full History with Mamba for Temporal Imitation Learning

Yulin Zhou, Yuankai Lin, Fanzhe Peng, Jiahui Chen, Kaiji Huang, Hua Yang, Zhouping Yin — arXiv 2025, arXiv 2505.12410

用 Mamba 线性时间编码完整轨迹历史为不断演化的压缩状态，解决长时程感知歧义。

Abstract. Standard imitation learning (IL) methods have achieved considerable success in robotics, yet often rely on the Markov assumption, which falters in long-horizon tasks where history is crucial for resolving perceptual ambiguity. This limitation stems not only from a conceptual gap but also from a fundamental computational barrier: prevailing architectures like Transformers are often constrained by quadratic complexity, rendering the processing of long, high-dimensional observation sequences infeasible. To overcome this dual challenge, we introduce Mamba Temporal Imitation Learning (MTIL). Our approach represents a new paradigm for robotic learning, which we frame as a practical synthesis of World Model and Dynamical System concepts. By leveraging the linear-time recurrent dynamics of State Space Models (SSMs), MTIL learns an implicit, action-oriented world model that efficiently encodes the entire trajectory history into a compressed, evolving state. This allows the policy to be conditioned on a comprehensive temporal context, transcending the confines of Markovian approaches. Through extensive experiments on simulated benchmarks (ACT, Robomimic, LIBERO) and on challenging real-world tasks, MTIL demonstrates superior performance against SOTA methods like ACT and Diffusion Policy, particularly in resolving long-term temporal ambiguities. Our findings not only affirm the necessity of full temporal context but also validate MTIL as a powerful and a computationally feasible approach for learning long-horizon, non-Markovian behaviors from high-dimensional observations.

18. Learning Long-Context Diffusion Policies via Past-Token Prediction

Marcel Torne, Andy Tang, Yuejiang Liu, Chelsea Finn — arXiv 2025, arXiv 2505.09561

PTP：让扩散策略同时预测过去动作 token 以正则化对历史的依赖（copycat 的反面），多阶段训练加速 10 倍，测试时自验证。

Abstract. Reasoning over long sequences of observations and actions is essential for many robotic tasks. Yet, learning effective long-context policies from demonstrations remains challenging. As context length increases, training becomes increasingly expensive due to rising memory demands, and policy performance often degrades as a result of spurious correlations. Recent methods typically sidestep these issues by truncating context length, discarding historical information that may be critical for subsequent decisions. In this paper, we propose an alternative approach that explicitly regularizes the retention of past information. We first revisit the copycat problem in imitation learning and identify an opposite challenge in recent diffusion policies: rather than over-relying on prior actions, they often fail to capture essential dependencies between past and future actions. To address this, we introduce Past-Token Prediction (PTP), an auxiliary task in which the policy learns to predict past action tokens alongside future ones. This regularization significantly improves temporal modeling in the policy head, with minimal reliance on visual representations. Building on this observation, we further introduce a multistage training strategy: pre-train the visual encoder with short contexts, and fine-tune the policy head using cached long-context embeddings. This strategy preserves the benefits of PTP while greatly reducing memory and computational

overhead. Finally, we extend PTP into a self-verification mechanism at test time, enabling the policy to score and select candidates consistent with past actions during inference. Experiments across four real-world and six simulated tasks demonstrate that our proposed method improves the performance of long-context diffusion policies by 3x and accelerates policy training by more than 10x.

19. SAM2Act: Integrating Visual Foundation Model with A Memory Architecture for Robotic Manipulation

Haoquan Fang, Markus Grotz, Wilbert Pumacay, Yi Ru Wang, Dieter Fox, Ranjay Krishna, Jiafei Duan — arXiv 2025, arXiv 2501.18564

SAM2 式记忆库 + 编码器 + 注意力赋予多视角策略空间记忆；附 MemoryBench，记忆任务 94.3%。

Abstract. Robotic manipulation systems operating in diverse, dynamic environments must exhibit three critical abilities: multitask interaction, generalization to unseen scenarios, and spatial memory. While significant progress has been made in robotic manipulation, existing approaches often fall short in generalization to complex environmental variations and addressing memory-dependent tasks. To bridge this gap, we introduce SAM2Act, a multi-view robotic transformer-based policy that leverages multi-resolution upsampling with visual representations from large-scale foundation model. SAM2Act achieves a state-of-the-art average success rate of 86.8% across 18 tasks in the RLBench benchmark, and demonstrates robust generalization on The Colosseum benchmark, with only a 4.3% performance gap under diverse environmental perturbations. Building on this foundation, we propose SAM2Act+, a memory-based architecture inspired by SAM2, which incorporates a memory bank, an encoder, and an attention mechanism to enhance spatial memory. To address the need for evaluating memory-dependent tasks, we introduce MemoryBench, a novel benchmark designed to assess spatial memory and action recall in robotic manipulation. SAM2Act+ achieves an average success rate of 94.3% on memory-based tasks in MemoryBench, significantly outperforming existing approaches and pushing the boundaries of memory-based robotic systems. Project page: sam2act.github.io.

20. Learning Memory Mechanisms for Decision Making through Demonstrations

William Yue, Bo Liu, Peter Stone — arXiv 2024, arXiv 2411.07954

把「第 p 步事件在第 q 步被回忆」作为记忆依赖对加入演示，AttentionTuner 用它监督 Transformer 的注意力。

Abstract. In Partially Observable Markov Decision Processes, integrating an agent's history into memory poses a significant challenge for decision-making. Traditional imitation learning, relying on observation-action pairs for expert demonstrations, fails to capture the expert's memory mechanisms used in decision-making. To capture memory processes as demonstrations, we introduce the concept of memory dependency pairs MATH9 indicating that events at time MATH10 are recalled for decision-making at time MATH11. We introduce AttentionTuner to leverage memory dependency pairs in Transformers and find significant improvements across several tasks compared to standard Transformers when evaluated on Memory Gym and the Long-term Memory Benchmark. Code is available at https://github.com/WilliamYue37/AttentionTuner.

双系统、智能体与符号记忆 / Dual-System, Agentic and Symbolic Memory

高层用 VLM / LLM / 代码维护文本、图或进度指针，低层 VLA 执行。RoboMME 与 RoboMemArena 的榜首（PonderPounce、HyMeS、BATON）都在这里：计数与过程记忆目前符号状态更稳。AGM 的论断「可靠记忆靠有纪律的状态更新而非容量」与 EventVLA 的 NMS + 冷却、TRACE 的门控写入是同一原则。

1. AGM: Achievement-Grounded Memory for Closed-Loop Agents with Frozen VLA Policies

Hongbo Gao, Zeyu Ni, Xin Wen, Siyu Xu, Ruifeng Li — arXiv 2026, arXiv 2608.29537

> **arXiv 2608.29537** · arXiv 2026 · 提交 2026-08-30 · 分类 `agentic` / 双系统、智能体与符号记忆
> *Hongbo Gao, Zeyu Ni, Xin Wen, Siyu Xu, Ruifeng Li*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

AGM 给冻结的 VLA 套一个极简闭环：任务展开成 T = 2N+1 个带类型的子目标加一个进度指针，只有抓取或放置被物理证据核实（冻结 CoTracker3 的上升点迹、冻结 SigLIP 的释放前后跨视角对比）后指针才前进，否则保持或回滚。RoboMME Counting 上 PickXTimes 100.0%、BinFill 84.0%，四任务平均 55.96%，比最强记忆基线 MemER（48.83%）高 7.1 点，只训练 2.43M 参数的验证头。最该记住的是表 10：同一策略、同一子目标、同一指针，写入等证据时 PickXTimes 100.0%，按尝试写入只有 32.0%。

解决什么问题

冻结 VLA 开环执行动作块、不跟踪进度，在重复计数任务里阶段感知混叠：方块放进盒子后，当前图像无法说明这是第几次。外部记忆看似直接解法，但论文发现在相同冻结策略与子目标接口下，”每次尝试就前进”的进度记忆可能比没有进度记忆更差：抓空、滑落、放偏被当成完成写入后，瞬时失败变成持久的任务状态错误。问题从”存什么”转成”何时有资格写”。MemER 检索关键帧、SAM2Act 加空间记忆，扩展的是存储与检索，没有回答写入的正当性。

方法

- **记忆结构**：M_t = (G, p_t)。G = {(g_k, c_k)} 在一回合内固定，c_k ∈ {grasp, place-rev, place-irrev, other}，平面放置可经重抓恢复，容器放置释放后不可恢复。p_t 是唯一在线更新的状态，冻结策略只看 g_{p_t}。容量是一个整数，不存历史。
- **何时验证**：夹爪开口宽度经滞回状态机产生事件 $e_t \in \{\varnothing, G^+$ 负载抓取, R^+ 负载释放, R^0 空释放$\}$（仿真阈值：闭合进入 0.030、退出 0.035、空闭合 0.008、连续 5 帧确认）。只有 G^+ 配抓取子目标、R^+ 配放置子目标才进入验证。
- **抓取验证（零训练）**：抓取瞬间末端位置投影到前视图，撒 6×6 查询网格（半宽 22 px），缓存到末端抬升 0.03 m 为止的片段，CoTracker3 追踪；保留终帧可见度 ≥ 0.5 的轨迹，取上升位移最大的 1/3 的中位数 r^G，r^G ≥ 4 px 判成功。依据是物体是否随夹爪一起动。
- **放置验证（2.43M 头）**：R^+ 前后各聚合 K = 4 帧的前视与腕视图，冻结 SigLIP 编码，特征 $f\_n = [z\_f^-; z\_f^+; z\_f^+ - z\_f^-; z\_w^-; z\_w^+; z\_{prop}; z\_q]$（4626 维，z_q 为去掉重复序号的子目标文本），MLP 4626→512→128→2。可恢复放置阈值 κ = 0.5，不可恢复容器放置 κ = 0.05：误拒绝会多放一个物体毁掉计数，所以宁可误接受。释放后沉降 30 / 40 帧取样，训练与部署严格同一时刻，错开 20 帧 BinFill 从 86 / 82% 掉到 66%。
- **指针转移**：成功 → p+1；失败且可回滚 → 回到同一重复的抓取子目标；失败且不可回滚 → 保持。连续拒绝 2–3 次后强制前进防死锁；终止前用不依赖学习头的色相–区域重叠测试复查最后一次可恢复放置。
- **训练**：验证头用 RoboMME 训练集 788 个事件（590 成功 / 198 失败）训练，负样本主要来自”按尝试前进”的采集控制器；标签由环境谓词构造，评测时不可用。类别加权交叉熵，AdamW lr 1e-3，300 步全批。π0.5（3.35B）、SigLIP（203.2M）、CoTracker3（25.4M）全部冻结。

关键数字

基准 / 协议	对比项	数字
RoboMME Counting 官方 50 回合测试集，1300 步预算，须在恰好 N 次后按停止键；AGM 为两次独立运行均值	冻结 π0.5 / π0.5 + 过去动作 / SAM2Act+ / SimpleSG + Qwen3-VL-4B / SimpleSG + Gemini-2.5-Pro / MemER / AGM，四任务平均	28.78 / 29.09 / 35.33 / 44.60 / 39.00 / 48.83 / **55.96%**
同上，PickXTimes、BinFill	π0.5 / MemER / AGM	42.89、30.00 / 79.33、56.67 / **100.00、84.00%**
SwingXTimes、StopCube（子目标不锚定夹爪）	AGM = 冻结 π0.5	35.56、6.67%，按设计退回基策略
按重复次数 N = 1…5，仿真	π0.5：PickXTimes / BinFill；π0.5 + AGM：PickXTimes / BinFill	100 → 0 / 90 → 0；**全部 100** / 100, 95, 84.4, 62.5, 66.7
跨骨干（LoRA 子目标 π0，规则与阈值不改）	π0 单独 / π0 + AGM，PickXTimes、BinFill 均值	接近 0 / 44.0、8.0%
真机 AgileX PiPER，每个 N 10 次（每任务 50 次）	π0.5 → π0.5 + AGM：PickXTimes 各 N；BinFill 各 N	100, 20, 10, 0, 0 → **全部 100**；100, 10, 0, 0, 0 → 100, 100, 80, 70, 60
表 10：谁决定当前子目标（同一检查点）	核实后写入 / 按尝试写入 / 整条指令无指针	PickXTimes **100.0** / 32.0 / 36.0；BinFill 84.0 / 82.0 / 40.0
表 3 消融（n = 50）	完整 / 抓取验证改纯本体 / 阈值不分可恢复性	PickXTimes 100 / 98 / 98；BinFill 84 / 78 / 74
指令解析器替代基准分解	PickXTimes / BinFill	98.0 / 82.0（基准分解 100.0 / 86.0）
验证头留出 138 事件	总体准确率 / place-rev / place-irrev	91.3% / 100% / 75%，且 place-irrev 的 4 个失败样本 argmax 全部漏判
外挂参数	SimpleSG、MemER 4B LoRA VLM vs AGM	231M 冻结 + 2.43M 可训练

局限与注意

- **结论强于证据**：”纪律胜于容量”只在两个夹爪锚定任务上成立，SwingXTimes、StopCube 上 AGM 一无所加。表 10 更诚实：BinFill 上按尝试写入 82.0 几乎等于核实写入 84.0，入盒即成功，验证无事可做，该任务的 10 点增益来自阈值分开而非验证本身。
- **验证头在关键分支上不可靠**：place-irrev 留出 4 个失败全部漏判，作者坦言该分支不靠判别，只靠 $\kappa = 0.05$ 在极端置信时拒绝。BinFill 的成绩主要来自分解 + 指针 + 事件检测。
- **强先验**：子目标序列线性、有类型、可恢复性由类型静态决定，不能表达分支或乱序计划。死锁上限本身就是无验证写入，只是罕见且有界。
- **基线对等**：MemER、SimpleSG 数字来自官方评测但骨干与训练预算不同，π0.5 + 过去动作是唯一同骨干的记忆对照。真机 π0.5 / π0 是作者在 194 条示教上全量微调的，与仿真官方检查点不是同一模型。
- **代价与规模**：每次事件同步调用 CoTracker3 与 SigLIP 服务，验证延迟未报告；仿真只有两个 seed 均值、无方差；真机每格 10 次，一台 6-DoF 平台、一种物体。

与核心论文的关系

- **与 EventVLA 共享事件触发写入的哲学，存的东西相反**：EventVLA 用 KEM 头从 VLA 隐状态预测哪一帧是关键帧（阈值 0.55、NMS 加冷却、5 槽 FIFO 原始图像），写入后由 VLM 自行解读；AGM 用夹爪状态机定位事件、用冻结感知模型判定成败，写入的只是一个整数指针，策略完全冻结。EventVLA 记"看过什么"，AGM 记"做成了什么"；前者代价是吞吐 2.91 → 0.94 Hz，后者是每次事件的同步验证调用。AGM 的表 10（尝试写入 32.0 vs 核实写入 100.0）量化了 EventVLA 未测的错误写入代价。
- **与 TRACE**：TRACE 用本体轨迹的 path signature 寻址 latent 槽、每步门控写入、靠三项稳定器损失防塌缩；AGM 也从本体（夹爪宽度）出发，但只用它决定"何时看"，写入内容由视觉证据决定，零 latent。AGM 的纯本体验证消融（PickXTimes 98、BinFill 78）说明本体信号足以定位事件但不足以判定成败，与 TRACE 完全依赖本体寻址形成对照；任务不同，只作设计提示。
- **与 SAI**：SAI 的 30 步 GRU 是策略内部的隐式历史，AGM 的指针是策略外部的显式状态；两者都针对过早行动类错误（SAI 过早释放 64.5% → 16.1%，AGM 过早计数），路径一内一外。

关联阅读

- RoboMME（arXiv 2603.04639）：Counting 基准及 14 种 π0.5 记忆变体，AGM 的评测场。
- MemER（arXiv 2510.20328）：关键帧检索加 4B VLM 的最强记忆基线（48.83%）。
- CoTracker3（arXiv 2410.11831）：零训练抓取验证所用的冻结点迹跟踪器。
- Inner Monologue（arXiv 2207.05608）：把环境反馈注入语言规划闭环的先例，AGM 把反馈收窄为夹爪事件触发的验证。
- SAM2Act（arXiv 2501.18564）：视觉基础模型加空间记忆的策略，表 1 中的对照。

2. PonderPounce: A Pretrained MLLM as an Episode Context Engine for Robot Control

Suhwan Choi, Jaeyoon Jung, Sungkyung Kim, Yunsung Lee, Youngjae Yu — arXiv 2026, arXiv 2608.24115

复用 MLLM 原生因果上下文作为回合记忆（Ponder），异步只把最新认知 token 传给 VLA（Pounce）；RoboMME 60.83%（9B），p50 78 ms 刷新 / 25 ms 动作。

Abstract. Multimodal large language models (MLLMs) can integrate long visual histories, reason under partial observability, and infer behavior from a few examples. Yet vision-language-action (VLA) models generally inherit pretrained representations without using this contextual capacity as episode memory. Memory-dependent policies address this gap through purpose-built history mechanisms. PonderPounce instead reuses an MLLM's native causal context as robot memory. Ponder, a System2 MLLM, accumulates episode observations, demonstrations, and prior cognition in its native causal context and can generate subgoal text and demonstration reasoning for internal use. Pounce, a System1 VLA, receives the current observation, instruction, and proprioception directly; through the Ponder–Pounce interface, it asynchronously receives only the newest continuous cognition token and its age. Both are jointly trained end to end without a purpose-built memory module or separate bridge pretraining. Optimized serving achieves p50 latencies of 78ms for cognition refresh and 25ms for action-model invocation, supporting 20Hz action playback. On RoboMME with base-scale training data, PonderPounce reaches 60.83% with 9B and 50.04% with 0.8B under the same Pounce architecture and interface, versus 44.51% for FrameSamp+Modul and 17.93% for the current-observation π_{0.5}. With 9x data, it reaches 75.54% versus 57.88% for FrameSamp+Modul. On RoboCasa-DC, the same interface learns from action supervision alone and reaches 12.5% versus 11.6% for the strongest published demonstration-conditioned baseline, falling to 8.6% when cognition is replaced by a learned null state.

3. Don’t Drop the BATON: Long-Horizon Robot Manipulation via Agentic Subtask Exploration and Transition-aware Memory

Bingxin Xu, Yuzhang Shang, Emilio Ferrara — arXiv 2026, arXiv 2608.16889

把子任务作为探索单位使成本从 T^K 变为 T·K，配转移感知记忆（验证器管调用转移、交接恢复入口状态、前瞻选后继可继承的策略）；RoboMemArena 任务成功 +11.6。

Abstract. Long-horizon robot manipulation chains many contact-rich skills into one multi-stage task. Vision-language-action (VLA) models increasingly master the individual skills, yet the chain still fails: errors compound beyond the policy’s ability to correct, and one subtask silently constrains the next. A promising recipe freezes the VLA and puts an LLM agent in charge: it plans in language, moves in free space with analytic primitives, invokes the VLA only for contact-rich segments, and writes adaptation into language memory. Applied to long horizons, it breaks twice. (1) Competence comes from whole-task exploration at test time, whose cost is multiplicative in stages: if one stage needs T episodes, a K-stage task needs about T^K, and a failure does not reveal which stage caused it. (2) It has no representation of transitions: the VLA primitive carries an exit but no entry condition, so a subtask can succeed in a form its successor cannot use. We present BATON. Against (1), BATON makes the subtask the unit of exploration: each is explored in the cheap short-horizon regime and its solution stored in memory; a long-horizon trajectory is then composed from these solutions rather than discovered whole. Cost becomes additive (T*K) and every failure is attributed to a single stage. Against (2), BATON equips exploration with a transition-aware memory. Within a subtask, a verifier agent governs the invocation transition: the VLA is called only after the wrist view confirms the scene is ready. Across subtasks, a handoff transition restores an entry state disturbed by the predecessor’s residue, and a lookahead transition selects the strategy whose outcome the successor can inherit. No parameters are updated. On the long-horizon benchmark RoboMemArena, BATON improves task success by 11.6% and cumulative success by 14.9% over the SoTA.

4. Skills in Weights, Memory in Code: Hybrid Learning for Memory-Dependent Robot Manipulation

Yunhao Zhao, Zhenyang Ni, Haoyang Chen, Ruohan Zhang, Qi Zhu — arXiv 2026, arXiv 2608.09410

> **arXiv 2608.09410** · arXiv 2026 · 提交 2026-08-10 · 分类 `agentic` / 双系统、智能体与符号记忆
> *Yunhao Zhao, Zhenyang Ni, Haoyang Chen, Ruohan Zhang, Qi Zhu*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

HyMeS 把记忆从 VLA 权重里拿出来：π0.5 只管马尔可夫的运动技能，编码智能体（Opus 4.8）在代码空间维护符号任务状态（计划、阶段、绑定、计数器、latch），并把它翻译成可微约束注入 flow-matching 去噪过程。RoboMemArena 修正后的 12 任务协议（160 episodes）上，同一份 π0.5 权重的 TSR 从 41.3% 升到 60.1%，比基准自带的双系统 PrediMem 高 14.5 点；SO-101 真机 TSR 25.7% → 57.1%。

解决什么问题

观测别名：任务状态 z_t（开头亮过随即拿走的提示卡、已按了几次按钮）无法从当前观测恢复，马尔可夫策略只能在「再按一次」与「抬起停止」之间平均（Fig. 1）。作者批评端到端记忆 VLA（MemoryVLA、EventVLA、MemoryWAM）把离散组合的记忆逻辑学进权重，要求示教覆盖组合空间，换个目标颜色或次数就得重采数据，latent

缓冲区又不可读不可改；批评推理时引导（VLS、DynaGuide、PPGuide）只用当前观测算引导，不维护 episode 级记忆，且策略执行结果不回流。

方法

权重微调后全程冻结，四个部分：

技能在权重。 π0.5 用标准 flow-matching 目标在示教集 D 上微调（Eq. 6–7），D 只需覆盖运动技能。仿真用 RoboMemArena 训好的 π0.5 检查点（OpenPI），真机用 LeRobot 在 SO-101 示教上微调。

记忆在代码。 程序 $P = (C, V, U)$：约束选择、事件验证、记忆更新三组规则。符号状态 $s_t = (plan, \rho_t, B_t, N_t, F_t)$：计划、当前阶段、任务绑定、重复事件计数、持久交互状态。学习是 rollout 驱动的改码：第 n 次 rollout 产出符号执行轨迹 $\xi_n = ((s_t, e_t, R_t))$ 和基准给出的分阶段判定 b_n，智能体对照 PACE 记录的切换证据与 b_n 定位失败阶段，诊断是约束奖励写错还是验证规则过严/过松，然后 $P^{(n+1)} = Edit(P^{(n)}, \xi_n, b_n)$（Eq. 10），不用梯度和专家动作标签。开发集上最好的程序保留为 P^*，评测时冻结：智能体仍在 episode 开头和每次阶段切换后，依据指令、观测和 SAM + DINOv2 关键点实例化阶段计划、验证规则与约束奖励，但不再反思改码。

引导 VLA。 当前阶段选出可微约束 $R_\rho(a, o_t; s_t)$，如到达阶段 $R_{reach} = -\|ee(a) - \Gamma(target(s_t), o_t)\|^2$，Γ 用开放词汇检测 + mask 精化 + 深度反投影把符号目标落到场景关键点（沿用 VLS）。约束梯度加到速度场 $\hat{v} = v_\theta + \lambda_t \nabla R$（Eq. 12）；$\lambda_t$ 随约束残差自进入阶段以来的归一化下降量做 sigmoid 衰减（Eq. 13），记忆决定选哪个行为，接近交互区后运动策略主导接触。

验证阶段完成（PACE）。 本体感知检测动作相关运动模式并用符号 latch 保留持久事件；Qwen3-VL-8B 看当前阶段描述和最近至多 5 帧 RGB 判断完成。每次查询取或 $d_r = e_{prop} \vee e_{vlm}$，最近 $w = 5$ 次里至少 $k = 3$ 次为真才触发阶段事件，切换后投票清零；事件接受后 U 更新绑定、计数器和持久状态，再激活下一阶段约束。不用特权环境状态，不训任务专用完成判别器。

关键数字

修正协议 12 任务、160 episodes：transferring 与 counting 每任务 10 个，occlusion 每任务 15 个，sequence 每任务 20 个；PrediMem 在同一协议上重测。CSR 为子目标平均完成率，TSR 为整任务成功率。

设置	指标	π0.5	PrediMem	HyMeS
Overall（160 ep）	CSR / TSR	52.5 / 41.3	61.7 / 45.6	**66.2 / 60.1**
Transferring（3 任务）	CSR / TSR	70.0 / 66.7	86.7 / 76.7	**93.3 / 93.3**
Counting（3 任务）	CSR / TSR	39.2 / 20.0	**72.2** / 36.7	60.3 / **50.0**
Sequence（2 任务）	CSR / TSR	52.5 / 40.0	70.0 / 50.0	**73.8 / 62.5**
Occlusion（4 任务）	CSR / TSR	50.4 / 40.0	38.3 / 31.7	**50.6 / 47.0**

SO-101 真机（每方法 35 次，同一份示教与权重）：Observe-and-pick-up 2/10 → 7/10，Number-Guided Button Pressing 7/15 → 11/15，Put-Back Block 0/10 → 2/10；总 TSR 25.7% → 57.1%。

消融（六任务子集、跨四类、同一权重与 episode）：

变体	程序	PACE 证据	CSR	TSR
One-shot P^(0)	无 rollout 学习	视觉 + 本体	63.3	53.3
Vision-only PACE	学习	视觉	55.8	50.0

变体	程序	PACE 证据	CSR	TSR
Proprio-only PACE	学习	本体	63.0	63.3
Full HyMeS	学习	视觉 + 本体	**71.8**	**71.7**

rollout 学习带来 +8.5 CSR / +18.4 TSR；只用视觉验证掉 16.0 / 21.7，只用本体掉 8.8 / 8.4。论文未报告推理延迟、每 episode 的智能体调用次数和开发阶段用了多少 rollout。

局限与注意

作者自陈：只处理能压成紧凑符号状态的历史；成功率仍受运动能力和阶段验证限制，Fig. 4 里记忆与约束都对，失败出在预抓取偏移、按钮抖动多按一次、按太浅没被 PACE 认出；周期性 VLM 查询和智能体调用增加延迟与成本。

再补四条。评测协议由作者筛选，只留「π0.5 有技能但缺历史」的 12 个任务，正是符号记忆最占便宜的子集，PrediMem 的劣势要打折看。开发阶段改码依赖基准返回的分阶段判定 b_n，这是仿真才有的特权反馈，真机上 P 如何迭代没有交代。counting 类 CSR 60.3 低于 PrediMem 的 72.2，作者归因于早期阶段的引导干扰，说明梯度引导会伤运动策略；occlusion 类 CSR 仅与 π0.5 持平（50.6 vs 50.4）。整套流程要求策略能在去噪循环里加梯度、有深度做反投影、有闭源大模型做智能体，迁移到 ACT 类确定性头并不直接。

与核心论文的关系

HyMeS 正是 EventVLA 批评的那种双系统，作者也承认延迟与成本。EventVLA 把稀疏证据记忆做进权重（关键帧证据头 + 5 槽 FIFO 原始帧）端到端训练；HyMeS 的反驳是端到端让示教预算随历史配置数增长，它只随可复用技能数增长。两者协议不同、数字不可比，但可按信息类型划界：多个物体的颜色、位置这类高熵细节符号状态表达不了；计数、阶段、对象-位置绑定则是符号状态的主场——HyMeS 对 π0.5 的 TSR 增益按 counting +30.0、transferring +26.6、sequence +22.5 排序，occlusion 只有 +7.0。防错误写入上两家一致：EventVLA 用 NMS + 冷却，HyMeS 用 3-of-5 投票 + 切换清零 + 符号 latch。

TRACE 的记忆是固定 K 槽 latent，用路径签名寻址、门控写入，作为适配器插进 ACT/Diffusion Policy，从示教里学，分支配置必须出现在示教里。HyMeS 的记忆是每步可打印的代码状态，失败能归因到记忆更新、运动执行或事件验证之一，latent 槽做不到。代价是 TRACE 以策略频率运行、不需要大模型，HyMeS 需要 Opus 4.8、Qwen3-VL-8B、SAM/DINOv2、深度反投影，还要求策略可微引导。「技能在权重、记忆在代码」这一立场目前赢在计数与过程记忆，输在延迟、遮挡下的视觉验证和对特权阶段反馈的依赖。

关联阅读

- RoboMemArena / PrediMem（2605.10921）：评测基准与主要对照，PrediMem 是基准自带的 VLM 规划器 + 关键帧记忆双系统。
- VLS（2602.03973）：HyMeS 沿用的引导接口（VLM 合成阶段奖励、梯度注入去噪），HyMeS 把奖励改为由符号记忆条件化。
- Harness VLA（2607.08448）：最接近的先行者，用失败规则把冻结 VLA 编排成可重试原语，记忆停在原语粒度。
- AGM（2608.29537）：同为冻结 VLA 上的进度记忆，只在物理证据验证后推进指针，用 2.43M 参数验证头替代大模型，是 PACE 的轻量替代。
- MemoryVLA（2508.19236）：HyMeS 点名批评的端到端感知-认知记忆路线代表。

5. OnEvoMemory: Evolving Memory through Online Robot Rollouts for Pretrained Robot Policies

Zhongxi Chen, Shenqi Zong — arXiv 2026, arXiv 2608.08749

价值引导的记忆模块：从离线演示初始化、由在线 rollout 成败学习该保留哪些经验与转移。

Abstract. Long-horizon robot manipulation requires policies to track completed subtasks and critical interaction events. However, existing memory mechanisms heavily rely on external models or predefined update rules. To address this, we propose OnEvoMemory, a value-guided memory module for pretrained robot policies. It maintains recent context, high-value experiences, and salient transitions, while learning which experiences should be retained from trajectory outcomes. Offline demonstrations initialize the memory prior, whereas successful and unsuccessful online rollouts refine memory selection, helping the policy recognize task-stage transitions and avoid repeating completed subtasks. Experiments on long-horizon manipulation benchmarks show that OnEvoMemory improves the performance of the base VLA policy through both offline initialization and online memory evolution.

6. SkillMemo: Expert-guided Skill Memory Framework for Compositional Embodied Manipulation

Changyuan Wang, Chubin Zhang, Zhenyu Wu, Runhao Li, Angyuan Ma, Ke Chao, Yinan Liang, Xiuwei Xu et al. — arXiv 2026, arXiv 2608.05970

MoE 门控隐式切分轨迹为原子技能，技能级情景记忆以键值对检索并融合进当前门控分布，提升组合泛化。

Abstract. Embodied visuomotor models, including Diffusion Policy (DP) and Vision-Language-Action (VLA) models, have demonstrated promising performance on robotic manipulation benchmarks. However, their potential remains fundamentally constrained by the scarcity of large-scale embodied trajectory datasets, leading to insufficient compositional generalization in out-of-distribution (OOD) scenarios with limited capability to capture reusable skill structures. To address this limitation, we propose Skill-Based Memory (SkillMemo) framework that implicitly decomposes long-horizon demonstrations into latent atomic skills and integrates skill-level features into a dynamic episodic memory bank for solving compositional tasks. Specifically, we first introduce an expert-guided trajectory segmentation module built upon a Mixture-of-Experts (MoE) architecture, which implicitly partitions trajectories into distinct skill primitives represented by learned gating coefficients. We further design a skill-level episodic memory architecture that stores compact skill representations as retrievable key-value pairs. During inference, the memory bank retrieves the most relevant skill primitives which are subsequently fused with the model's current gating distribution, providing a robust contextual prior to refine action predictions. Extensive experiments on the simulation benchmark and real-world manipulation tasks demonstrate that SkillMemo consistently enhances both DP and VLA backbones, achieving state-of-the-art performance and outperforming MATH12, while exhibiting strong compositional generalization to unseen task configurations.

7. Explicit Language Memory for Long-Horizon Planning in Vision-Language-Action Models

Houze Xu, Jizhong Li, Ziyi Ye — arXiv 2026, arXiv 2608.04765

高层 VLM 把离散观测递归写成带时间逻辑的文本记忆并更新子任务指令，低层 VLA 执行；可解释的语义决策记录。

Abstract. Vision-language-action (VLA) models provide a unified paradigm for connecting visual perception, language understanding, and robotic control. However, existing VLA models still face major challenges in long-horizon tasks: sparse expert demonstrations constrain cross-task compositional generalization; the non-Markovian nature of long-horizon tasks makes it difficult for policies conditioned only on current observations to maintain temporal consistency; limited closed-loop error correction allows execution errors to accumulate; and end-to-end action fine-tuning may weaken the high-level semantic representations of vision-language model (VLM) backbones.

To address these issues, we propose a hierarchical long-horizon VLA architecture with an explicit language-memory module. The central idea is to convert discrete temporal observations into a coherent textual memory sequence with temporal logic. The system is decoupled into a high-level VLM and a low-level VLA: the high-level VLM performs semantic reasoning through a visual question answering training paradigm, while the low-level VLA executes precise continuous control conditioned on subtask instructions and visual observations. The high-level VLM recursively updates both language memory and subtask instructions using the previous memory as a contextual anchor, enabling persistent temporal tracking and dynamic correction during long-horizon execution. We evaluate the proposed method in multiple simulation environments and conduct sim-to-real experiments on a real robotic platform. The results demonstrate that explicit language memory improves the success rate and robustness of VLA models on complex long-horizon tasks while providing an interpretable semantic account of the decision process.

8. ChainVLA: Chaining Vision-Language-Action Queries through a Unified Execution State for Long-Horizon Manipulation

Yuzhi Huang, Weijue Bu, Ziyi Xiong, Jie Wu, Fanding Huang, Jingyan Jiang, Zhi Wang — arXiv 2026, arXiv 2608.02326

跨查询链接的执行状态：循环工作状态 + 稀疏事件记忆（Progress Context）与上一预测的未执行尾部（Motion Tail）；RMBench 62.8%，去掉两者分别掉到 3.0 / 11.2。

Abstract. Humans perform long-horizon manipulation by retaining knowledge of what earlier actions have established while continuously adapting the motion underway. By contrast, action-chunked vision-language-action (VLA) policies repeatedly replan from the current input at each query. Existing methods preserve either long-term task evidence through memory or short-term motion through action reuse and ensembling, leaving the cross-query handoff incomplete. We introduce ChainVLA, a 1.2B-parameter VLA policy that chains successive queries through a joint and revisable execution state. Progress Context combines a recurrent Working State with sparse event memory to carry observation-derived task progress, while Motion Tail feeds the preceding prediction's unexecuted continuation into state construction and action generation. Together, the two components condition a decoder that regenerates each action horizon under the latest observation, allowing the carried state to guide the next prediction without fixing it. ChainVLA reaches 62.8% average success on RMBench and 98.8% across four LIBERO suites, while removing Motion Tail or Progress Context reduces RMBench success to 11.2% and 3.0%, respectively. These asymmetric ablations are consistent with motion continuity helping preserve the observation stream from which task progress is inferred.

9. Harness VLA: Steering Frozen VLAs into Reliable Manipulation Primitives via Memory-Guided Agents

Yixian Zhang, Huanming Zhang, Feng Gao, Xiao Li, Zhihao Liu, Chunyang Zhu, Jiaxing Qiu, Yuchen Yan et al. — arXiv 2026, arXiv 2607.08448

把冻结 VLA 当作可重试的接触密集原语，与少量解析原语由记忆引导的智能体组合；从执行轨迹学原语的适用范围。

Abstract. Language-conditioned manipulation requires both precise contact-rich control and robust reasoning over language, scenes, and long horizons. End-to-end Vision-Language-Action (VLA) models provide strong local visuomotor skills, but they are trained on in-distribution task trajectories and often fail under deployment perturbations such as semantic retargeting, goal re-binding, spatial-layout shifts, and unstable local contacts. LLM

coding agents provide complementary semantic and compositional reasoning, but purely analytic primitives struggle with irregular grasping, constrained placement, and articulated-object interaction. We present Harness VLA, a memory-augmented agentic framework that exposes a frozen VLA as a retryable contact-rich primitive and composes it with a small fixed library of analytic primitives for grounding, staging, transport, navigation, and release. Rather than expanding the skill library, the harness learns the operating range of these fixed primitives from task-specific execution traces, global success rules, and failure models. By lifting semantic re-grounding, non-contact execution, and VLA re-staging to the planner while reserving the frozen VLA for local contact-rich phases, Harness VLA extends pretrained VLAs beyond their original trajectory distribution without finetuning. Across perturbed tabletop, household kitchen, and clean-to-randomized bimanual manipulation, Harness VLA improves over the strongest relevant baselines by 38.6 and 25.4 percentage points on LIBERO-Pro and RoboCasa365, respectively, and reaches 58.4% on RoboTwin C2R. Code is available at https://github.com/RLinf/RPent.

10. GeneralVLA-2: Geometry-Aware Reconstruction and Governed Memory for Robot Planning

Haoyu Wang, Guoqing Ma, Zeyu Zhang, Yandong Guo, Boxin Shi, Hao Tang — arXiv 2026, arXiv 2606.17480

几何感知重建 + 受治理的长期记忆（质量/置信/生命周期/冲突元数据与精度导向检索）；记忆治理在代码基准上评测。

Abstract. Generalist vision-language-action systems need object-centric 3D evidence and reusable manipulation experience to plan reliable robot trajectories. GeneralVLA provides a hierarchical interface for converting language and RGB-D observations into 3D end-effector paths, but two bottlenecks remain. First, monocular SAM3D-style object reconstruction can hallucinate pose and unseen geometry, while manipulation benefits from stable object shape when calibrated multi-view observations are available. Second, the original KnowledgeBank mainly retrieves semantically similar snippets and appends new knowledge, which makes it difficult to control memory quality, conflicts, confidence, and geometric relevance. To address the first challenge, we introduce GeoFuse-MV3D, a geometry-prior-guided MV-SAM3D reconstruction branch that verifies external geometry cues with input-view masks, applies soft visual-hull support, performs axis-wise refinement, and fuses only geometry while preserving appearance. To address the second challenge, we upgrade KnowledgeBank into a governed long-term memory system with explicit quality, confidence, lifecycle, verifier, and conflict metadata, together with precision-oriented retrieval. Finally, we evaluate the reconstruction branch on GSO-30 and the memory module on Terminal-Bench 2.0 and SWE-Bench Verified; GeoFuse-MV3D improves over the MV-SAM3D baseline by reducing CD and LPIPS by 2.20% and 2.02% while increasing PSNR and SSIM by 2.36% and 1.03%, and KnowledgeBank improves over ReasoningBank by 4.53% on Terminal-Bench SR and 3.73% on SWE-Bench resolve rate, while reducing AS by 4.95% and 5.65%, respectively. Code: https://github.com/AIGeeksGroup/GeneralVLA-2. Website: https://aigeeksgroup.github.io/GeneralVLA-2.

11. CodeGraphVLP: Code-as-Planner Meets Semantic-Graph State for Non-Markovian Vision-Language-Action Models

Khoa Vo, Sieu Tran, Taisei Hanyu, Yuki Ikebe, Duy Nguyen, Nghi D. Q. Bui, Minh Vu, Anthony Gunderman et al. — arXiv 2026, arXiv 2604.22238

持久语义图状态 + 代码规划器 + 进度引导的视觉-语言提示，构造抑制杂乱的观测让 VLA 聚焦关键证据；比 VLM-in-the-loop 规划延迟更低。

Abstract. Vision-Language-Action (VLA) models promise generalist robot manipulation, but are typically trained and deployed as short-horizon policies that assume the latest observation is sufficient for action reasoning. This assumption breaks in non-Markovian long-horizon tasks, where task-relevant evidence can be occluded or appear only earlier in the trajectory, and where clutter and distractors make fine-grained visual grounding brittle. We present CodeGraphVLP, a hierarchical framework that enables reliable long-horizon manipulation by combining a persistent semantic-graph state with an executable code-based planner and progress-guided visual-language prompting. The semantic-graph maintains task-relevant entities and relations under partial observability. The synthesized planner executes over this semantic-graph to perform efficient progress checks and outputs a subtask instruction together with subtask-relevant objects. We use these outputs to construct clutter-suppressed observations that focus the VLA executor on critical evidence. On real-world non-Markovian tasks, CodeGraphVLP improves task completion over strong VLA baselines and history-enabled variants while substantially lowering planning latency compared to VLM-in-the-loop planning. We also conduct extensive ablation studies to confirm the contributions of each component.

12. HELM: Harness-Enhanced Long-horizon Memory for Vision-Language-Action Manipulation

Zijian Zeng, Fei Ding, Huiming Yang, Xianwei Li — arXiv 2026, arXiv 2604.18791

记忆缺口/验证缺口/恢复缺口三分法：CLIP 索引关键帧的情景记忆 + 学习的状态验证器 + 回滚重规划控制器；LIBERO-LONG 58.4→81.5，而加长上下文只 +5.4。

Abstract. Vision-Language-Action (VLA) models fail systematically on long-horizon manipulation tasks despite strong short-horizon performance. We show that this failure is not resolved by extending context length alone in the current reactive execution setting; instead, it stems from three recurring execution-loop deficiencies: the memory gap, the verification gap, and the recovery gap. We present HELM, a model-agnostic framework that addresses these deficiencies with three components: an Episodic Memory Module (EMM) that retrieves key task history via CLIP-indexed keyframes, a learned State Verifier (SV) that predicts action failure before execution from observation, action, subgoal, and memory-conditioned context, and a Harness Controller (HC) that performs rollback and replanning. The SV is the core learning contribution: it consistently outperforms rule-based feasibility checks and ensemble uncertainty baselines, and its effectiveness depends critically on access to episodic memory. On LIBERO-LONG, HELM improves task success rate by 23.1 percentage points over OpenVLA (58.4% to 81.5%), while extending the context window to H=32 yields only a 5.4-point gain and same-budget LoRA adaptation remains 12.2 points below HELM. HELM also improves long-horizon performance on CALVIN and substantially boosts recovery success under controlled perturbations. Ablations and mechanism analyses isolate the contribution of each component, and we release LIBERO-Recovery as a perturbation-injection protocol for evaluating failure recovery in long-horizon manipulation.

13. MEM: Multi-Scale Embodied Memory for Vision Language Action Models

Marcel Torne, Karl Pertsch, Homer Walke, Kyle Vedder, Suraj Nair, Brian Ichter, Allen Z. Ren, Haohuan Wang et al. — arXiv 2026, arXiv 2603.03596

> **arXiv 2603.03596** · arXiv 2026 · 提交 2026-03-04 · 分类 `agentic` / 双系统、智能体与符号记忆
> *Marcel Torne, Karl Pertsch, Homer Walke, Kyle Vedder, Suraj Nair, Brian Ichter, Allen Z. Ren, Haohuan Wang, Jiaming Tang, Kyle Stachowicz, Karan Dhabalia, Michael Equi, Quan Vuong, Jost Tobias Springenberg, Sergey Levine, Chelsea Finn,*

> *Danny Driess*
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一句话定位

MEM（π0.6-MEM）用两种模态分担两种时间尺度：视频编码器把最近几秒到约一分钟的稠密观测压成与单帧同样多的 token 给低层策略，高层策略用自然语言摘要 m_t 记录并压缩已完成的语义事件，支撑整理厨房这类长达 15 分钟的任务。论文没有结果表格，最可引用的硬数字是两条上下文自适应增益：捡筷子 +11 个百分点、开冰箱 +62 个百分点（每策略每任务 10 次 rollout）。

解决什么问题

十几分钟的任务无法把全部历史观测塞进上下文，而单一模态的压缩各有代价：纯本体感受、点迹或纯语言丢掉纠正滑落抓取所需的空间细节；关键帧为了推理可行必须激进稀疏化，丢掉估计动力学所需的密集采样；多帧平均池化在需要记住多个杯子位置或袋里还剩几件时失效。另一个被点名的失败是朴素语言记忆：直接拼接过去所有子任务指令，训练时每条指令只出现一次，推理时反复失败会产生"pick up bowl"连续三次的序列，形成训练–推理分布偏移。

方法

- **分解**：π(a_{t:t+H}, l_{t+1}, m_{t+1} | o_{t–T:t}, m_t, g) ≈ π_LL(a | o_{t–K:t}, l_{t+1}, g) · π_HL(l_{t+1}, m_{t+1} | o_t, m_t, g)，K ≪ T。高层同时输出子任务指令 l 和更新后的语言记忆 m，这是与以往高低层分解的区别。
- **语言长时记忆**：m_t 是自然语言摘要。训练标签由离线 LLM 生成：输入逐子任务标注及成功/失败标志，要求只保留对后续执行仍有用的信息并压缩（"三只碗放进右上柜"而非逐一描述颜色）。失败尝试不写入，子任务真正成功后才更新。
- **视频短时记忆**：把 VLM 自带的 ViT 改成视频编码器。每 4 层加一次同一 patch 跨时间的因果注意力（时空可分离，复杂度 $O(Kn^2 + nK^2)$ 而非 $O(n^2K^2)$），加固定正弦时间位置编码且 e(0) = 0，不引入新参数，K = 1 时与原 ViT 一致；上层丢掉过去时刻的 token，只把当前帧表示送入骨干，token 数与无记忆 VLA 相同。过去本体状态经线性投影成 K 个连续 token。
- **容量与写入**：预训练 6 帧（5 帧过去 + 当前），步长 1 s；后训练可扩到 18 帧 / 54 s。视频记忆每步滑动写入，语言记忆由高层策略每次调用时决定是否改写。
- **骨干与训练**：π0.6，Gemma3-4B VLM（SigLIP 400M + Gemma 4B）加 860M 流匹配动作专家，动作专家梯度不回流 VLM；448×448，最多 4 路相机。预训练混合遥操作示教、策略 rollout、人类纠正、视觉语言与视频字幕数据；真机推理用 RTC。监督只有低层动作损失加高层文本预测，无记忆专用损失。

关键数字

设置 / 协议	对比项	数字
评测协议	每策略每任务（或每份食谱）10 次 rollout，均值 ± 标准误；结果全部以柱状图给出	未列精确数值
长程任务：Recipe Setup（42 份食谱训练，5 份在未见厨房与未见物体上评测，每份 6–7 分）、Clean Kitchen（平均 8 个子任务，按进度计分）	π0.6 无记忆 / 仅视频 / 仅文本 / 朴素拼接文本 + 视频 / MEM	MEM 最高，去掉任一模态明显下降，朴素拼接显著差于模型摘要；数值未给

设置 / 协议	对比项	数字
上下文自适应（人类纠正数据微调）	捡筷子（桌面高度 OOD）/ 开冰箱（≤ 4 次抓取算成功）	有记忆比无记忆 +11 / +62 个百分点
核心记忆能力 6 任务（Swap 3 Mugs、Find Object、Unpack Groceries、Scoop Coffee、Grilled Cheese、Window Cleaning）	π0.6 无记忆 / Pool Memory / Proprio Memory / MEM（均在 π0.6 上重实现，MEM 去掉语言记忆）	无记忆在 Find Object 25%（四抽屉随机）、Scoop Coffee 50%；Pool 在多杯位置与剩余计数上差；Proprio 只在记自身状态时有效；MEM 唯一全面强
预训练是否含记忆	仅后训练加视频编码器 vs 完整 MEM	仅后训练明显更低，但仍优于 Pool Memory
无记忆任务 7 项（Table Bussing、Shirt Folding、Box Building 等）	π0.6 vs MEM	持平，无退化
延迟（4 路相机、单 H100）	逐帧编码 vs 视频编码器，实时阈值 300 ms	逐帧随帧数增长越过阈值，视频编码器 16 帧内保持在阈值下

局限与注意

- **几乎没有可引用的数字**：主结果全是柱状图，没有表格和每任务成功率，每格只有 10 次 rollout。作为基线只能被复现而无法被引用，EventVLA 的真机对比正是这么做的。
- **基线对等性**：Pool Memory 与 Proprio Memory 是作者在 π0.6 上重实现的代理，不是原系统；没有与任何外部记忆 VLA 的直接对比；语言记忆消融只在两个长程任务上。
- **数据与算力壁垒**：效果被归因于大规模多样预训练，仅后训练加记忆”明显更差”，复现门槛极高；语言记忆标签依赖离线 LLM 与逐子任务标注，成本未报告。
- **未评估**：没有记忆长度与性能的曲线；没有摘要错误率，而漏记一个已完成步骤会直接改写任务状态；高层调用频率与整体延迟未量化；”15 分钟”是任务时长，不是被测量的记忆窗口。

与核心论文的关系

- **EventVLA 的真机基线**：EventVLA 在 π0.5 骨干上复现了 π_MEM 作对照，四个真机任务成功率 50 / – / 30 / 40%，EventVLA 为 90 / 60 / 90 / 75%。那是 EventVLA 作者的复现而非本文的 π0.6 系统，本文也没有可对照的数字。
- **对比 EventVLA**：MEM 短时层存稠密视频（每步滑动写入、不做选择、由 ViT 时间注意力隐式压缩、容量 6–18 帧），EventVLA 存稀疏原始关键帧（KEM 头预测事件触发写入、5 槽 FIFO、直接喂视觉编码器）。两者都用原始图像而非 latent 作载体，但 EventVLA 在”何时写”上显式，MEM 在”如何压”上显式。MEM 长时层是文本，与 EventVLA、TRACE 都不同，是双系统与符号记忆家族的代表。
- **与 TRACE**：TRACE 用 path signature 从本体轨迹寻址固定 K 槽 latent；MEM 的 Proprio Memory 基线正是”只用本体历史”的对照，结论是它只在记自身状态时有效、记环境状态时失效。两者任务集不同不能直接推论，但提示了本体寻址记忆的边界。
- **与 SAI**：SAI 的 30 步 GRU 历史 token 是最轻的稠密短时记忆；MEM 的视频编码器是其重量级版本，多了跨时间注意力与大规模预训练。

关联阅读

- ContextVLA（arXiv 2510.04246）：多帧平均池化成单 token，MEM 的 Pool Memory 基线原型。
- MemER（arXiv 2510.20328）：关键帧检索加语言指令接口，另一条稀疏化路线。

- OneTwoVLA（arXiv 2505.11917）：以自然语言表示记忆的先例，MEM 语言记忆的近亲。
- Past-Token Prediction（arXiv 2505.09561）：同一作者群针对长上下文扩散策略因果混淆的辅助目标，MEM 声称靠数据多样性绕过。
- TA-VLA（arXiv 2509.07962）：纯本体感受历史，MEM 的 Proprio Memory 基线原型。

14. Notes-to-Self: Scratchpad Augmented VLAs for Memory Dependent Manipulation Tasks

Sanjay Haresh, Daniel Dijkman, Apratim Bhattacharyya, Roland Memisevic — arXiv 2026, arXiv 2602.21013

语言草稿板给 VLA 空间与时间记忆：记物体位置、追踪计划与子目标进度；在 ClevrSkills / MemoryBench 上提升泛化。

Abstract. Many dexterous manipulation tasks are non-markovian in nature, yet little attention has been paid to this fact in the recent upsurge of the vision-language-action (VLA) paradigm. Although they are successful in bringing internet-scale semantic understanding to robotics, existing VLAs are primarily “stateless” and struggle with memory-dependent long horizon tasks. In this work, we explore a way to impart both spatial and temporal memory to a VLA by incorporating a language scratchpad. The scratchpad makes it possible to memorize task-specific information, such as object positions, and it allows the model to keep track of a plan and progress towards subgoals within that plan. We evaluate this approach on a split of memory-dependent tasks from the ClevrSkills environment, on MemoryBench, as well as on a challenging real-world pick-and-place task. We show that incorporating a language scratchpad significantly improves generalization on these tasks for both non-recurrent and recurrent models.

15. Action-Sketcher: From Reasoning to Action via Visual Sketches for Long-Horizon Robotic Manipulation

Huajie Tan, Peterson Co, Yijie Xu, Shanyu Rong, Yuheng Ji, Cheng Chi, Xiansheng Chen, Qiongyu Zhang et al. — arXiv 2026, arXiv 2601.01618

See-Think-Sketch-Act 循环：在当前视图上画点/框/箭头的视觉草图外化空间意图；EventVLA 把它归入双系统路线。

Abstract. Long-horizon robotic manipulation is increasingly important for real-world deployment, requiring spatial disambiguation in complex layouts and temporal resilience under dynamic interaction. However, existing end-to-end and hierarchical Vision-Language-Action (VLA) policies often rely on text-only cues while keeping plan intent latent, which undermines referential grounding in cluttered or underspecified scenes, impedes effective task decomposition of long-horizon goals with close-loop interaction, and limits causal explanation by obscuring the rationale behind action choices. To address these issues, we first introduce Visual Sketch, an implausible visual intermediate that renders points, boxes, arrows, and typed relations in the robot’s current views to externalize spatial intent, connect language to scene geometry. Building on Visual Sketch, we present Action-Sketcher, a VLA framework that operates in a cyclic See-Think-Sketch-Act workflow coordinated by adaptive token-gated strategy for reasoning triggers, sketch revision, and action issuance, thereby supporting reactive corrections and human interaction while preserving real-time action prediction. To enable scalable training and evaluation, we curate diverse corpus with interleaved images, text, Visual Sketch supervision, and action sequences, and train Action-Sketcher with a multi-stage curriculum recipe that combines interleaved sequence alignment for modality unification, language-to-sketch consistency for precise linguistic grounding, and imitation learning augmented with sketch-to-action reinforcement for robustness. Extensive experiments on cluttered scenes and multi-object tasks, in simulation and on real-world tasks, show improved long-horizon success, stronger robustness to dynamic scene

changes, and enhanced interpretability via editable sketches and step-wise plans. Project website: https://action-sketcher.github.io

16. EchoVLA: Robotic Vision-Language-Action Model with Synergistic Declarative Memory for Mobile Manipulation

Min Lin, Xiwen Liang, Bingqian Lin, Jingzhi Liu, Zijian Jiao, Kehan Li, Ziang Yan, Yu Sun et al. — arXiv 2025, arXiv 2511.18112

场景记忆（空间-语义地图）+ 情景记忆（任务经验）的协同陈述性记忆，指导移动操作的底盘-手臂扩散策略；附 MoMani 基准。

Abstract. Recent progress in Vision-Language-Action (VLA) models has enabled embodied agents to interpret multimodal instructions and perform complex tasks. However, existing VLAs are mostly confined to short-horizon, table-top manipulation, lacking the memory and reasoning capability required for mobile manipulation, where agents must coordinate navigation and manipulation under changing spatial contexts. In this work, we present EchoVLA, a memory-aware VLA model for mobile manipulation. EchoVLA incorporates a synergistic declarative memory inspired by the human brain, consisting of a scene memory that maintains a collection of spatial-semantic maps and an episodic memory that stores task-level experiences with multimodal contextual features. The two memories are individually stored, updated, and retrieved based on current observations, task history, and instructions, and their retrieved representations are fused via coarse- and fine-grained attention to guide base-arm diffusion policies. To support large-scale training, we further introduce MoMani, an automated benchmark that generates expert-level trajectories through multimodal large language model (MLLM)-guided planning and feedback-driven refinement, supplemented with real-robot demonstrations. Comprehensive simulated and real-world results demonstrate that EchoVLA substantially improves overall performance, e.g., it achieves the highest success rates of 0.52 on manipulation/navigation tasks and 0.31 on mobile manipulation tasks in simulation, exceeding the strong baseline MATH13 by +0.20 and +0.11, respectively.

17. MAP-VLA: Memory-Augmented Prompting for Vision-Language-Action Model in Robotic Manipulation

Runhao Li, Wenkai Guo, Zhenyu Wu, Changyuan Wang, Haoyuan Deng, Zhenyu Weng, Yap-Peng Tan, Ziwei Wang — arXiv 2025, arXiv 2511.09516

从演示构建阶段级软提示记忆库，执行时按轨迹相似度检索并注入冻结 VLA；TRACE 的「检索提示记忆」对照即来源于此。

Abstract. Pre-trained Vision-Language-Action (VLA) models have achieved remarkable success in improving robustness and generalization for end-to-end robotic manipulation. However, these models struggle with long-horizon tasks due to their lack of memory and reliance solely on immediate sensory inputs. To address this limitation, we propose Memory-Augmented Prompting for Vision-Language-Action model (MAP-VLA), a novel framework that empowers pre-trained VLA models with demonstration-derived memory prompts to augment action generation for long-horizon robotic manipulation tasks. To achieve this, MAP-VLA first constructs a memory library from historical demonstrations, where each memory unit captures information about a specific stage of a task. These memory units are implemented as learnable soft prompts optimized through prompt tuning. Then, during real-time task execution, MAP-VLA retrieves relevant memory through trajectory similarity matching and dynamically integrates it into the VLA model for augmented action generation. Importantly, this prompt tuning and

retrieval augmentation approach operates as a plug-and-play module for a frozen VLA model, offering a lightweight and flexible solution to improve task performance. Experimental results show that MAP-VLA delivers up to 7.0% absolute performance gains in the simulation benchmark and 25.0% on real robot evaluations for long-horizon tasks, surpassing the current state-of-the-art methods.

18. ExpReS-VLA: Specializing Vision-Language-Action Models Through Experience Replay and Retrieval

Shahram Najam Syed, Yatharth Ahuja, Arthur Jakobsson, Jeff Ichnowski — arXiv 2025, arXiv 2511.06202

压缩经验回放 + 检索增强在设备上快速专精 VLA，存储省 97%，12 条演示 31 秒适应；偏「经验记忆」而非回合内记忆。

Abstract. Vision-Language-Action (VLA) models like OpenVLA demonstrate impressive zero-shot generalization across robotic manipulation tasks but struggle to adapt to specific deployment environments where consistent high performance on a limited set of tasks is more valuable than broad generalization. We present EXPierence replayed, REtrieval augmented, Specialized VLA (ExpReS-VLA), a method that enables rapid on-device adaptation of pre-trained VLAs to target domains while preventing catastrophic forgetting through compressed experience replay and retrieval-augmented generation. Our approach maintains a memory-efficient buffer by storing extracted embeddings from OpenVLA's frozen vision backbone, reducing storage requirements by 97% compared to raw image-action pairs. During deployment, ExpReS-VLA retrieves the MATH14 most similar past experiences using cosine similarity to augment training batches, while a prioritized experience replay buffer preserves recently successful trajectories. To leverage failed attempts, we introduce Thresholded Hybrid Contrastive Loss (THCL), enabling the model to learn from both successful and unsuccessful demonstrations. Experiments on the LIBERO benchmark show improvements from 82.6% to 93.1% on spatial reasoning and 61% to 72.3% on long-horizon tasks over base OpenVLA, with gains across architectures including MATH15 (+3.2 points) and OpenVLA-OFT (+1.7 points). Physical robot experiments across five tasks demonstrate 98% success on both in-distribution and out-of-distribution conditions, improving from 84.7% and 32% respectively for naive fine-tuning. Adaptation completes in 31 seconds using 12 demonstrations on a single RTX 5090.

19. MemER: Scaling Up Memory for Robot Control via Experience Retrieval

Ajay Sridhar, Jennifer Pan, Satvik Sharma, Chelsea Finn — arXiv 2025, arXiv 2510.20328

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> *Ajay Sridhar, Jennifer Pan, Satvik Sharma, Chelsea Finn*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

MemER（ICLR 2026）是双系统记忆：高层 Qwen2.5-VL-7B 以约 1 Hz 看最近 8 帧（2 Hz 采样，约 4 s）加至多 8 张已选关键帧，输出低层 $\pi_{0.5}$ 要执行的文本子任务，同时从近期帧里提名值得记住的帧；提名索引经 1D 单链接聚类（d = 5）取中位数固化为关键帧，只增不删。三个需要数分钟记忆的 DROID 真机任务、每任务 20 trials，全部 >90%，与人类给子任务的上界持平。最该记的数字：Counting 任务错误舀取次数 MemER 1，No History 61，Long History（32 帧）12。

解决什么问题

给通用 VLA 加分钟级视觉记忆。直接拉长历史有两个代价：算力——32 帧（约 16 s）就要 1 s 推理，已到闭环可容忍的上限；脆弱——长上下文与专家动作之间的伪相关在策略自身的状态分布下失效，误差随历史变长复合。无差别下采样则塞进无关或重复的帧。已有的上下文扩展（Past-Token Prediction、SAM2Act）最多两打帧或 N = 10 最近帧，MemER 要从上千帧的整段 episode 里挑任务相关帧，且必须微调开源 VLM：API VLM 延迟 10–15 s，零样本也挑不出机器人相关的关键帧。

方法

分解：π(A_t | o_0:t) = π_l(A_t | l_t, q_t, l't) · *π_h(l'_t, J_t* | *I*{t–N+1:t}, K_t)（Eq. 3）。记忆只存在于高层。

高层：Qwen2.5-VL-7B-Instruct，冻结视觉编码器与投影层只训 LLM，4500 步、batch 256、96 H200 GPU 时，三任务合训一个模型。输入：任务指令 + 已选关键帧 K_t（实践 |K_t| ≤ 8）+ 最近 N = 8 帧视频（第三人称 + 腕部两路相机）；输出 JSON，含 current_subtask 与 keyframe_positions（1–8 的索引）。

写入与固化：每步提名的候选帧索引 J_t 汇入有序列表（保留重复），相邻索引距离 ≤ d = 5 的归为一簇，每簇取中位数为代表帧，代表帧集合即 K_t。重复提名让中位数偏向被多次选中的帧；早于 t – N + 1 – d 的簇不再重算。没有删除机制。

低层：$\pi_{0.5}$ 的 DROID 检查点微调，条件于子任务文本 + 当前图像 + 关节状态，马尔可夫；三任务共用一个策略，每任务 50 条长程演示 + 10–15 条干预演示（从常见失败状态遥操作回到分布内），18000 步、batch 128、48 H200 GPU 时；输出 15 步动作 chunk（15 Hz），开环执行 8 步。

标注：子任务由操作员采集时按键切分。关键帧半自动：取子任务边界帧为候选，每类子任务人工定一条规则（取首帧、末帧或不取，如 “look inside the bin” 取末帧，”reset scooper position” 不取），每段至多一帧。

模型合并：θ = 0.2 θ_pre + 0.8 θ_ft（α = 0.8），找回微调后丢掉的对低层卡顿、重试的鲁棒性（Fig. 6 右）。

部署：π_h 约 1 Hz、π_l 约 2 Hz，各在独立服务器异步运行；图像 320×180，15 Hz 流按 2 Hz 入队。

关键数字

Franka + 平行夹爪，ZED 第三人称 + miniZED 腕相机；每任务 20 trials。Object Search 每 trial 找三个物体，每列满分 60；Dust & Replace 每列满分 20。

指标	MemER	No History	Short History (8 帧)	Long History (32 帧)	Human HL
Object Search 取回次数 ↑	59	32	38	47	58
Object Search 最优路径次数 ↑	57	25	31	41	58
Counting 错误舀取数 ↓	1	61	26	12	0
Dust & Replace 擦下层 / 擦上层 ↑	20 / 19	5 / 4	14 / 14	11 / 11	19 / 19
Dust & Replace 放回下层 / 上层 ↑	18 / 20	5 / 7	11 / 12	12 / 12	18 / 17

模态消融（同协议）：Short History + Text 取回 40 / 最优 28 / 错舀 10 / 擦 16, 16 / 放回 7, 10；MemER + Text 59 / 49 / 13 / 20, 18 / 17, 20。加文本反而更差，尤其 Counting。

离线子任务预测（轨迹准确率 / 边界准确率，Object Search、Counting、Dust & Replace）：MemER 0.80/0.76、0.67/0.65、0.87/0.86；GPT-5 0.15/0.16、0.43/0.47、0.67/0.63；Gemini Robotics-ER 1.5 0.21/0.23、0.13/0.14、0.19/0.22。API 模型在线部署因 10–15 s 延迟全部失败。

局限与注意

论文自述：关键帧只增不删，小时级任务会撑爆；吞吐受 VLM 与 1 Hz / 2 Hz 调度限制，反应慢；记忆只有视觉；单一本体。

我的读法： - 记忆只在高层，低层 $\pi_{0.5}$ 马尔可夫、只收文本子任务，凡是子任务词表表达不了的信息（物体的精确位置）都到不了执行器。UniMem 正是攻这一点：TableClean 13%、TapScoopPour 7%；EventVLA 的复现里 RMBench 8.7%（低于无记忆 $\pi_{0.5}$ 的 10.4%）、RoboTwin-MeM 10.5%。 - 子任务词表封闭且逐任务写死，高层实际是在一个小动作列表上做选择；关键帧规则也是逐子任务手定的首帧/末帧。监督是子任务级的，KEMO 批评的就是这个。 - 所有基线共用同一低层、同一分层结构，只改高层输入；没有端到端记忆 VLA 基线，没有公开基准，三个任务自建、各 20 trials，59 vs 58 在噪声内。 - Dust & Replace 测试用训练 17 个物体中低层能抓的 9 个，Object Search 训练测试同 15 个物体，数字略偏乐观。 - 「文本有害」与 UniMem 的结论相反，但两者存的文本不同：MemER 存高层自己预测、可能出错的子任务，UniMem 存分类器判定的事件名。 - 成本：7B VLM 独占一台服务器以 1 Hz 运行，每次查询 2 路 × 8 帧 + ≤ 8 张关键帧；训练 96 + 48 H200 GPU 时。

与核心论文的关系

EventVLA 把 MemER 当双系统基线：RMBench 8.7%（$\pi_{0.5}$ 无记忆 10.4%，EventVLA 仅锚帧 67.8%），RoboTwin-MeM 10.5%（EventVLA 75.2%）。这与 MemER 自报的 >90% 不矛盾，是基准性质不同：RoboTwin-MeM 要求低层看到瞬态证据（哪只杯子），文本子任务带不动这种信息，除非词表预先编码了它。存什么：两者都存原图关键帧，但 MemER 只给 VLM 规划器看，EventVLA 把帧拼进执行 VLA 的视觉编码器；容量 MemER ≤ 8 张 + 每相机 8 张近期帧且不删，EventVLA 5 槽 FIFO + 锚帧。何时写：MemER 由 VLM 从刚滑出 8 帧窗口的帧里回顾式提名，再用 d = 5 单链接聚类 + 中位数去抖；EventVLA 的 KEM 头前瞻式预测未来 50 步哪些会成为关键帧，用 1-D NMS + 冷却去抖。监督：MemER 是子任务边界上的首帧/末帧规则，EventVLA 是 Qwen3-VL-235B 软标签。

与 TRACE 处于两个极端：TRACE 是 latent、无标签、每步门控写入 4/6 槽、挂在 ACT/DP 上，路径签名提供「我在轨迹哪一段」的进度信号；MemER 是显式图像 + 文本、1 Hz、7B VLM，进度靠近期帧加关键帧推断。与 SAI：SAI 的层级在两个智能体（两臂）之间而非抽象层级之间，其 30 步 GRU 历史 token 相当于 MemER 证明不够用的 Short History（错舀 26 次）；MemER 每任务 10–15 条从失败状态出发的干预演示，与 SAI 第三阶段的 DAgger 干预是同一类数据。

关联阅读

- Hi Robot（2502.19417）：MemER 直接沿用的分层 VLA 模板，No History 基线即此。
- Past-Token Prediction（2505.09561）：Counting 任务的出处，也是「长上下文引入伪相关」论证的来源。
- SAM2Act（2501.18564）：N = 10 最近帧的记忆架构，MemER 对比的短上下文代表。
- UniMem（2608.22869）：把记忆直接给低层，真机 80.0 vs MemER 43.5。
- EventVLA（2606.20092）：把 MemER 当双系统基线，RoboTwin-MeM 10.5%。

世界模型 / 视频动作模型中的记忆 / Memory inside World and Video Action Models

记忆服务于预测未来，预测再服务于动作。MemoryWAM 的「近期帧 + 事件边界锚帧 + gist token」与 EventVLA 的锚帧 + 事件帧同构。

1. MemoryWAM: Efficient World Action Modeling with Persistent Memory

Sizhe Yang, Juncheng Mu, Tianming Wei, Chenhao Lu, Xiaofan Li, Linning Xu, Zhengrong Xue, Zhecheng Yuan et al. — arXiv 2026, arXiv 2606.20562

世界动作模型的持久记忆：近期帧 + 事件边界锚帧 + 长程 gist token 三层，定制注意力兼顾细节与压缩；与 EventVLA 的锚帧 + 事件帧同构。

Abstract. Robust robotic manipulation in the real world requires not only an understanding of the current observation, but also memory and dynamics modeling. World action models (WAMs) possess these capabilities by jointly modeling visual foresight and actions conditioned on both current and historical observations, making them a promising paradigm for robotic manipulation. However, existing WAMs face a fundamental trade-off: methods with efficient inference typically condition only on a bounded window of recent observations and therefore struggle in non-Markovian environments, whereas methods that preserve long histories incur time and space costs that grow substantially with sequence length. To address this challenge, we introduce MemoryWAM, a world action model with efficient persistent memory. MemoryWAM uses a hybrid memory design that combines recent frames, event-boundary anchor frames, and compact gist tokens that summarize long-range history. A tailored attention mechanism enables retrieval of both detailed short-term context and compressed long-term context, supporting memory-dependent decision-making with reduced inference latency and GPU memory usage. Across long-horizon, memory-dependent manipulation tasks in both simulation and the real world, MemoryWAM outperforms strong vision-language-action (VLA) and WAM baselines while maintaining favorable computational efficiency.

2. MemoryVAM: Integrating Memory into Video Action Model for Robot Manipulation

Yuxin Jiang, Chang Yu, Yunuo Chen, Xiang Feng, Yin Yang, Nishank Gite, Chenfanfu Jiang — arXiv 2026, arXiv 2606.20679

视频动作模型的情景记忆：Perceiver 压缩器把逐帧 CLIP 嵌入压成记忆 token，Cue Gate 估计任务完成度；LIBERO-Mem 5→42.5%。

Abstract. Video-world-model policies learn action-relevant representations by predicting future observations. However, they condition on only a short observation window, which renders long-horizon manipulation non-Markovian when the correct action depends on earlier events that are no longer visible. We present MemoryVAM, an episodic memory mechanism for video-world-model policies. We employ a Recap-Cue (RC) module, in which a Perceiver-based Recap Compressor maps per-frame CLIP embeddings into compact memory tokens, and a lightweight Cue Gate estimates task completion from memory and language. These tokens are injected into both the video backbone and the action decoder, aligning policy imagination with episode progress and conditioning actions on history. Our model trains the memory module with video prediction, a delta-reconstruction auxiliary loss, and episode-boundary supervision, requiring no per-frame progress labels. The same mechanism applies to UNet and Diffusion Transformer (DiT) backbones by changing only the cross-attention injection interface. On LIBERO-Mem, our model improves average success from 5% to 42.5%. On real robots, it achieves 78.3% success on counting tasks, 80.0% on spatial recall, and 75.0% on sequential tracking. Project page: https://MemoryVAM.github.io/

3. TriVLA: A Triple-System-Based Unified Vision-Language-Action Model with Episodic World Modeling for General Robot Control

Zhenyang Liu, Yongchong Gu, Sixiao Zheng, Yanwei Fu, Xiangyang Xue, Yu-Gang Jiang — arXiv 2025, arXiv 2507.01424

三系统（VLM 语义 + 视频扩散动态感知 + 流匹配策略）形式化的情景世界模型，约 36 Hz。

Abstract. Recent advances in vision-language models (VLMs) have enabled robots to follow open-ended instructions and demonstrate impressive commonsense reasoning. However, current vision-language-action (VLA) frameworks primarily rely on static representations and limited temporal context, restricting agents to short-horizon, reactive behaviors and hindering robust generalization in dynamic embodied environments. Inspired by cognitive neuroscience theories of episodic memory, we propose, to our knowledge, one of the first formalized episodic world models in VLA, enabling embodied robots to accumulate, recall, and predict sequential experiences. As an instantiation of this concept, our unified TriVLA realizes the episodic world model through a triple-system architecture: integrating multimodal grounding from a pretrained VLM (System 2) and temporally rich dynamics perception from a video diffusion model (System 3). This enables the agent to accumulate and recall sequential experiences, interpret current contexts, and predict future environmental evolution. Guided by episodic representations that span both the past and anticipated future, the downstream policy (System 1) generates coherent, context-aware action sequences through flow-matching and cross-modal attention mechanisms. Experimental results show that TriVLA operates efficiently at approximately 36 Hz and consistently outperforms baseline models on standard benchmarks and challenging real-world manipulation tasks. It demonstrates strong long-horizon planning and open-ended intent understanding, showcasing the advantages of episodic world model-inspired reasoning for robust, generalizable robot intelligence. Project Page: https://zhenyangliu.github.io/TriVLA/.

多机器人协作与搭档记忆（SAI 一族） / Multi-Robot Collaboration and Partner Memory

搭档的相位是一个只能从历史推断的隐变量，所以多机器人协作是同一个非马尔可夫问题的另一张脸。目前的工作几乎都在解数据采集（HATS、Duet、Tri-Manual），还没有人把显式记忆模块用于搭档状态估计——这是一个明确的空白。

1. Making two action heads agree: coordination mechanisms and a runtime collapse certificate for flow-matching policies

Jinhui Sun, Wei Zhou, Bowen Yang, Xinliang Xiao, Li Yang — arXiv 2026, arXiv 2608.15748

双表征流匹配策略两分支的协调机制与运行时塌缩证书；「两个动作头选到不同模态」的形式化。

Abstract. A dual-representation flow-matching policy decodes each predicted motion into joint and end-effector spaces, and the residual between the two kinematically equivalent decodings provides a physically interpretable runtime signal. On multimodal tasks, however, independently sampled branches may choose different valid modes, causing false alarms. We study how to coordinate the two branches and at what cost. Across two robot environments and a non-robotic testbed, the tested mechanisms fall into four classes. An auxiliary latent shared by both branches but absent from the flow-matching construction is erased at the population optimum, a provable dead end confirmed within a prespecified 2% equivalence band. Sharing source noise can coordinate or anti-coordinate: its effect changes sign with the representation map and tracks the alignment of decoder mode basins. Consistency regularization gives intermediate coordination but reduces the valid-pair rate, while training-supported discrete partitions achieve near-ceiling coordination robustly. We further derive a chance-corrected coordination bound based only on each branch's Gini-Simpson diversity, yielding an attainable region and a label-free certificate that separates coordination from collapse when zero mismatch is ambiguous. On LIBERO-Plus, benign multimodality adds 1.57 percentage points of false alarms to the residual, which remains the strongest evaluated failure signal; the preregistered token intervention does not meet its false-alarm criterion or produce a seed-robust

detection change. Code, models, and per-run configurations are available at https://github.com/kimo423/dual-head-coordination.

2. AutoIntervene: Calibrated Intervention for Action-Chunking Imitation Learning Policies

Jinhe Tang, Weiming Zhi — arXiv 2026, arXiv 2608.07065

> **arXiv 2608.07065** · arXiv 2026 · 提交 2026-08-07 · 分类 `multi` / 多机器人协作与搭档记忆（SAI 一族）
> *Jinhe Tang, Weiming Zhi*
> arXiv · PDF · 仓库内英文 PDF · 中文翻译 PDF

一句话定位

AutoIntervene 让机器人自己决定何时把控制权交给操作者、何时收回：把当前多视角视觉嵌入和策略即将执行的动作块，与成功示教构成的视觉-动作支持记忆做检索比对，阶段局部支持掉线就交人，全局支持恢复就交回，两个方向的阈值分别从留出示教的分位数校准。七个真机双臂任务上，两轮干预后平均成功率 30.9% → 80.0%，累计额外操作者控制时间 122.9 s；人工决定切换的同一流程只到 68.6%、用时 179.9 s。

解决什么问题

动作块策略（ACT、Diffusion Policy、Flow Matching）被感知误差或执行漂移带出示教分布后，仍会输出平滑却与状态不符的动作块——抓歪、子任务卡住，却不会停。前人各缺一块：DART 靠噪声注入扩大覆盖，部署时没有检测；DAgger/HG-DAgger 要操作者全程盯着并手动决定接管与交回；机器人门控的 LazyDAgger（策略-专家动作差异）和 RND-DAgger（状态新颖度）用同一种信号管两个方向，恢复期间差异或新颖度仍高，前者交不回去、后者刚交回又被触发；运行时监控器（Sentinel、FAIL-Detect、PATCH、Rewind-IL）只做检测、暂停或回退，不把恢复段变成训练数据。

方法

支持记忆。 用训练当前策略的全部成功轨迹建库：每条轨迹每个起点 u 产出一条 $m_{i,u} = (E_{i,u}, A_{i,u})$，即多视角视觉嵌入（策略自带的 DINOv3 ConvNeXt-Base 编码器）加从 u 起的 $H_r = 40$ 步记录动作块。查询 $Q_t = (E_t, A_t)$ 是当前观测嵌入加策略预测动作块的前 H_r 步，动作按左右臂分两组。视觉相似度取跨视角余弦相似度的最小值（Eq. 1），任一相机不匹配都压低分数。

阶段局部 vs 全局支持。 操作者控制时（$\beta = op$）策略在后台继续预测，检索整库：恢复可能改变了物体状态或完成了部分任务，策略应能从任何合法阶段重新获得支持，这是全局支持。策略控制时（$\beta = pol$）相似观测可能出现在需要不同动作的阶段，整库检索会匹配错阶段，于是接管时先全库匹配一次，选匹配最强的 $J = 16$ 条轨迹，每条给一个从匹配点起最多 $B = 40$ 条的前向窗口，之后只在窗口内检索；每执行 $U_{win} = 6$ 个策略动作，窗口按新匹配条目单调前移（Eq. 4），这是阶段局部支持。检索取相似度最高的 $K = 16$ 条邻居，对每组动作算按维度标准差归一化的欧氏距离（Eq. 2），每组取最近的 $M = 3$ 条，风险由平均距离最大的那组决定；视觉支持 s 取同批参考的相似度均值，动作风险 $\bar{r}$ 取最近 $W = 5$ 次评估的均值，视觉支持不平滑以便立即捕捉对应关系的丢失。

分位数校准双阈值。 每任务留出 6 条成功示教作 D_{cal}，不进训练也不进记忆库。每轮部署前分别在 pol 和 op 两种检索范围下对 D_{cal} 跑同样的支持评估，阈值取经验分位数：$\theta_s = Q_{\alpha_s}(S_{cal})$ 为视觉支持下限，$\theta_r = Q_{1-\alpha_r}(R_{cal})$ 为动作风险上限（Eq. 5）；策略侧 $(\alpha_s, \alpha_r) = (0.05, 0.05)$，操作者侧 $(0.30, 0.30)$。提案被接受需同时 $s \geq \theta_s$

且 $\bar{r} \le \theta\_r$。策略控制下连续 $L\_pol = 2$ 次被拒（含所有窗口走到末尾，重复抓失败靠这条兜底）就交给操作者；操作者控制下后台提案连续 $L\_op = 2$ 次被接受就交回。策略侧评估 5 Hz，操作者侧 30 Hz。

干预段回流训练。 每段操作者控制（从交人到交回或 episode 结束）作为独立干预轨迹保存，只保留成功 rollout 里的段，每轮每任务取 5 个成功 rollout。下一轮 $D^{(k+1)} = D^{(k)} \cup \Delta D^{(k)}$，采样按 $\lambda\_mix = 2/3$ 混合（旧数据 : 新干预 = 2 : 1）做标准行为克隆；再用新策略的编码器重建记忆库、在 D_cal 上重新校准阈值、重新部署。

关键数字

两台 AgileX PiPER-X 6-DoF 臂，三路 RGB（俯视 + 双腕），28 维状态（14 关节/夹爪 + 14 力矩），14 维动作，H = 100。每任务 36 条初始示教中 30 条训练、6 条校准；除特别说明，每个成功率来自 25 次无辅助真机 rollout。

七任务主基准	Initial	Human R1	Human R2	AutoIntervene R1	AutoIntervene R2	Additional Full Data
平均成功率	30.9%	45.7%	68.6%	59.4%	**80.0%**	56.0%
时间	977.4 s（示教总时长）	+77.7 s	+179.9 s（累计）	+66.1 s	+122.9 s（累计）	1442.9 s（示教总时长，含初始）

AutoIntervene R2 逐任务（%）：Peg Disassembly 16 → 72、Potato Transfer 44 → 76、Towel Folding 56 → 96、Towel Bagging 24 → 60、Lidded Box Packing 8 → 100、Plant Sorting 24 → 72、Towel Box Packing 44 → 84；Human R2 在 Potato Transfer（80）和 Towel Folding（100）反超。相对 Additional Full Data 新增的示教时间，AutoIntervene 少用约 74%。

长时程任务（三轮，%）	Initial	R1	R2	R3	Additional Full Data
Two-Towel Box Packing（ACT）	28	44	64	88	52
Towels-and-Cable Bagging（ACT）	8	20	28	48	28
Two-Towel Box Packing，Diffusion Policy 头	32	40	56	92	44
Two-Towel Box Packing，Flow Matching 头	32	36	48	80	40

切换质量（Lidded Box Packing 开盖阶段，同一 20,000 步检查点，10 次 5 cm 箱体平移扰动 + 10 次正常 rollout，最多 5 个切换周期）：

方法	Cut-in 召回	Cut-in 精度	Cut-out 召回	Cut-out 精度	正常 rollout 误触发率
AutoIntervene	1.00	1.00	1.00	1.00	0.00
LazyDAgger	0.40	1.00	0.00	N/A	0.80
RND-DAgger（W_rec = 5）	0.80	0.16	0.00	0.00	0.90
RND-DAgger（W_rec = 30）	0.80	0.16	0.75	0.12	0.90
去掉视觉支持	0.00	N/A	N/A	N/A	0.00
去掉动作风险	0.90	1.00	0.56	0.56	0.00

另一组消融把目标物体降低而外观几乎不变，使标称抓取失败：10 次 rollout 里带窗口前移的 cut-in 召回 1.00，去掉窗口前移只有 0.30。

局限与注意

论文没有单列局限，结论只说未来要把在线校准扩展到更多任务、扰动和操作者。自己补几点：

- 「不手调分数阈值」是把手调挪到了尾部比例 α（0.05 / 0.30）、持久长度 L = 2、窗口 B、邻居数 K、M 等超参上；校准集每任务只有 6 条轨迹，5% 分位数的估计方差论文未报告。
- 切换质量对比只在一个任务、一种扰动（5 cm 平移）、各 10 次 rollout 上做，LazyDAgger/RND-DAgger 是作者移植到 ACT 部署环境的实现。
- 25 次 rollout 的粒度是 4%，Human R2 与 AutoIntervene R2 多项任务只差一两次；Human 基线的操作者人数未说明，其控制时间也受人的反应速度影响。
- 支持记忆只由成功数据构成，评估的是「提案是否落在示教支持内」而非任务是否在推进：在支持内但停滞的行为（重复抓失败）只能靠窗口走到末尾这一间接机制触发。
- 只保留成功 rollout 的干预段；每轮干预数据量由「5 个成功 rollout」这个人为设定决定，不由支持分数自适应。

与核心论文的关系

SAI 的阶段三是在两台耦合移动操作臂协调失败附近对 Robot A 做稀疏 DAgger 式人工干预，干预数据约占基线集的 30%，论文承认恢复没有饱和、干预多少和在哪里干预仍靠人。AutoIntervene 把「何时交人、何时交回」自动化，并用支持记忆定义「哪里」：阶段局部支持跌破阈值就是干预点，全局支持恢复就是交回点。它没有回答「多少」——每轮 5 个成功 rollout 仍是人定的；也没有涉足多机器人，实验是单控制器双臂，但按臂分组、由支持最弱的组决定风险的设计，可以映射为 SAI 里只对 Robot A 门控。

与 TRACE 要分清两种「记忆」。TRACE 记的是本 episode 已执行的历史：路径签名寻址 K 槽 latent，解决延迟证据下的分支歧义。AutoIntervene 的支持记忆是跨 episode 的：训练示教的（视觉嵌入, 动作块）对，回答「成功执行在这里长什么样」，不存本 episode 历史；它对历史的唯一利用是单调前移的检索窗口，一个只进不退的阶段指针，论文在动机处直接引用 TRACE 和 SCIL 说明视觉别名。两者殊途同归于「有纪律的状态更新」：TRACE 门控写入，AutoIntervene 用分方向阈值 + 持久计数 + 不回退窗口做滞回，Table V 里 RND-DAgger 单阈值管双向的失败正是缺少这种纪律的反例。

关联阅读

- LazyDAgger（2104.00053）：基于动作差异、带滞回带的机器人门控 DAgger，直接基线，Table V 显示其交回控制失败。
- Rewind-IL（2604.16683）：同组，校准的块间差异检测失败后回退到语义验证过的安全状态，「检测后重生」而非「检测后交人」。
- PATCH（2606.16690）：同组，以当前动作块为条件的 latent patch 新颖度监控，做暂停/恢复，同样把动作块纳入监控信号。
- TriPilot-FF（2602.09888）：同组的力反馈遥操作系统，AutoIntervene 的 leader-follower 接管接口基于它。
- IntervenGen（2405.01472）：从少量人工干预自动生成大量纠正数据，与 AutoIntervene 互补：一个决定何时采干预，一个把每次干预放大。

3. Tri-Manual Visuomotor Imitation Learning of Robot Policies

James Zhao, Mingyuan Ba, Weiming Zhi — arXiv 2026, arXiv 2607.25731

单操作者为三臂系统演示：DATS 按任务顺序与手臂使用约束离线重排演示时序，训练单一同步三臂策略。

Abstract. Bimanual teleoperation provides an effective way to collect robot demonstrations, but it assumes that the operator and robot have matching numbers of simultaneous control channels. This assumption breaks for tri-manual systems: the robot can coordinate three arms concurrently, whereas a single operator can continuously control only two. Pairwise mode switching may therefore record otherwise independent motions sequentially, causing behaviour cloning to reproduce delays imposed by the interface rather than required by the task. We present TriManPolicy, a tri-manual imitation learning system that allows one operator to demonstrate behaviours for three arms. Its central component is Dependency-Aware Tri-Arm Scheduling (DATS). The key idea is to preserve the demonstrated arm motions while reconsidering when they occur. DATS retimes demonstrations offline by preserving local sensorimotor segments of fixed duration and repositioning them according to constraints on task order and arm usage that are reviewed by a human. The resulting data train a single synchronous policy for all three arms, while deployment requires neither the dependency graph nor the scheduler. Across six challenging tasks performed in the real world, policies trained on demonstrations retimed by DATS exhibit more efficient coordination while maintaining comparable observed task success. Offline analysis further shows that DATS changes the supervision across arms rather than merely removing idle periods. Project videos and additional material are available at https://aus.bot/trimanpolicy/.

4. Duet: Dual-Robot Understanding via Efficient Teaching

Yiqi Zhao, Ruohai Ge, Celina Shiyu Wang, Junjie Ye, Muchen Xu, Minhao Li, Sergey Zakharov, Basile Van Hoorick et al. — arXiv 2026, arXiv 2606.20990

异构双机器人（Unitree G1 + Vega1）协作：人-人演示预训练 + 少量双机器人遥操微调，人类数据采集加速 5.4 倍。

Abstract. Dual-robot collaboration enables tasks that exceed the reach and payload of a single robot, such as collaboratively transporting objects across environments and executing coordinated handovers. Data acquisition is the primary bottleneck for training these systems. To this end, we introduce DUET, a dual-robot learning framework for mobile manipulation. For efficient data collection, we create a unified dual-embodiment synchronized VR-based teleoperation system for in-domain heterogeneous robot data collection. We further develop a complementary tracking pipeline that records human-human coordination and collaborative mobile manipulation priors. To allow efficient learning, we introduce an Action Chunking Transformer based architecture that first pretrains collaborative policies on efficient human-human demonstrations, before finetuning them on a minimal set of real-robot teleoperation trajectories. We develop a benchmark of four collaborative tasks to evaluate our framework using a Unitree G1 humanoid and a Dexmate Vega1 mobile manipulator. The results demonstrate that harnessing human priors not only yields superior task performance compared to baselines trained only on robot data, but also reduces the total human effort required for data collection. Our human data collection pipeline achieves 5.4x acceleration on average from teleoperation, but we perform equally or better than robot-only data trained policies across all tasks. Our project page is available at https://zhaoy37.github.io/Duet/.

5. HATS: A Human-Agent Teleoperation System for Multi-Arm Data Collection

Zesen Lin, Jian-Jian Jiang, Haoming Cen, Xiao-Ming Wu, Dandan Zhang, Wei-Shi Zheng — arXiv 2026, arXiv 2606.16491

单人 + MLLM 智能体的多臂遥操作：两臂人控、两臂智能体控，语音纠正；数据效率与双人专家团队相当。与 SAI 相邻编号，同样在解「单操作者采多机器人数据」。

Abstract. Many real-world manipulation scenarios, such as handling complex collaborative tasks and dealing with large workspaces, require coordination of more than two robotic arms. Consequently, an effective multi-arm teleoperation system is required to collect demonstrations for training coordinated multi-arm manipulation policies. However, existing teleoperation frameworks mainly focus on single-operator or multi-operator setups, facing a practical trade-off between the cognitive load placed on a single operator and the coordination cost incurred by multiple operators. To address this problem, we introduce HATS, a human-agent teleoperation system that enables a single human operator, assisted by an MLLM-based agent, to collect data for multi-arm manipulation tasks. Our system decouples the control space: two primary arms are directly teleoperated by the human, while two assistive arms are controlled by a training-free agent that handles sub-tasks. In addition, the human operator can use voice commands to prevent collisions and correct assistive arm behaviors during execution. Extensive evaluations demonstrate that HATS achieves data collection efficiency and success rates comparable to expert dual-human teams. Moreover, downstream policy evaluations demonstrate the efficacy and quality of the data collected through HATS.

6. TriPilot-FF: Coordinated Whole-Body Teleoperation with Force Feedback

Zihao Li, Yanan Zhou, Ranpeng Qiu, Hangyu Wu, Guoqiang Ren, Weiming Zhi — arXiv 2026, arXiv 2602.09888

TRACE 与 SAI 的数据采集系统：脚踏板 + lidar 触觉的全身遥操作，臂侧力反射，反馈信号并入 ACT 策略。

Abstract. Mobile manipulators broaden the operational envelope for robot manipulation. However, the whole-body teleoperation of such robots remains a problem: operators must coordinate a wheeled base and two arms while reasoning about obstacles and contact. Existing interfaces are predominantly hand-centric (e.g., VR controllers and joysticks), leaving foot-operated channels underexplored for continuous base control. We present TriPilot-FF, an open-source whole-body teleoperation system for a custom bimanual mobile manipulator that introduces a foot-operated pedal with lidar-driven pedal haptics, coupled with upper-body bimanual leader-follower teleoperation. Using only a low-cost base-mounted lidar, TriPilot-FF renders a resistive pedal cue from proximity-to-obstacle signals in the commanded direction, shaping operator commands toward collision-averse behaviour without an explicit collision-avoidance controller. The system also supports arm-side force reflection for contact awareness and provides real-time force and visual guidance of bimanual manipulability to prompt mobile base repositioning, thereby improving reach. We demonstrate the capability of TriPilot-FF to effectively ``co-pilot'' the human operator over long time-horizons and tasks requiring precise mobile base movement and coordination. Finally, we incorporate teleoperation feedback signals into an Action Chunking with Transformers (ACT) policy and demonstrate improved performance when the additional information is available. We release the pedal device design, full software stack, and conduct extensive real-world evaluations on a bimanual wheeled platform. The project page of TriPilot-FF is http://bit.ly/46H3ZJT.

7. Rethinking Bimanual Robotic Manipulation: Learning with Decoupled Interaction Framework

Jian-Jian Jiang, Xiao-Ming Wu, Yi-Xiang He, Ling-An Zeng, Yi-Lin Wei, Dandan Zhang, Wei-Shi Zheng — ICCV 2025, arXiv 2503.09186

每臂独立模型 + 选择性交互模块的解耦双臂框架，RoboTwin +23.5%，可扩展到多智能体；单机版的「去中心化 + 局部交互」。

Abstract. Bimanual robotic manipulation is an emerging and critical topic in the robotics community. Previous works primarily rely on integrated control models that take the perceptions and states of both arms as inputs to directly predict their actions. However, we think bimanual manipulation involves not only coordinated tasks but also various uncoordinated tasks that do not require explicit cooperation during execution, such as grasping objects with the closest hand, which integrated control frameworks ignore to consider due to their enforced cooperation in the early inputs. In this paper, we propose a novel decoupled interaction framework that considers the characteristics of different tasks in bimanual manipulation. The key insight of our framework is to assign an independent model to each arm to enhance the learning of uncoordinated tasks, while introducing a selective interaction module that adaptively learns weights from its own arm to improve the learning of coordinated tasks. Extensive experiments on seven tasks in the RoboTwin dataset demonstrate that: (1) Our framework achieves outstanding performance, with a 23.5% boost over the SOTA method. (2) Our framework is flexible and can be seamlessly integrated into existing methods. (3) Our framework can be effectively extended to multi-agent manipulation tasks, achieving a 28% boost over the integrated control SOTA. (4) The performance boost stems from the decoupled design itself, surpassing the SOTA by 16.5% in success rate with only 1/6 of the model size.

8. IntervenGen: Interventional Data Generation for Robust and Data-Efficient Robot Imitation Learning

Ryan Hoque, Ajay Mandlekar, Caelan Garrett, Ken Goldberg, Dieter Fox — IROS 2024, arXiv 2405.01472

从少量人类干预自动生成大量纠正数据，鲁棒性提升至 39 倍；直接可补 SAI 阶段三的干预覆盖不足。

Abstract. Imitation learning is a promising paradigm for training robot control policies, but these policies can suffer from distribution shift, where the conditions at evaluation time differ from those in the training data. A popular approach for increasing policy robustness to distribution shift is interactive imitation learning (i.e., DAgger and variants), where a human operator provides corrective interventions during policy rollouts. However, collecting a sufficient amount of interventions to cover the distribution of policy mistakes can be burdensome for human operators. We propose IntervenGen (I-Gen), a novel data generation system that can autonomously produce a large set of corrective interventions with rich coverage of the state space from a small number of human interventions. We apply I-Gen to 4 simulated environments and 1 physical environment with object pose estimation error and show that it can increase policy robustness by up to 39x with only 10 human interventions. Videos and more results are available at https://sites.google.com/view/intervengen2024.

9. Mobile ALOHA: Learning Bimanual Mobile Manipulation with Low-Cost Whole-Body Teleoperation

Zipeng Fu, Tony Z. Zhao, Chelsea Finn — CoRL 2024, arXiv 2401.02117

低成本全身遥操作的双臂移动操作系统，静态 ALOHA 数据协同训练；SAI 的硬件与任务范式源头之一。

Abstract. Imitation learning from human demonstrations has shown impressive performance in robotics. However, most results focus on table-top manipulation, lacking the mobility and dexterity necessary for generally useful tasks. In this work, we develop a system for imitating mobile manipulation tasks that are bimanual and require whole-body control. We first present Mobile ALOHA, a low-cost and whole-body teleoperation system for data collection. It augments the ALOHA system with a mobile base, and a whole-body teleoperation interface. Using data collected with Mobile ALOHA, we then perform supervised behavior cloning and find that co-training with existing static ALOHA datasets boosts performance on mobile manipulation tasks. With 50 demonstrations for each task, co-training can increase success rates by up to 90%, allowing Mobile ALOHA to autonomously

complete complex mobile manipulation tasks such as sauteing and serving a piece of shrimp, opening a two-door wall cabinet to store heavy cooking pots, calling and entering an elevator, and lightly rinsing a used pan using a kitchen faucet. Project website: https://mobile-aloha.github.io

背景：记忆机制与轨迹描述子 / Background: Memory Mechanisms and Trajectory Descriptors

理解三篇核心论文需要的最少背景：外部记忆与记忆 token 的源头（MANN、RMT、Titans）、路径签名（Signatory、CILO）、交互式模仿（DAgger）。

1. Titans: Learning to Memorize at Test Time

Ali Behrouz, Peilin Zhong, Vahab Mirrokni — arXiv 2025, arXiv 2501.00663

测试时学习记忆的神经长期记忆模块，注意力为短期、神经记忆为长期；latent 记忆的 LLM 侧最新参照。

Abstract. Over more than a decade there has been an extensive research effort on how to effectively utilize recurrent models and attention. While recurrent models aim to compress the data into a fixed-size memory (called hidden state), attention allows attending to the entire context window, capturing the direct dependencies of all tokens. This more accurate modeling of dependencies, however, comes with a quadratic cost, limiting the model to a fixed-length context. We present a new neural long-term memory module that learns to memorize historical context and helps attention to attend to the current context while utilizing long past information. We show that this neural memory has the advantage of fast parallelizable training while maintaining a fast inference. From a memory perspective, we argue that attention due to its limited context but accurate dependency modeling performs as a short-term memory, while neural memory due to its ability to memorize the data, acts as a long-term, more persistent, memory. Based on these two modules, we introduce a new family of architectures, called Titans, and present three variants to address how one can effectively incorporate memory into this architecture. Our experimental results on language modeling, common-sense reasoning, genomics, and time series tasks show that Titans are more effective than Transformers and recent modern linear recurrent models. They further can effectively scale to larger than 2M context window size with higher accuracy in needle-in-haystack tasks compared to baselines.

2. Explorative Imitation Learning: A Path Signature Approach for Continuous Environments

Nathan Gavenski, Juarez Monteiro, Felipe Meneguzzi, Michael Luck, Odinaldo Rodrigues — arXiv 2024, arXiv 2407.04856

CILO：模仿学习中用路径签名做轨迹的非参数表征以自动编码约束；签名进入模仿学习的先例。

Abstract. Some imitation learning methods combine behavioural cloning with self-supervision to infer actions from state pairs. However, most rely on a large number of expert trajectories to increase generalisation and human intervention to capture key aspects of the problem, such as domain constraints. In this paper, we propose Continuous Imitation Learning from Observation (CILO), a new method augmenting imitation learning with two important features: (i) exploration, allowing for more diverse state transitions, requiring less expert trajectories and resulting in fewer training iterations; and (ii) path signatures, allowing for automatic encoding of constraints, through the creation of non-parametric representations of agents and expert trajectories. We compared CILO with a baseline and two leading imitation learning methods in five environments. It had the best overall performance of all methods in all environments, outperforming the expert in two of them.

3. Recurrent Memory Transformer

Aydar Bulatov, Yuri Kuratov, Mikhail S. Burtsev — NeurIPS 2022, arXiv 2207.06881

递归记忆 Transformer：用记忆 token 在段之间传递信息；EventVLA 归入循环范式的源头之一。

Abstract. Transformer-based models show their effectiveness across multiple domains and tasks. The self-attention allows to combine information from all sequence elements into context-aware representations. However, global and local information has to be stored mostly in the same element-wise representations. Moreover, the length of an input sequence is limited by quadratic computational complexity of self-attention. In this work, we propose and study a memory-augmented segment-level recurrent Transformer (RMT). Memory allows to store and process local and global information as well as to pass information between segments of the long sequence with the help of recurrence. We implement a memory mechanism with no changes to Transformer model by adding special memory tokens to the input or output sequence. Then the model is trained to control both memory operations and sequence representations processing. Results of experiments show that RMT performs on par with the Transformer-XL on language modeling for smaller memory sizes and outperforms it for tasks that require longer sequence processing. We show that adding memory tokens to Tr-XL is able to improve its performance. This makes Recurrent Memory Transformer a promising architecture for applications that require learning of long-term dependencies and general purpose in memory processing, such as algorithmic tasks and reasoning.

4. Signatory: differentiable computations of the signature and logsignature transforms, on both CPU and GPU

Patrick Kidger, Terry Lyons — ICLR 2021, arXiv 2001.00706

路径签名 / 对数签名的 GPU 可微计算库，TRACE 轨迹键的计算基础。

Abstract. Signatory is a library for calculating and performing functionality related to the signature and logsignature transforms. The focus is on machine learning, and as such includes features such as CPU parallelism, GPU support, and backpropagation. To our knowledge it is the first GPU-capable library for these operations. Signatory implements new features not available in previous libraries, such as efficient precomputation strategies. Furthermore, several novel algorithmic improvements are introduced, producing substantial real-world speedups even on the CPU without parallelism. The library operates as a Python wrapper around C++, and is compatible with the PyTorch ecosystem. It may be installed directly via \texttt{pip}. Source code, documentation, examples, benchmarks and tests may be found at \texttt{\url{https://github.com/patrick-kidger/signatory}}. The license is Apache-2.0.

5. One-shot Learning with Memory-Augmented Neural Networks

Adam Santoro, Sergey Bartunov, Matthew Botvinick, Daan Wierstra, Timothy Lillicrap — ICML 2016, arXiv 1605.06065

内容寻址外部记忆的经典（NTM 系），TRACE 表 2 的 LRU 外部记忆对照即受其启发。

Abstract. Despite recent breakthroughs in the applications of deep neural networks, one setting that presents a persistent challenge is that of “one-shot learning.” Traditional gradient-based networks require a lot of data to learn, often through extensive iterative training. When new data is encountered, the models must inefficiently relearn their parameters to adequately incorporate the new information without catastrophic interference. Architectures with

augmented memory capacities, such as Neural Turing Machines (NTMs), offer the ability to quickly encode and retrieve new information, and hence can potentially obviate the downsides of conventional models. Here, we demonstrate the ability of a memory-augmented neural network to rapidly assimilate new data, and leverage this data to make accurate predictions after only a few samples. We also introduce a new method for accessing an external memory that focuses on memory content, unlike previous methods that additionally use memory location-based focusing mechanisms.

6. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning

Stephane Ross, Geoffrey J. Gordon, J. Andrew Bagnell — AISTATS 2011, arXiv 1011.0686

交互式模仿学习的理论基础，SAI 阶段三与 AutoIntervene 的方法源头。

Abstract. Sequential prediction problems such as imitation learning, where future observations depend on previous predictions (actions), violate the common i.i.d. assumptions made in statistical learning. This leads to poor performance in theory and often in practice. Some recent approaches provide stronger guarantees in this setting, but remain somewhat unsatisfactory as they train either non-stationary or stochastic policies and require a large number of iterations. In this paper, we propose a new iterative algorithm, which trains a stationary deterministic policy, that can be seen as a no regret algorithm in an online learning setting. We show that any such no regret algorithm, combined with additional reduction assumptions, must find a policy with good performance under the distribution of observations it induces in such sequential settings. We demonstrate that this new approach outperforms previous approaches on two challenging imitation learning problems and a benchmark sequence labeling problem.

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Beamer 讲稿版幻灯片	`slides/awesome_memory_vla_deck.pdf`
BibTeX	`awesome_memory_vla.bib`

复现：`python3 scripts/build_manifest.py && python3 scripts/make_notes.py && python3 scripts/make_bib.py && python3 scripts/make_readme.py && python3 scripts/build_pdfs.py`。翻译：

`bash scripts/translate_core.sh <KEY>` （DeepSeek）或 `bash scripts/translate_fallback_google.sh` （免 key）。